

Probability and Statistics

Introduction

Importance Area

- Demography
- Business
- Policy making
- Public Health
- Astronomy
- Environment

Limitations

- It is not cent percent accurate.
- It does not deal with individuals.
- It does not explain qualitative variables.
- Statics can be misused.

Data Source

- Primary Source
- Secondary Source

Graphical Representation

- Bar Graph
 - Single Bar Graph
 - Multiple Bar Graph
 - Subdivided Bar Graph
- Pie Chart
- Line Graph
- Histogram
 - frequency curve
 - frequency polygon
- Ogive curve

Bar Graph

Age	17	18	19	20	22	24
Frequency	20	22	24	30	15	7

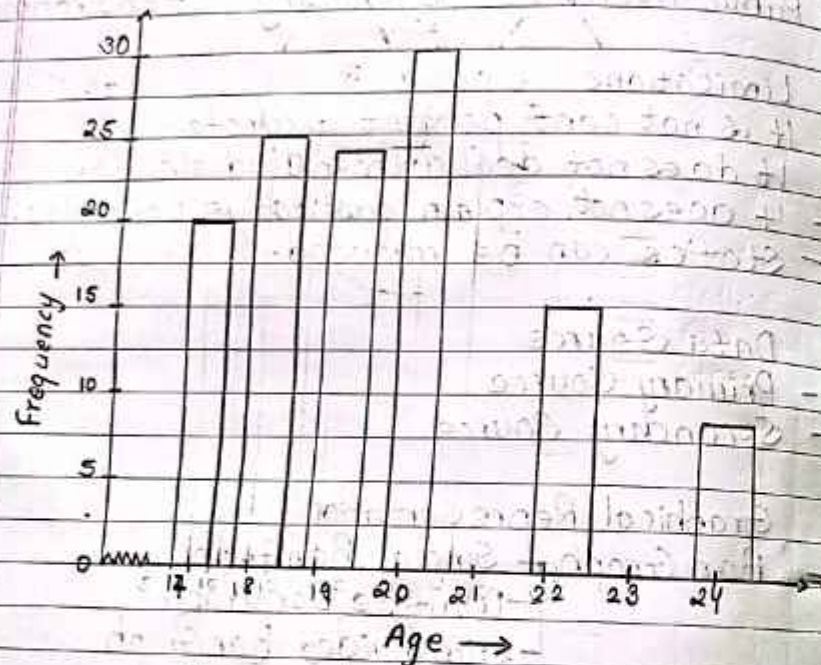
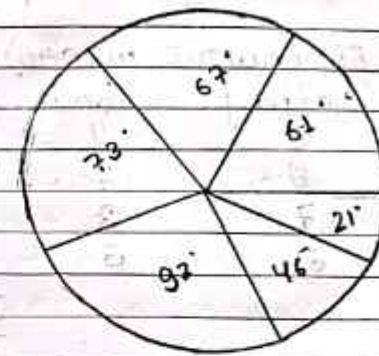


fig: Age Vs Frequency

Pie Chart



Age	Frequency	Degree	
17	20	61°	$\frac{20}{118} \times 360^\circ$
18	22	67°	
19	24	73°	
20	30	92°	
22	15	46°	
24	7	21°	
	118		

Multiple Bar Graph

- Use in time series comparison of variables

Year	Expenditure in (000)			
	Rent	clothing	food	other
2021	10	5	4	5
2022	11	9	7	6
2023	15	7	3	8
2024	18	6	6	9

☐ Rent
☐ clothing
☐ food
☐ other

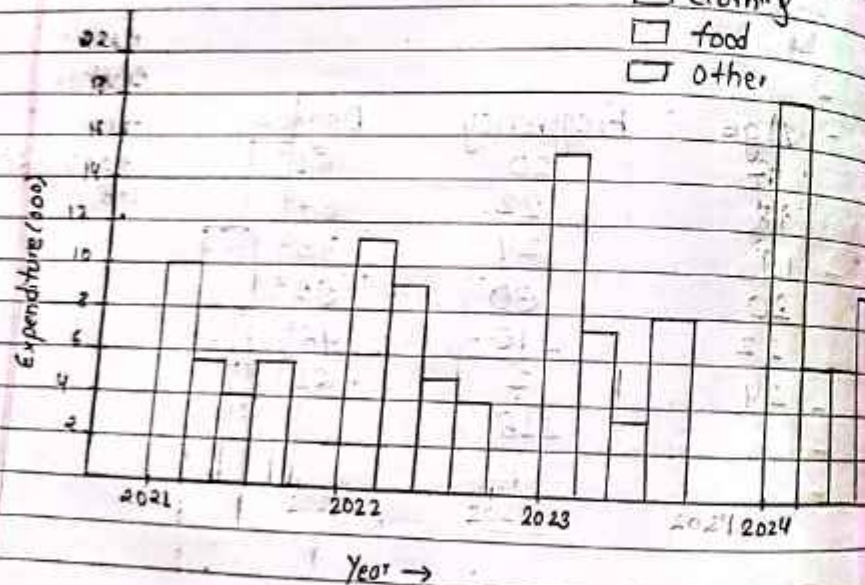


Fig:

Sub-divided Bar Graph

- Used in comparison between variables at a point of time.

Student	Expenditure in (000)				Total
	Rent	clothing	food	other	
A	10	5	4	5	24
B	11	9	7	6	33
C	15	7	3	8	33
D	18	6	6	9	39

Rent - ☐
 Clothing - ☐
 food - ☐
 other - ☐

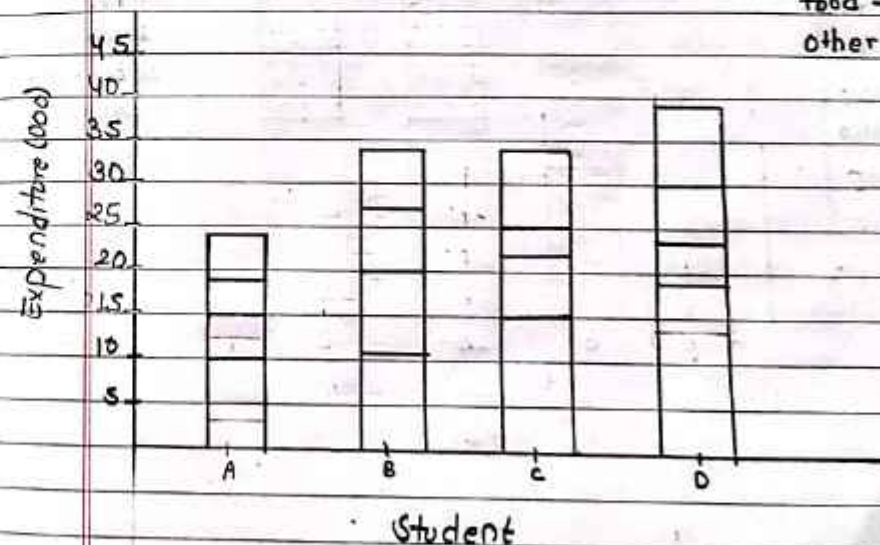
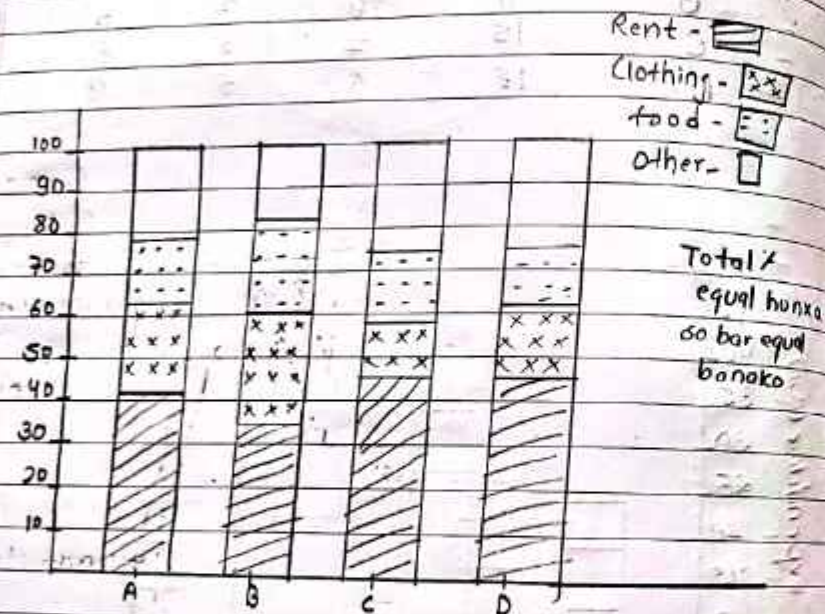


Fig:

Percentage Sub-divided Bar Graph

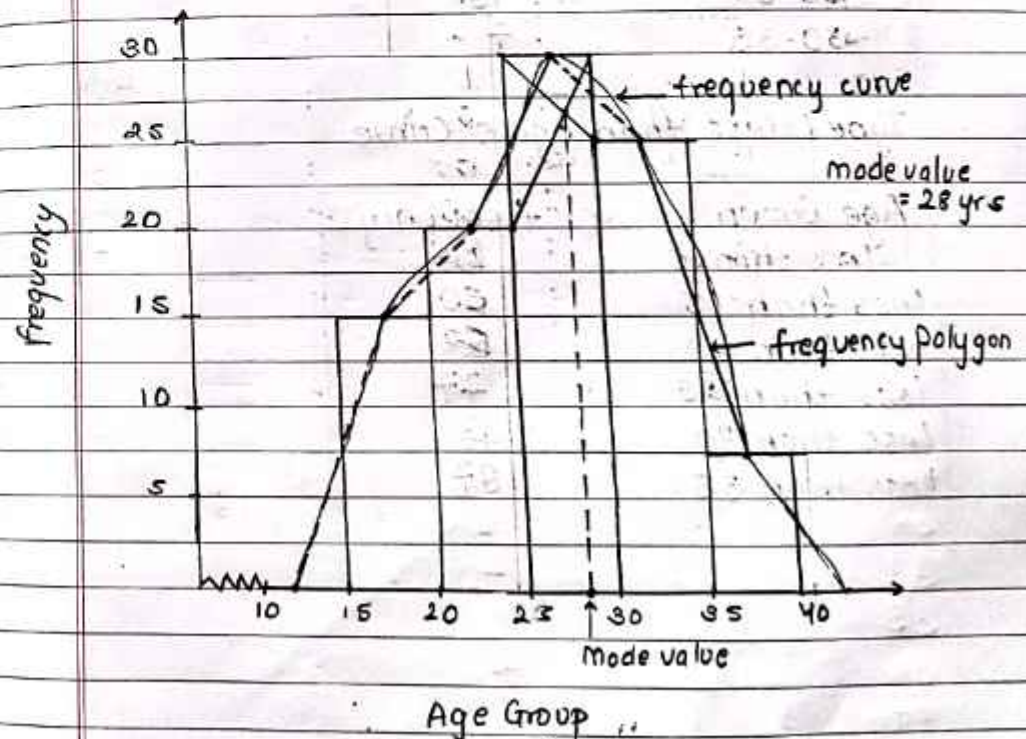
Student	Expenditure (000)			
	Rent	clothing	food	other
A	$\frac{10 \times 100}{41} = 24$	21	17	22
B	33	27	22	18
C	46	21	9	24
D	46	15	15	24



Histogram

- for continuous series - exclusive
- To calculate mode value

Age Group	frequency
15-20	15
20-25	20
25-30	30
30-35	25
35-40	10



Ogive Curve

- Cumulative Curve
- Use in Continuous series
- Used to calculate median value
- Types
 - More than ogive curve
 - ~~more~~ less than ogive curve

Age group	frequency
5-10	10
10-15	20
15-20	22
20-25	25
25-30	13
30-35	7

Type 1: less than ogive curve

Age Group	frequency
less than 10	10
less than 15	30
less than 20	52
less than 25	77
less than 30	90
less than 35	97

2: More than ogive curve

Age Group	frequency
More than 5	97
More than 10	87
More than 15	67
More than 20	45
More than 25	20
More than 30	7

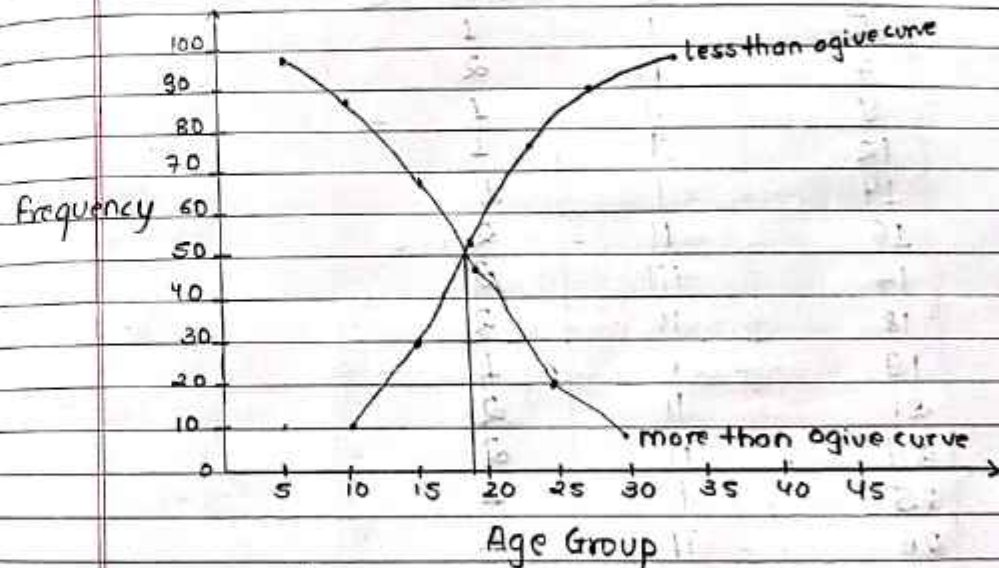


Fig. Ogive curve

Median age = 18 years

Frequency distribution

Marks of Student

46, 12, 14, 16, 17, 21, 34, 35, 40, 42, 3, 7, 9, 2, 17, 18, 24, 3, 7, 18, 4, 40, 37, 36, 32, 29, 26, 39, 41, 5, 3, 2, 36, 26, 16, 17, 19, 21, 24, 25.

Marks	Tally	Frequency
2		2
3		3
4		1
5		1
7		2
9		1
12		1
14		1
16		2
17		3
18		2
19		1
21		2
24		2
25		1
26		2
29		1
32		1
34		1
35		1
36		2
37		1
39		1
40		2

41		1
42		2
46		1

Continuous Series

$K = \text{No. of intervals}$

$$= 1 + 3.322 \log N$$

$$= 13.25 \approx 13$$

$N = \text{Sample Size} = 40$

$W = \text{Width of Class interval}$

$$= \frac{\text{largest} - \text{smallest}}{K}$$

$$= \frac{46 - 2}{13}$$

$$= 3.38$$

Exclusive form:

Marks	Tally	frequency
0-4		5
4-8		4
8-12		1
12-16		2
16-20		5
20-24		2
24-28		4
28-32		1
32-36		3
36-40		4
40-44		3
44-48		1

classmate
Date _____
Page _____

Inclusive form	Tally	Frequency
Marks		5
0-3		4
4-7		1
8-11	I	2
12-15	II	8
16-19		2
20-23	II	5
24-27		1
28-31	I	3
32-35		4
36-39		4
40-43		1
44-47	I	

Correction factor (c)

$$= \frac{4-3}{2} = \frac{1}{2} = 0.5$$

= lower limit of second class - 1
upper limit of first class

Updated limit

Upper limit = old U.L + c

Lower limit = old L.L - c

0-3
4-7
8-11
12-15
16-19
20-23
24-27
28-31
32-35
36-39
40-43
44-47

Exclusive form

Marks	Tally	Frequency
0-3.5		5
3.5-7.5		4
7.5-11.5	I	1
11.5-15.5	II	2
15.5-19.5		8
19.5-23.5	II	2
23.5-27.5		5
27.5-31.5	I	1
31.5-35.5		3
35.5-39.5		4
39.5-43.5		4
43.5-47.5	I	1

0-3.5
3.5-7.5
7.5-11.5
11.5-15.5
15.5-19.5
19.5-23.5
23.5-27.5
27.5-31.5
31.5-35.5
35.5-39.5
39.5-43.5
43.5-47.5

Q. From the following set of data, draw histogram, frequency curve, frequency polygon and ogive curve. Also find out mode value and median value.

Monthly expenditure (in 000)

36, 44, 74, 72, 42, 46, 54, 53, 84, 31, 62, 54, 58, 34, 82, 78, 74, 72, 71, 69, 80, 41, 54, 52, 59, 64, 67, 69, 70, 72, 82, 80, 47, 39, 34, 98, 47, 40, 54, 56, 42, 59, 62, 75, 81.

Soln Here

$$N = \text{Sample Size} = 45$$

$$k = 1 + 3.322 \log N$$

$$= 13.64 \approx 14$$

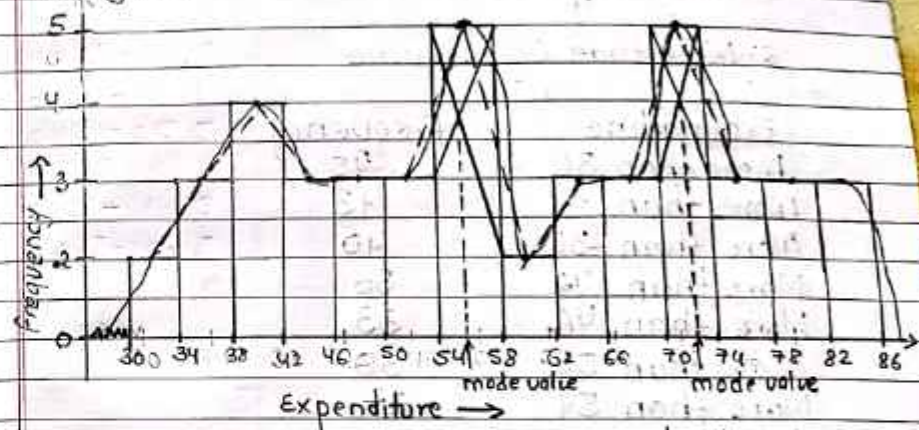
$$w = \frac{\text{largest} - \text{smallest}}{k}$$

$$= \frac{84 - 31}{14} = 3.7857 \approx 4$$

Expenditure	Tally	Frequency
30-34		2
34-38		3
38-42		4
42-46		3
46-50		3
50-54		3
54-58		5
58-62		2
62-66		3

66-70		3
70-74		5
74-78		3
78-82		3
82-86		3

Histogram



Ogive Curve

Type I: less than ogive curve

Expenditure	frequency
less than 34	2
less than 38	5
less than 42	9
less than 46	12
less than 50	15
less than 54	18
less than 58	23

Less than 62	25
less than 66	28
less than 70	31
less than 74	36
less than 78	39
less than 82	42
less than 86	45

2. More than Ogive Curve

Expenditure	Frequency
More than 80	45
More than 34	43
More than 38	40
More than 42	36
More than 46	33
More than 50	30
More than 54	27
More than 58	22
More than 62	20
More than 66	17
More than 70	14
More than 74	9
More than 78	6
More than 82	3

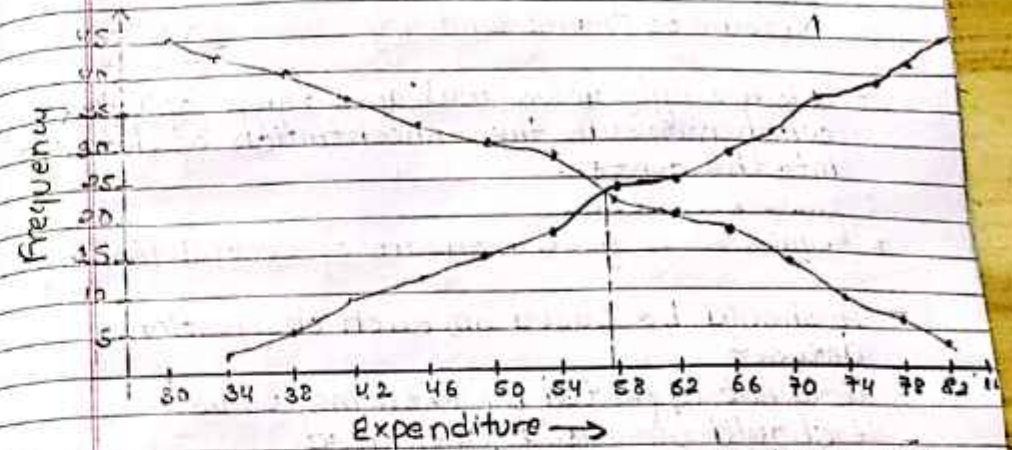


Fig: Ogive curve

Chapter 2 Measure of Central Tendency

- It is a single value with in a range of value which measure the concentration of data set into the center.

* Requisties of good measure of Central Tendency

- It should be based on each observation of dataset.
- It is not affected by extreme value.
- It should be easily calculated.

* Types of measure of Central Tendency

1. Arithmetic mean
2. Median
3. Mode
4. Geometric mean
5. Harmonic mean

1. Arithmetic mean

→ It is the sum of observation divided by total number of observation. It is denoted by $\bar{X}$.

* Formula to calculate Arithmetic mean

1. Individual Series

$$\bar{X} = \frac{\sum X}{N}$$

For large dataset

$$\bar{X} = A + \frac{\sum d}{N}$$

$$d = A - X$$

A = Assumed Value

2. Discrete Series

$$\bar{X} = \frac{\sum fx}{N}$$

$$N = \sum f$$

f = frequency

X = observation

$$\bar{X} = A + \frac{\sum fd}{N}$$

3. Continuous Series

$$\bar{X} = \frac{\sum fm}{N}$$

$$N = \sum f$$

m = Mid-value

$$\bar{X} = A + \frac{\sum fd'}{N} \quad \text{where,}$$

A = Assume value

M = Mid value

h = width of class

$$d' = \frac{A - M}{h}$$

Q

Expenditure (000)	Frequency	Mid-value	fm
30-34	2	32	64
34-38	3	36	108
38-42	4	40	160
42-46	3	44	132
46-50	3	48	144
50-54	3	52	156
54-58	5	56	280
58-62	2	60	120
62-66	3	64	192
66-70	3	68	204
	$\sum f = 31$		$\sum fm = 1560$

Continuous Series

$$\bar{X} = \frac{\sum fx}{N} = \frac{1560}{31} = 50.32 \times 1000$$

$$= 50,320$$

∴ The average expenditure per month is Rs 50,320

Discrete Series

Marks	frequency	fx
2	2	4
3	3	9
4	1	4
5	1	5
7	2	14
9	1	9
12	1	12
14	1	14
16	2	32
17	3	51
N = 17		Σfx = 154

$$\bar{X} = \frac{\sum fx}{N} = \frac{154}{17} = 9.0588$$

Individual Series

$$\bar{X} = \frac{\sum X}{N}$$

$$= \frac{2+3+4+5+7+9+12+14+16+17}{10}$$

$$= \frac{90}{10} = 9$$

Alternative Continuous

Expenditure (000) (x)	frequency (f)	Mid-value (m)	fm	d = $\frac{A-m}{h}$	fd
30-34	2	32	64	4	8
34-38	3	36	108	3	9
38-42	4	40	160	2	8
42-46	8	44	352	1	8
46-50	3	48	144	0	0
50-54	3	52	156	-1	-3
54-58	5	56	280	-2	-10
58-62	2	60	120	-3	-6
62-66	3	64	192	-4	-12
66-70	3	68	204	-5	-15
Σf = 31			Σfm = 1560		Σfd = (-18)

$$\bar{X} = A + \frac{\sum fd}{N}$$

$$= 48 + \frac{(-18)}{31}$$

$$= 47.419 \times 1000$$

$$= 47,419$$

∴ The average expenditure per month is Rs 47,419

Combine Mean

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

N = population size, n = sample size

- * mean age ($\bar{X}_1$) = 35
- mean weight ($\bar{X}_2$) = 56
- Size (n_1) = 40
- Size (n_2) = 50

$$\begin{aligned}\bar{X}_{12} &= \frac{40 \times 35 + 50 \times 56}{40 + 50} \\ &= 46.6667\end{aligned}$$

9.

Nationality	Size (n)	Average income (000)
Nepali	400	700
Indian	500	760
Pakistani	440	900
Other	460	540

Combine mean of income of Nationality = ?

$$\begin{aligned}\bar{X}_{12} &= \frac{400 \times 700 + 500 \times 760 + 440 \times 900 + 460 \times 540}{400 + 500 + 440 + 460} \\ &= 724.67 \times 1000 \\ &= 724,667\end{aligned}$$

* Weighted Arithmetic Mean

$$\begin{aligned}\bar{X}_w &= \frac{X_1 w_1 + X_2 w_2 + X_3 w_3 + \dots + X_n w_n}{w_1 + w_2 + w_3 + \dots + w_n} \\ &= \frac{\sum_{i=1}^n X_i w_i}{\sum_{i=1}^n w_i}\end{aligned}$$

Weight	X	Variable
0.50	100	Rent
0.20	220	Clothing
0.30	400	Other

$$\begin{aligned}\bar{X}_w &= \frac{100 \times 0.50 + 220 \times 0.20 + 400 \times 0.30}{0.50 + 0.20 + 0.30} \\ &= 214\end{aligned}$$

* Birth weight of 30 children (in kg)

2.0, 2.1, 3.0, 3.1, 2.7, 2.8, 3.5, 3.1, 3.7, 4.0,
2.3, 2.5, 3.5, 4.2, 3.7, 3.2, 2.7, 2.5, 2.7,
3.8, 3.1, 3.0, 2.6, 2.8, 2.9, 3.5, 4.1, 3.9,
2.8, 2.2

Soln

N = sample size = 30

$$\begin{aligned}K &= 1 + 3.322 \log N \\ &= 12.2988 \approx 12\end{aligned}$$

$$w = \frac{\text{largest} - \text{smallest}}{k}$$

$$= \frac{4.2 - 2.0}{12}$$

$$= 0.18$$

$$\approx 0.2$$

Birth weight	Tally	Frequency	Middle (cm)	mf
2.0 - 2.2		2	2.1	4.2
2.2 - 2.4		2	2.3	4.6
2.4 - 2.6		2	2.5	5.0
2.6 - 2.8		4	2.7	10.8
2.8 - 3.0		4	2.9	11.6
3.0 - 3.2		5	3.1	15.5
3.2 - 3.4		1	3.3	3.3
3.4 - 3.6		3	3.5	10.5
3.6 - 3.8		2	3.7	7.4
3.8 - 4.0		2	3.9	7.8
4.0 - 4.2		2	4.1	8.2
4.2 - 4.4		1	4.3	4.3
		N=30		Emf=98.4

So

$$\bar{x} = \frac{\sum mf}{N}$$

$$= \frac{98.4}{30}$$

$$= 3.1067 \text{ kg}$$

Age of Students

12, 14, 17, 13, 20, 14, 16, 15, 17, 19, 18, 20, 22, 23, 15, 17, 16, 19, 10, 11, 13, 17, 16, 20, 24, 23, 20, 24, 22, 20

Soln

Age of Students

$$N = \text{Sample Size} = 30$$

$$k = 1 + 3.322 \log N$$

$$= 1 + 3.322 \log (30)$$

$$k = 1 + 1.2988 \approx 1.2988$$

$$w = \frac{\text{largest} - \text{smallest}}{k}$$

$$= \frac{24 - 10}{12.2988}$$

$$= 1.1667 \approx 1.1667$$

Discrete table

Age of Students	Tally	frequency	fx
10		1	10
11		1	11
12		1	12
13		2	26
14		2	28
15		2	30
16		3	48
17		4	68
18		1	18
19		2	38
20		5	100
22		2	44
23		2	46

24 11 2 48
 $f = 30$ $\Sigma fx = 527$

$$\bar{X} = \frac{\Sigma fx}{f}$$

$$= \frac{527}{30} = 17.57$$

Continuous table

Age (X)	frequency f	Mid-Value m	fm
10-12	2	11	22
12-14	3	13	39
14-16	4	15	60
16-18	7	17	119
18-20	3	19	57
20-22	5	21	105
22-24	4	23	92
24-26	2	25	50
	$\Sigma f = 30$		$\Sigma fm = 544$

$$\text{Mean} = \frac{\Sigma fm}{\Sigma f}$$

$$= \frac{544}{30}$$

$$= 18.13$$

$$\approx 18$$

Advantages

- It is use for further treatment
- It is based on all observation.

Disadvantages

- It is affected by extreme value.
- It is not suitable for open class interval.

x Median

- It is the value of variable which divide distribution into two equal parts.
- It is calculated as +

Individual Series Discrete Series Continuous Series

$$Md = \left(\frac{N+1}{2} \right)^{th} \text{ term}$$

$$Md = \left(\frac{N+1}{2} \right)^{th} \text{ term}$$

$$Md = \left(\frac{N}{2} \right)^{th} \text{ term}$$

$$Md = L + \left(\frac{N}{2} - c.f \right) \times h$$

$$f$$

L = Lower Limit

c.f = cumulative frequency of preceding class

f = frequency of same class

h = width of class

* Weight of farm workers (in kg)

47.8, 50.2, 70.4, 60.2, 42.8, 76.3, 57.4, 44.8,
67.8, 71.3

Arranging in ascending order

42.8, 44.8, 47.8, 50.2, 57.4, 60.2, 67.8, 70.4,
71.3, 76.3

$$Md = \left(\frac{N+1}{2} \right)^{th} \text{ term}$$

$$= \left(\frac{10+1}{2} \right)^{th} \text{ term}$$

$$= \frac{11}{2}^{th} \text{ term}$$

$$= 5.5^{th} \text{ term}$$

$$\begin{aligned} Md &= 5^{th} \text{ term} + (6^{th} \text{ term} - 5^{th} \text{ term}) \times 0.5 \\ &= 57.4 + (60.2 - 57.4) \times 0.5 \\ &= 58.8 \end{aligned}$$

The median value of weight of farm workers is 58.8 kg.

Marks No. of Students

30	4
40	7
5	6
50	8
70	10
80	2

Arranging in ascending order

Marks No. of stds Cf (Li)

30	4	4
35	6	10
40	7	17
50	8	25
70	10	35
80	2	37

$$N = 37$$

$$Md = \left(\frac{N+1}{2} \right)^{th} \text{ term}$$

$$= \left(\frac{37+1}{2} \right)^{th} \text{ term}$$

$$= 19^{th} \text{ term}$$

19th term in table (c.f) is 50.

Marks (x)	frequency (f)	c.f
Below 45	4	4
45-55	7	11
55-65	6	17
65-75	8	25
75-85	10	35
Above 85	2	37

$N = 37$

$$Md = \left(\frac{N}{2} \right)^{th} \text{ term}$$

$$= \left(\frac{37}{2} \right)^{th} \text{ term}$$

$$= 18.5^{th} \text{ term}$$

$$Md = 65 + \frac{18.5 - 17}{8} \times 10$$

$$= 66.875$$

* Advantages

- It is suitable for open class interval.
- It is not affected by extreme values.
- It is based on all observation.

* Disadvantage

- It is not use in further mathematical treatment.

* Partitions

- Quartiles $\rightarrow$ Q_1, Q_2, Q_3

- Deciles $\rightarrow$ $D_1, D_2, D_3, \dots, D_9$

- Percentiles $\rightarrow$ $P_1, \dots, P_{99}$

Median $\rightarrow Q_2 \rightarrow D_5 \rightarrow P_{50}$

Partation	Individual Series	discrete Series	continuous Series
1. Q_i ($i = 1, 2, 3$)	$Q_i = i \left(\frac{N+1}{4} \right)^{th} \text{ term}$	$Q_i = i \left(\frac{N+1}{4} \right)^{th} \text{ term}$	$Q_i = i \left(\frac{N}{4} \right)^{th} \text{ term}$ $Q_i = L + i \left(\frac{N}{4} \right) - c.f$ f
2. D_i ($i = 1, 2, \dots, 9$)	$D_i = i \left(\frac{N+1}{10} \right)^{th} \text{ term}$	$D_i = i \left(\frac{N+1}{10} \right)^{th} \text{ term}$	$D_i = i \left(\frac{N}{10} \right)^{th} \text{ term}$ $D_i = L + i \left(\frac{N}{10} \right) - c.f$ f
3. P_i ($i = 1, 2, \dots, 99$)	$P_i = i \left(\frac{N+1}{100} \right)^{th} \text{ term}$	$P_i = i \left(\frac{N+1}{100} \right)^{th} \text{ term}$	$P_i = i \left(\frac{N}{100} \right)^{th} \text{ term}$ $P_i = L + i \left(\frac{N}{100} \right) - c.f$ f

* Weight

47.8, 50.2, 70.4, 60.2, 42.8, 76.3, 57.4, 44.8, 67.8, 71.3

Soln

$$Q_1 = 1\left(\frac{10+1}{4}\right)^{th} \text{ term}$$

$$= \left(\frac{11}{4}\right)^{th} \text{ term}$$

$$= 2.75^{th} \text{ term}$$

$$Q_1 = 2^{nd} \text{ term} + (3^{rd} \text{ term} - 2^{nd} \text{ term}) \times 0.75$$

$$= 44.8 + (47.8 - 44.8) \times 0.75$$

$$= 47.05$$

$$Q_3 = 3\left(\frac{N+1}{4}\right)^{th} \text{ term}$$

$$= 3 \times \frac{11}{4}^{th} \text{ term}$$

$$= 8.25^{th} \text{ term}$$

$$Q_3 = 8^{th} \text{ term} + (9^{th} \text{ term} - 8^{th} \text{ term}) \times 0.25$$

$$= 70.4 + (71.3 - 70.4) \times 0.25$$

$$= 70.625$$

* Mode

→ The value of variable having highest frequency. It is denoted by M_o .

For continuous Series

$$M_o = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

f_1 = frequency of same class

f_0 = frequency of preceding class

f_2 = " " " successive "

Q. Marks

(x)

Below 45

45-55

55-65

65-75

75-85

Above 85

Class with highest frequency = 75-85

$$L = 75$$

$$f_1 = 10$$

$$f_0 = 8$$

$$f_2 = 2$$

Now

$$M_o = 75 + \frac{10 - 8}{2 \times 10 - 8 - 2} \times 10$$

$$= 77$$

Q. Class	frequency	mid-value	cf	fm
10-20	10	15	10	150
20-30	? x	25	10+x	25x
30-40	20	35	30+x	700
40-50	? y	45	30+x+y	45x
50-60	10	55	40+x+y	550
$M_d = 45$	$\bar{X} = ?$	$M_o = ?$		
$N = 80$				

Soln

$$M_d = L + \frac{(N/2) - c.f}{f} \times h$$

$$45 = 40 + \frac{80 - (30+x)}{2} \times 10$$

$$45y = 40 + 40 - 30 - x \times 10 \Rightarrow 45 = 40 + 10 - x \times 10$$

$$45y = (50 - x) \times 10$$

$$45y = 500 - 10x$$

$$10x + 45y = 500 \dots \textcircled{i}$$

$$\Rightarrow 45y = 40y + 100 - 10x$$

$$\Rightarrow 10x + 5y = 100 \dots \textcircled{ii}$$

$$N = 40 + x + y$$

$$80 = 40 + x + y$$

$$\therefore x + y = 40 \dots \textcircled{iii}$$

Solving (i) & (ii)

$$x = 37.14 - 20$$

$$y = 2.85 \quad 60$$

Class	frequency	Mid-value	fm
10-20	10	15	150
20-30	20	25	500
30-40	20	35	700
40-50	60	45	2700
50-60	10	55	550
	$f_m = 80$		$\Sigma f_m = 3600$

$$\text{Mean } (\bar{X}) = \frac{\Sigma f_m}{f_m}$$

$$= \frac{3600}{80}$$

$$= 45$$

Again

Class with highest frequency = 40-50

$$L = 40$$

$$f_1 = 60$$

$$f_0 = 20$$

$$f_2 = 10$$

Now,

$$M_o = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

$$= 40 + \frac{60 - 20}{2 \times 60 - 20 - 10} \times 10$$

$$= 40 + \frac{40}{120 - 20 - 10} \times 10$$

$$= 40 + \frac{40}{90} \times 10$$

$$= 40 + \frac{40}{9}$$

$$= \frac{360 + 40}{9}$$

$$= \frac{400}{9}$$

$$= 44.44$$

Geometric Mean

→ Let $X_1, X_2, \dots, X_n$ be a sample of a variable then the geometric mean of the variable is calculated as:-

$$G.M = (X_1 \cdot X_2 \cdot \dots \cdot X_n)^{1/n}$$

$$\log G.M = \frac{1}{n} \log (X_1 \cdot X_2 \cdot \dots \cdot X_n)$$

$$= \frac{1}{n} \sum_{i=1}^n \log X_i$$

$$\therefore G.M = \text{Antilog} \left(\frac{1}{n} \sum_{i=1}^n \log X_i \right) \text{ for individual series}$$

For discrete Series

$$G.M = \text{Antilog} \left[\frac{1}{N} \sum_{i=1}^n f \log X_i \right]$$

$$N = \sum f$$

For Continuous Series

$$G.M = \text{Antilog} \left(\frac{1}{N} \sum_{i=1}^n f \log M_i \right)$$

$$N = \sum f$$

M = Mid-Value

f = frequency

Q.	X	f	logX	flogX
	28	5	3.135	15.65
	27	7	3.29	28.03
	37	3	3.61	10.83
	40	10	3.68	36.8
	42	5	3.73	18.65
	33	4	3.49	13.96
	20	2	2.99	5.98

$$\Sigma f = 36$$

$$\Sigma f \log X = 124.9$$

Now, $G.M = \text{Antilog} \left[\frac{1}{N} \Sigma f \log X \right]$

$$= \text{Antilog} \left[\frac{1}{36} \times 124.9 \right]$$

$$= 32.11$$

* Harmonic Mean [Average Speed / distance / Velocity]

→ It is the reciprocal of A.M. It is denoted by H.M or $\bar{X}_{HM}$.

It is calculated as:
for individual series

$$H.M = \frac{n}{\Sigma \frac{1}{x}}$$

For discrete series

$$H.M = \frac{N}{\Sigma f \left(\frac{1}{x} \right)}$$

For Continuous Series

$$H.M = \frac{N}{\Sigma f \left(\frac{1}{M} \right)}$$

$$M = \text{Mid-Value}$$

where, $N = \Sigma f$ = Total no. of obs
 M = Mid-Value

Q.	X	f	$\frac{1}{x}$	$f \frac{1}{x}$
	28	5	0.0485	0.215
	27	7	0.0370	0.259
	37	3	0.027	0.081
	40	10	0.025	0.25
	42	5	0.0238	0.12
	33	4	0.0303	0.12
	20	2	0.05	0.1

$$\Sigma f = 36$$

$$\Sigma \frac{1}{x} = 0.2866$$

$$\Sigma f \frac{1}{x} = 1.145$$

So, $H.M = \frac{N}{\Sigma f \left(\frac{1}{x} \right)}$

$$H.M = \frac{36}{1.145}$$

$$= 31.44$$

Chapter-3

Measure of Dispersion

- It is one of the descriptive statistics which measure the scatterness of the distribution from central value.

* Objective of measure of dispersion

- To test the reliability and validity of central value.
- To test deviation of data from central value.
- It is used to compare two or more data set.

* Characteristics of Measure of Dispersion

1. It should be least affected by extreme value.
2. It should be on all observation.
3. It should be rigidly define.
4. It should not fluctuate from sample to sample taken from same population.

* Types of Measure of Dispersion

1. Absolute measure of dispersion
2. Relative measure of dispersion
 - Coefficient of Range
 - Coefficient of Q.D
 - Coefficient of variation C.V
 - Range, Standard Deviation, Quartile Deviation

1. Range

$$\text{Range} = L - S$$

where L = Largest Value
 S = Smallest value

2. Q.D

$$Q.D = \frac{Q_3 - Q_1}{2}$$

3. Coeff. of Range

$$\text{Range} = \frac{L - S}{L + S}$$

4. Coeff. of Q.D

$$\text{Coeff. of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

* Standard Deviation (S.D)

- It is the square root of deviation of value from it's center value.
- It is denoted by ' σ '

* For individual series

$$i) \sigma = \sqrt{\frac{1}{n} \sum (x - \bar{x})^2}$$

$$ii) \sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

For change of origin

$$iii) \sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$$

$$d = x - A$$

A = Assumed Value

for discrete Series

$$i) \sigma = \sqrt{\frac{1}{N} \sum f(x - \bar{x})^2}$$

$$ii) \sigma = \sqrt{\frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2}$$

For change of origin

$$iii) \sigma = \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N}\right)^2}$$

$$d = x - A$$

for Continuous Series

$$i) \sigma = \sqrt{\frac{1}{N} \sum f(M - \bar{x})^2}$$

$$ii) \sigma = \sqrt{\frac{\sum fM^2}{N} - \left(\frac{\sum fM}{N}\right)^2}$$

For change of origin

$$iii) \sigma = \sqrt{\frac{\sum fd'^2}{N} - \left(\frac{\sum fd'}{N}\right)^2} \times h$$

$$\text{where, } d' = \frac{M - A}{h}$$

* Variance (σ^2)

→ It is defined as the square of S.D.

* Coefficient of Variance (C.V) → Consistent / Variable / Reliable / Uniformity

→ It is unit less measure of dispersion. It is calculated by using S.D (σ) and Mean ($\bar{x}$). It is always in %.

$$C.V = \frac{\sigma}{\bar{x}} \times 100\%$$

C.V greater less consistent / uniformity / Reliable
C.V greater greater variability.

Group A		Group B	
Income (000)	Frequency	Income (000)	Frequency
15-20	7	10-20	12
20-25	9	20-30	17
25-30	15	30-40	25
30-35	5	40-50	7
35-40	26	50-60	9
40-45	8	60-70	6

- Which group has uniform income?
- " " " " higher variability in income?

Soln

For group A

Income(000)	Frequency ^(f)	Mid-Value ^(m)	fm
15-20	7	17.5	122.5
20-25	9	22.5	202.5
25-30	15	27.5	412.5
30-35	5	32.5	162.5
35-40	26	37.5	975
40-45	8	42.5	340

$$\Sigma f = 65$$

$$\Sigma fm = 2002.5$$

Then

$$\text{Mean}(\bar{x}) = \frac{\Sigma fm}{\Sigma f}$$

$$= \frac{2002.5}{65}$$

$$= 30.8077$$

$m_1 - \bar{x}$	$(m_1 - \bar{x})^2$	$f(m_1 - \bar{x})^2$
-13.31	177.15	1240.05
-8.31	69.0561	621.50
-3.31	10.9561	164.34
1.69	2.8561	14.2805
6.69	44.7561	1163.5
11.69	136.6561	409.98

$$\Sigma f(m_1 - \bar{x})^2 = 3613.6505$$

$$\sigma = \sqrt{\frac{1}{N} \Sigma f(m_1 - \bar{x})^2}$$

$$= \sqrt{\frac{1}{65} \times 3613.6505}$$

$$= 7.4562$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100\%$$

$$= \frac{7.4562}{30.8077} \times 100\%$$

$$= 24.2\%$$

for group B

Income(000)	Frequency ^(f)	Mid-Value ^(m)	fm
10-20	12	15	180
20-30	17	25	425
30-40	25	35	875
40-50	7	45	315
50-60	9	55	495
60-70	6	65	390

$$\Sigma f = 76$$

$$\Sigma fm = 2680$$

Then

$$\text{Mean} \bar{x} = \frac{\Sigma fm}{\Sigma f}$$

$$= \frac{2680}{76}$$

$$= 35.2632$$

$m_2 - \bar{x}$	$(m_2 - \bar{x})^2$	$f(m_2 - \bar{x})^2$
-20.26	410.46	4925.52
-10.26	105.26	1789.42
-0.26	0.0676	1.69
9.74	94.86	664.02
18.74	350.66	3506.94
29.74	884.46	5306.76

$$\Sigma f(m_2 - \bar{x})^2 = 16194.35$$

$$\sigma = \sqrt{\frac{1}{N} \sum f(m-\bar{x})^2}$$

$$= \sqrt{\frac{1}{76} \times 16194.35}$$

$$= 14.59$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100\%$$

$$= \frac{14.59}{35.26} \times 100\%$$

$$= 41.39\%$$

$\therefore$ Group A has uniform income.

$\therefore$ Group B has higher variability in income.

Q. Find Average Marks of the students from the following data.

Combine S.D

For two

$\sigma_1, n_1, \bar{X}_1$

$\sigma_2, n_2, \bar{X}_2$

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

$$\text{Combine SD } (\sigma_{12}) = \sqrt{\frac{n_1(d_1^2 + \sigma_1^2) + n_2(d_2^2 + \sigma_2^2)}{n_1 + n_2}}$$

where

$$d_1 = \bar{X}_1 - \bar{X}_{12}$$

$$d_2 = \bar{X}_2 - \bar{X}_{12}$$

Q. $\sigma_1 = 4.5$

$\sigma_2 = 7.9$

$n_1 = 40$

$n_2 = 45$

$\bar{X}_1 = 20$

$\bar{X}_2 = 36$

$\sigma_{12} = ?$

Also find combine variance.

Soln

$$\bar{X}_{12} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

$$= \frac{40 \times 20 + 45 \times 36}{40 + 45} = 28.47$$

Then

$$d_1 = \bar{x}_1 - \bar{x}_2$$

$$= 20 - 28.47$$

$$= -8.47$$

$$d_2 = \bar{x}_2 - \bar{x}_2$$

$$= 36 - 28.47$$

$$= 7.53$$

$$\text{Combine S.D. } (\sigma_{12}) = \sqrt{\frac{n_1(d_1^2 + \sigma_1^2) + n_2(d_2^2 + \sigma_2^2)}{n_1 + n_2}}$$

$$= \sqrt{\frac{40((-8.47)^2 + 4.5^2) + 45(7.53^2 + 7.9^2)}{40 + 45}}$$

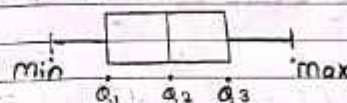
$$= 10.31$$

$$\text{Combine Variance } (\sigma^2) = (10.31)^2$$

$$= 106.34$$

* Box and Whisker Plot

→ It is a graphical representation of scatterness of the data using box and whisker.



It is based on five summary measure. They are: Minimum, Q_1 , Q_2 , Q_3 and Maximum.

Decision

1. If $(\text{Min} - Q_1)$ is less than $(\text{Maximum} - Q_3)$ then their distribution is Right Skewed and vice versa.
2. If $(\text{Min} - Q_1)$ is less than $(\text{Maximum} - Q_3)$ then their distribution is Right Skewed and vice versa.
3. If $(Q_2 - Q_1)$ is less than $(Q_3 - Q_2)$ then their distribution is Right Skewed and vice versa.

Chapter-4

Probability

Q. What is the probability of occurrence of sum of two face is 4 while rolling a dice?

Dice	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Total face = 12

Total event occurrence = $6^2 = 36$

Total favourable case = 3

$\therefore$ Probability of sum of faces is 4 = $\frac{\text{Total favourable case}}{\text{Total outcomes}}$

$$= \frac{3}{36}$$

$$= \frac{1}{12}$$

$$= \frac{1}{12}$$

Terminology

• Random Experiment:

→ A random experiment is a process or action that produces an outcome that can not be predicted with certainty in advance but follows a set of possible outcomes, such as flipping a coin or rolling a dice.

• Sample Space:

→ It is the set of all possible outcomes of a random experiment. For example: the sample space for flipping a coin is {Heads, Tails}.

• Trial and Event:

→ Trial is a single performance or execution of a random experiment. For eg. rolling a dice once.

→ Event is a specific outcome or a set of outcomes from the sample space. (e.g. rolling an even number on dice).

• Impossible and Sure event:

→ An event that cannot happen with probability of 0. (e.g. rolling a 7 on a standard 6-sided dice) is impossible event.

→ An event that is certain to happen, with a probability of 1.

* Mutually exclusive event:

→ Events that cannot happen at same time. If one event occurs, the other cannot. For eg: While flipping a coin, the events of getting heads and tails are mutually exclusive.

* Disjoint event:

→ The events where the occurrence of a event does not affect the other event or, the events that cannot occur simultaneously.

* Favourable Case:

→ An outcome or set of outcomes that satisfy the conditions of specific event. For eg: While rolling a dice, if the event is rolling an event number, the favourable cases are 2, 4, 6.

Axiomatic Probability

→ Let 'S' be a sample space of a random experiment. 'A' be an event in a sample S. Then, the prob. of occurrence of event 'A' in a sample space 'S' is denoted as $P(A)$. And must satisfy

- The probability of A i.e. $P(A)$ must be non-negative i.e. $P(A) \geq 0$
- Total probability is always unity i.e. $P(S) = 1$
- Prob. of impossible event is zero i.e. $P(\emptyset) = 0$

Addition Theorem of Probability:

Let A and B be two events in a sample space 'S' and these events are not disjoint then prob. of occurrence of A or B is given as:

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For three events A, B & C.

$$\begin{aligned} P(A \text{ or } B \text{ or } C) &= P(A \cup B \cup C) \\ &= P(A) + P(B) + P(C) - P(A \cap B) - \\ &\quad P(B \cap C) - P(A \cap C) + P(A \cap B \cap C) \end{aligned}$$

If these events are disjoint / independent / mutually exclusive

$$P(A \text{ or } B) = P(A) + P(B)$$

Q) In a bag contains 5 red balls, 4 black and 2 white balls. When two balls drawn at a random, what is the prob. of getting

- Both red balls
- Both black
- 1 red and 1 white
- 1 white and 1 black
- 1 red and 1 black

Here,

$$R = 5$$

$$B = 4$$

$$W = 2$$

$$\text{Total no. of balls} = 5 + 4 + 2 = 11$$

$$\text{No. of ball drawn} = 2$$

$$\begin{aligned} \text{Total Outcomes} &= {}^{11}C_2 = \frac{11!}{2!(11-2)!} \\ &= \frac{11 \times 10 \times 9!}{2! \times 9!} \\ &= 55 \end{aligned}$$

$$\begin{aligned} \text{a) Both red} \\ \text{favourable case} &= {}^5C_2 = \frac{5!}{2!3!} \\ &= \frac{5 \times 4 \times 3!}{2! \times 3!} \\ &= 10 \end{aligned}$$

$$\begin{aligned} \text{The prob. of getting both red balls} &= P(R, R) \\ &= \frac{\text{favourable Case}}{\text{Total Outcomes}} \end{aligned}$$

$$= \frac{10}{55}$$

$$= \frac{2}{11}$$

∴ The probability of getting both red balls is $\frac{2}{11}$.

b. Both Black

$$\text{favourable case} = {}^4C_2 = \frac{4!}{2!2!}$$

$$= \frac{4 \times 3 \times 2!}{2! \times 2!} = 6$$

$$\begin{aligned} \text{The probability of getting both black balls} &= P(B, B) \\ &= \frac{\text{favourable Case}}{\text{Total Outcomes}} \end{aligned}$$

$$= \frac{6}{55}$$

∴ The probability of getting both black balls is $\frac{6}{55}$.

c) 1 red and 1 white

$$\begin{aligned} \text{favourable case} &= {}^5C_1 \times {}^2C_1 \\ &= \frac{5!}{1!4!} \times \frac{2!}{1!1!} \\ &= 10 \end{aligned}$$

$$\begin{aligned} \text{Prob. of getting 1 white \& 1 black balls} &= P(W, B) \\ &= \frac{10}{55} \end{aligned}$$

$$= \frac{2}{11}$$

d) 1 white and 1 black
 Favourable case = ${}^2C_1 \times {}^4C_1$
 $= \frac{2!}{1!1!} \times \frac{4!}{3!1!}$
 $= 8$

Prob. of getting 1 white & 1 black = $P(W, B)$
 $= \frac{8}{55}$

e) 1 red and 1 black
 Favourable case = ${}^5C_1 \times {}^4C_1$
 $= \frac{5!}{4!1!} \times \frac{4!}{3!1!}$
 $= 20$

Prob. of getting 1 red & 1 black = $P(R, B)$
 $= \frac{20}{55}$
 $= \frac{4}{11}$

* Multiplicative Theorem

→ Let A and B be two events in a sample space 'S', then the prob. of occurrence of A and B is given as:-

$P(A \text{ and } B) = P(A \cap B) = P(A) \cdot P(B)$ if events are disjoint/independent

if these events are disjoint:-
 $\left[\begin{aligned} &= P(A) \cdot P(B|A) \\ &= P(B) \cdot P(A|B) \end{aligned} \right]$ conditional probability

for three events A, B and C

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$
 $= P(A \cap B) \cdot P(C)$
 $= P(A \cap B) \cdot P(C|A \cap B)$
 $= P(A) \cdot P(B|A) \cdot P(C|A \cap B)$

Q Three newspapers A, B and C are published in a certain city. It is estimated from a survey that 20% read A, 16% read B, 14% read C, 8% read A and B, 5% read A and C and 4% read B and C and 2% read all three newspaper. What are the probabilities that a normally chosen person.

- does not read newspaper
- does not read C.
- does not read A
- Read A not B
- Read B not C

Given

$$P(A) = 0.20$$

$$P(B) = 0.16$$

$$P(C) = 0.14$$

$$P(A \cap B) = 0.08$$

$$P(A \cap C) = 0.05$$

$$P(B \cap C) = 0.04$$

$$P(A \cap B \cap C) = 0.02$$

$$\text{Read newspaper} = P(A \cup B \cup C)$$

(at least 1 newspaper)

$$= P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B)$$

$$- P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$= 0.20 + 0.16 + 0.14 - 0.08 - 0.05 - 0.04 + 0.02$$

$$= 0.35$$

a) prob. of who does not read newspaper

$$= 1 - 0.35$$

$$= 0.65$$

b) does not read C

Total probability = 1

$$\text{does not read C} = 1 - P(C)$$

$$= 1 - 0.14$$

$$= 0.86$$

c) does not read A

$$\text{does not read A} = 1 - P(A)$$

$$= 1 - 0.20$$

$$= 0.8$$

d) Read A not B

$$= P(A) - P(A \cap B)$$

$$= 0.20 - 0.08$$

$$= 0.12$$

e) Read B not C

$$= P(B) - P(B \cap C)$$

$$= 0.16 - 0.04$$

$$= 0.12$$

* Bayes Theorem

Let $E_1, E_2, \dots, E_n$ be mutually disjoint events of a sample space S with $P(E_i) \neq 0$, i.e. $i = 1, 2, \dots, n$, then for any arbitrary event A which is subset of S such that $P(A) > 0$, we have

$$P(E_r/A) = \frac{P(E_r) \cdot P(A/E_r)}{\sum_{i=1}^n P(E_i) \cdot P(A/E_i)} \quad 1 \leq r \leq n$$

$$\text{prior prob} = P(E_r)$$

$$\text{Likelihood (present)} = P(A/E_r)$$

300 = defective on E_r lot

$$\text{Posterior prob} = P(E_r/A)$$

In General, Bayes Theorem used to obtain posterior probability using prior probability and likelihood of events.

Q. In a factory machines A, B and C manufacture respectively 25%, 35% and 40% of total of their total output 5%, 4% and 2% are defective bolts. A bolt is drawn at random.

- i) What is the prob. of that the bolt will be defected?
 ii) What is the prob. that the defective bolt was manufactured from machine A.

Given

$$P(A) = 0.25$$

$$P(B) = 0.35$$

$$P(C) = 0.40$$

$$P(D/A) = 0.05$$

$$P(D/B) = 0.04$$

$$P(D/C) = 0.02$$

$$i) \text{ Prob. of defectives} = P(D) = \sum_{i=1}^n P(A_i) \cdot P(D/A_i)$$

$$= (P(A) \times P(D/A)) + (P(B) \cdot P(D/B)) + (P(C) \cdot P(D/C))$$

$$= (0.25 \times 0.05) + (0.35 \times 0.04) + (0.40 \times 0.02)$$

$$= 0.0345 \approx 0.035$$

$$ii) P(A/D) = \frac{P(A \cap D)}{P(D)}$$

$$= \frac{P(A) \cdot P(D/A)}{P(D)}$$

$$= \frac{0.25 \times 0.05}{0.035}$$

$$= 0.357 \approx 0.36$$

Q. In a factory machines A, B and C manufacture respectively 60%, 20% and 20% of total of their total output 6%, 4% and 3% are defective bolts. A bolt is drawn at a random. What is the probability of that bolt will be defected? What is the prob. that the defective bolt was manufactured from machine B and C.

Given

$$P(A) = 0.6$$

$$P(B) = 0.2$$

$$P(C) = 0.2$$

$$P(D/A) = 0.06$$

$$P(D/B) = 0.04$$

$$P(D/C) = 0.03$$

$$i) \text{ Prob of defectives} = P(D)$$

$$= \sum_{i=1}^n P(A_i) \cdot P(D/A_i)$$

$$= (P(A) \times P(D/A)) + (P(B) \cdot P(D/B)) + (P(C) \cdot P(D/C))$$

$$= (0.6 \times 0.06) + (0.2 \times 0.04) + (0.2 \times 0.03)$$

$$= 0.05$$

$$ii) P(B/D) = \frac{P(B \cap D)}{P(D)}$$

$$= \frac{P(B) \cdot P(D/B)}{P(D)}$$

$$= \frac{0.2 \times 0.04}{0.05}$$

$$= 0.16$$

$$\begin{aligned}
 P(C|D) &= \frac{P(C \cap D)}{P(D)} \\
 &= \frac{P(C) \cdot P(D|C)}{P(D)} \\
 &= \frac{0.5 \times 0.03}{0.023} \\
 &= 0.12
 \end{aligned}$$

Q. There are three machine A, B and C producing 1000, 2000 and 3000 articles per hour respectively. These machines are known to be 1%, 2% and 3% defective respectively. One article is selected at a random from an hour production of the three machines and found to be defective. What is the probability that the article is produced from

- machine A
- machine B
- machine C

Soln

$$A = 1000 \quad P(D|A) = 0.01$$

$$B = 2000 \quad P(D|B) = 0.02$$

$$C = 3000 \quad P(D|C) = 0.03$$

i) prob. that article produced from machine A

$$\begin{aligned}
 P(A) &= \frac{1000}{1000+2000+3000} \\
 &= \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } P(B) &= \frac{2000}{6000} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } P(C) &= \frac{3000}{6000} \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total prob. of defectives} &= P(D) \\
 &= \sum_{i=1}^n P(A_i) \cdot P(D|A_i)
 \end{aligned}$$

$$\begin{aligned}
 &= (P(A) \times P(D|A)) + (P(B) \cdot P(D|B)) \\
 &\quad + (P(C) \cdot P(D|C))
 \end{aligned}$$

$$\begin{aligned}
 &= (0.17 \times 0.01) + (0.33 \times 0.02) + (0.5 \times 0.03) \\
 &= 0.023
 \end{aligned}$$

$$P(A|D) = \frac{P(A) \cdot P(D|A)}{P(D)}$$

$$= \frac{0.17 \times 0.01}{0.023}$$

$$= 0.07$$

$$P(B|D) = \frac{P(B) \cdot P(D|B)}{P(D)}$$

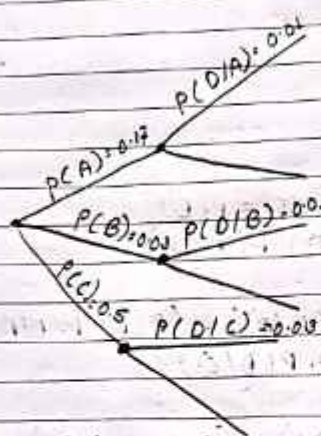
$$= \frac{0.33 \times 0.02}{0.023}$$

$$= 0.29$$

$$P(C|D) = \frac{P(C) \cdot P(D|C)}{P(D)}$$

$$= \frac{0.5 \times 0.03}{0.023}$$

$$= 0.65$$



$$P(A \cap D) = P(A) \cdot P(D|A)$$

$$P(B \cap D) = P(B) \cdot P(D|B)$$

$$P(C \cap D) = P(C) \cdot P(D|C)$$

$$\text{Total no. of prob} = P(A \cap D) + P(B \cap D) + P(C \cap D)$$

8. Two students A and B asked to solve a problem the probability of A solving is $\frac{2}{3}$ and that of B is $\frac{3}{4}$. Find the probability that

- ① Both can solve the problem
- ② A solve the problem but B cannot
- ③ B solve the problem but A cannot
- ④ None of them can solve

Soln

$$P(A) = \frac{2}{3}$$

$$P(\bar{A}) = 1 - \frac{2}{3} = 0.33$$

$$P(B) = \frac{3}{4}$$

$$P(\bar{B}) = 1 - \frac{3}{4} = 0.25$$

$$i) P(A \cap B) = P(A) \cdot P(B)$$

$$= \frac{2}{3} \times \frac{3}{4}$$

$$= \frac{1}{2} = 0.5$$

$$ii) P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$$

$$= 0.67 \times 0.25$$

$$= 0.17$$

$$iii) P(B \cap \bar{A}) = P(B) \cdot P(\bar{A})$$

$$= 0.75 \times 0.33$$

$$= 0.25$$

$$iv) P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B})$$

$$= 0.33 \times 0.25$$

$$= 0.08$$

* Random Variables (R.V)

- Discrete R.V $\rightarrow$ pmf $\rightarrow$ prob. mass function
 $\text{pmf} = \sum p(x) \rightarrow \text{pmf} = \text{Total prob} = 1$
- Continuous R.V $\rightarrow$ pdf $\rightarrow$ prob. density function

* Cumulative Distributions function

X	P(X)
-3	0.1
-1	0.2
0	0.15
1	0.2
2	0.1
3	0.15
5	0.05
8	0.05

Soln

X	P(X)	F(X)
-3	0.1	0.1
-1	0.2	0.3
0	0.15	0.45
1	0.2	0.65
2	0.1	0.75
3	0.15	0.90
5	0.05	0.95
8	0.05	1

Find compute the probability

- $P(X \leq 2)$
- $P(X \leq 5)$
- X is negative
- X is even

$$\begin{aligned}
 P(X \leq 2) &= 0.75 \\
 P(X \leq 5) &= 0.95 \\
 P(X \text{ is } -ve) &= 0.3 \\
 P(X \text{ is even}) &= 0.1 + 0.05 \\
 &= 0.15
 \end{aligned}$$

X	P(X)
-3	0.1
-1	0.2
0	a
1	0.3
3	0.15

Soln

X	P(X)
-3	0.1
-1	0.2
0	a
1	0.3
3	0.15

Total probability = 1

$$\begin{aligned}
 1 &= 0.1 + 0.2 + a + 0.3 + 0.15 \\
 1 &= a + 0.75 \\
 a &= 0.25
 \end{aligned}$$

Diff
 pdf & pms
 continuous, discrete & random

* Random Variable (R.V)

- Random Variable is define as a real value function in a sample space 'S' in a random experiment.
- Random Variable is denoted by capital letters i.e. X, Y etc and the value of a random variable is denoted by small letter.

i.e. let 'X' be a r.v in a sample space 'S' of a random experiment getting values $x_1, x_2, \dots, x_n$ where $x_1, x_2, \dots, x_n$ are values of R.V.

Types of R.V:

1. Discrete R.V:

A R.V is said to discrete R.V if it takes most countable values i.e. the real valued function defined on a discrete sample space.

for example:

- Number of calls in a plane in a minute.
- Number of Accidents in a road.

2. Continuous R.V:

→ A R.V is said to be continuous R.V if it can take all possible values in a certain interval for example:

- The average family income of worker in a factory lies in between 10 to 20 thousands.

Probability Mass function (pmf)

→ Let 'X' be a r.v in a sample space 'S' having values $x_1, x_2, \dots, x_n$ with their respective probability of occurrence is $p_1, p_2, \dots, p_n$.

The prob. of these values must satisfy the following condition:-

1. $P(X_i) \geq 0 \quad i=1, 2, \dots, n$
2. $\sum_{i=1}^n P(X_i) = 1$

The prob. can be written as:-
 $P(X_i)$ or $P(X = x_i)$ or p_i .

This 'p' is called pmf of a discrete R.V and the pair of values and prob. $\{x_i, P(x_i)\}$ is called prob. distribution of a R.V.

Probability Distribution function

Let 'X' be a discrete r.v of a sample space 'S' having probability. Then, the probability distribution function of r.v 'X' defined as:-

$$F(X_i) = F_X(x_i) = P(X \leq x_i) \\ = P(x_0) + P(x_1) + P(x_2) + \dots + P(x_i)$$

It is also known as cumulative prob.

The properties of Prob. distⁿ function are:

- i) $0 \leq F(X_i) \leq 1, i=0, 1, 2, \dots, \infty$

$$ii) F(-\infty) = \lim_{x_i \rightarrow -\infty} F(x_i) = 0$$

$$iii) F(\infty) = \lim_{x_i \rightarrow \infty} F(x_i) = 1$$

$$iv) P(a \leq X \leq b) = F(b) - F(a)$$

$$v) F(x_i) \leq F(x_j) \text{ if } x_i \leq x_j$$

Q Let 'X' be a rv is a sample space 'S'

Value of X(x)	-2	-1	0	1	2	3
P(X)	0.1	k	0.2	2k	0.3	k

1) Find value of k.

2) Find $P(X < 2)$, $P(-2 \leq X < 2)$, $P(0 \leq X < 2)$, $P(X \geq -2)$

3) Find $F(-2)$, $F(0)$, $F(3)$

4) Construct distribution function.

Soln

X	P(X)	F(X)	P(X)
-2	0.1	0.1	0.1
-1	k	$0.1 + k = 0.2$	0.1
0	0.2	$0.3 + k = 0.4$	0.2
1	2k	$0.3 + 3k = 0.6$	0.2
2	0.3	$0.6 + 3k = 0.9$	0.3
3	k	$0.6 + 4k = 1$	0.1

Total probability = 1

1)

$$1 = 0.6 + 4k$$

$$\text{or } 0.4 = 4k$$

$$\therefore k = 0.1$$

$$2) P(X < 2) = 0.1 + 0.1 + 0.2 + 0.2 = 0.6$$

$$P(-2 \leq X < 2) = 0.1 + 0.2 + 0.2 = 0.5$$

$$P(0 \leq X < 2) = 0.2 + 0.2 = 0.4$$

$$P(X \geq -2) = 0.1 + 0.1 + 0.2 + 0.2 + 0.3 + 0.1 = 1$$

$$3) F(-2) = P(X \leq -2) = 0.1$$

$$F(0) = P(X \leq 0)$$

$$= P(0) + P(-1) + P(-2)$$

$$= 0.2 + 0.1 + 0.1$$

$$= 0.4$$

$$F(3) = P(X \leq 3)$$

$$= P(3) + P(2) + P(1) + P(0) + P(-1) + P(-2)$$

$$= 0.1 + 0.3 + 0.2 + 0.2 + 0.1 + 0.1 = 1$$

X	-2	-1	0	1	2	3
F(X)	0.1	0.2	0.4	0.6	0.9	1

* Probability density function (pdf)

- Let 'X' be a continuous R.V in a sample space 'S' have value x. Let f(x) be a function of x considering small interval $(x - \frac{dx}{2}, x + \frac{dx}{2})$ then the prob. that

the continuous r.v 'X' lies in an interval $(x - \frac{dx}{2}, x + \frac{dx}{2})$ is given by $f_x(x)$ or f(x)

or $f(x=x)$ and calculated as:-

$$f(x) = \int_{x-dx}^{x+dx/2} f(x) dx$$

* Properties:-

i) $f(x) \geq 0$

ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

iii) Let 'X' be event lies in between α and β then
 $P(\alpha \leq X \leq \beta) = \int_{\alpha}^{\beta} f(x) dx$

Probability distribution function in continuous R.V.

$$F(X \leq x) = F(x) = f(X \leq x_i)$$

Q.1 A continuous R.V in a range $(-3, 3)$ is defined as:-

$$f(X=x) = \frac{1}{16} (x+3)^2, \text{ for } -3 \leq x < -1$$

$$= \frac{1}{16} (6-2x^2), \text{ for } -1 \leq x < 1$$

$$= \frac{1}{16} (3-x)^2, \text{ for } 1 \leq x \leq 3$$

1) Verify that the area under the curve is unity.

2) Determine $P(0.5 < X < 1)$

Soln

$$\text{pdf} = \int_{-3}^{-1} \frac{1}{16} (x+3)^2 dx + \int_{-1}^1 \frac{1}{16} (6-2x^2) dx + \int_1^3 \frac{1}{16} (3-x)^2 dx$$

$$= \frac{1}{16} \int_{-3}^{-1} x^2 + 6x + 9 dx + \frac{1}{16} \int_{-1}^1 6 - 2x^2 dx + \frac{1}{16} \int_1^3 9 - 6x + x^2 dx$$

$$= \frac{1}{16} \left[\frac{x^3}{3} + 6x^2 + 9x \right]_{-3}^{-1} + \frac{1}{16} \left[6x - \frac{2x^3}{3} \right]_{-1}^1 + \frac{1}{16} \left[9x - \frac{6x^2}{2} + \frac{x^3}{3} \right]_1^3$$

$$= \frac{1}{16} \left[\frac{-1}{3} + 3 - 9 - \left(\frac{-27}{3} + 6(-3)^2 + 9(-3) \right) \right] + \frac{1}{16} \left[\frac{6-2}{3} + 6 + \frac{2(-1)^3}{3} \right] + \frac{1}{16} \left[\frac{27-27}{3} + 8 - 9 + 3 - \frac{1}{3} \right]$$

$$= \frac{1}{16} \left[\frac{-1}{3} + 3 - 8 + 8 - 27 + 27 \right] + \frac{1}{16} \left[\frac{6-2}{3} + \frac{6-2}{3} \right] + \frac{1}{16} \left[8 - \frac{1}{3} \right]$$

$$= \frac{1}{6} + \frac{2}{3} + \frac{1}{6}$$

$$= \frac{6}{6}$$

$$= 1$$

$$ii) P(0.5 < X < 1) = \int_{0.5}^1 \frac{1}{16} (6 - 2x^2) dx$$

$$= \frac{1}{16} \int_{0.5}^1 (6 - 2x^2) dx$$

$$= \frac{1}{16} \left[6x - \frac{2x^3}{3} \right]_{0.5}^1$$

$$= \frac{1}{16} \left[\frac{6-2}{3} - \frac{3+0.25}{3} \right] =$$

$$= 0.15$$

2. A pdf of O.r.v is given as:

$$f(x) = \frac{x+2}{6}, 0 \leq x < 2$$

= 0 otherwise

Find

1. $P(X > 0)$

2. $F(1.5)$

3. $P(0 < X \leq 1)$

Soln

1. $P(X > 0)$

$$p.d.f = \int_{-\infty}^{\infty} f(x) dx$$

$$= \int_0^2 \frac{x+2}{6} dx + \int_{-\infty}^0 0 dx$$

$$= \frac{1}{6} \left[\frac{x^2}{2} + 2x \right]_0^2$$

$$= \frac{1}{6} \times 6$$

$$= 1$$

2. $F(1.5) = P(X \leq 1.5)$

$$= \int_0^{1.5} \frac{x+2}{6} dx$$

$$= \frac{1}{6} \left[\frac{x^2}{2} + 2x \right]_0^{1.5}$$

$$= \frac{11}{16} = 0.69$$

3) $P(0 < X \leq 1)$

$$\begin{aligned} p.d.f &= \int_0^1 \frac{x+2}{6} dx \\ &= \frac{1}{6} \left[\frac{x^2}{2} + 2x \right]_0^1 \\ &= \frac{5}{12} \end{aligned}$$

* Mathematical Expectation of R.V

Let 'X' be a r.v of a sample space 'S' having values $x_1, x_2, \dots, x_n$ with respective prob. $P_1, P_2, \dots, P_n$ then mathematical expectation of a r.v is given as:-

$$E(X) = \sum_{i=1}^n x_i \cdot P(X=x_i) = \text{Mean for discrete variable}$$

$$= \int_{-\infty}^{\infty} x f(x) dx \text{ for continuous variables}$$

Theorem-1

The mathematical expectation of sum of r.v is equal to the sum of their expectation.

$$\text{i.e. } E(X+Y+Z+\dots) = E(X) + E(Y) + E(Z) + \dots$$

Theorem-2

The mathematical expectation of product of number of independent R.V is equal to the product of their expectation.

$$E(XYZ\dots) = E(X) \cdot E(Y) \cdot E(Z) \cdot \dots$$

Note

* If 'a' be constant value then, $E(a)$ is unchanged.
i.e. $E(a) = a$

* If 'a' and 'b' are constant then,
 $E(ax+b) = aE(x) + b$

Variance

$$V(X) = E[X - E(X)]^2$$

$$= E[X^2 - 2X \cdot E(X) + (E(X))^2]$$

$$= E(X^2) - 2(E(X))^2 + (E(X))^2$$

$$= E(X^2) - (E(X))^2$$

$$\begin{aligned} E(X) &= \sum x_i \cdot P(X=x_i) \\ E(X^2) &= \sum x_i^2 \cdot P(X=x_i) \end{aligned}$$

for discrete

$$\int_{-\infty}^{\infty} x^2 f(x) dx$$

for continuous

Properties of Variance

* If 'a' and 'b' be constant, then,

$$V(aX+b) = V(aX) + V(b)$$

$$= a^2 V(X) + 0$$

$$= a^2 V(X)$$

* If a_1 and a_2 be two constant variables then, 'X' and 'Y' be two r.v.

$$V(a_1X + a_2Y) = a_1^2 V(X) + a_2^2 V(Y) + 2a_1a_2 \cdot \text{cov}(X, Y)$$

If 'X' and 'Y' are independent, $\text{cov}(X, Y) = 0$

$$\therefore V(a_1X + a_2Y) = a_1^2 V(X) + a_2^2 V(Y)$$

Q.	X	P(X)
	-2	0.1
	-1	k^2
	0	0.2
	1	$2k^2$
	2	0.3
	3	k^2

find the value of k

$$\rightarrow E(X)$$

$$\rightarrow V(X)$$

$$\rightarrow E(X+3)$$

$$\rightarrow V(X+3)$$

Soln

$$P(X) = 1$$

then,

$$P(X) = P(-2) + P(-1) + P(0) + P(1) + P(2) + P(3)$$

$$1 = 0.1 + k^2 + 0.2 + 2k^2 + 0.3 + k^2$$

$$\Rightarrow 1 = 0.6 + 4k^2$$

$$\Rightarrow 4k^2 = 0.4$$

$$\therefore k^2 = 0.1$$

X	P(X)	X^2
-2	0.1	4
-1	0.1	1
0	0.2	0
1	0.2	1
2	0.3	4
3	0.1	9

$$E(X) = \sum_{i=-2}^3 x_i P(X=x_i)$$

$$= -2 \times 0.1 - 1 \times 0.1 + 0 \times 0.2 + 1 \times 0.2 + 2 \times 0.3 + 3 \times 0.1$$

$$= 0.8$$

$$E(X^2) = \sum_{i=-2}^3 x_i^2 P(X=x_i)$$

$$= 4 \times 0.1 + 1 \times 0.1 + 0 \times 0.2 + 1 \times 0.2 + 4 \times 0.3 + 9 \times 0.1$$

$$= 2.8$$

$$V(X) = E(X^2) - (E(X))^2$$

$$= 2.8 - (0.8)^2$$

$$= 2.16$$

$$V(X+3) = V(X) + V(3)$$

$$= 2.16 + 0$$

$$= 2.16$$

$$E(X+3) = E(X) + E(3)$$

$$= 0.8 + 3$$

$$= 3.8$$

$$V(2X+3) = V(2X) + V(3)$$

$$= 2^2 V(X) + 0$$

$$= 4 \times 2.16$$

$$= 8.64$$

$$V(ax+b) = a^2 V(X)$$

$$+ V(b)$$

Covariance

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E[XY - X \cdot E(Y) - E(X) \cdot Y + E(X) \cdot E(Y)]$$

$$= E(XY) - E(X) \cdot E(Y) - E(X) \cdot E(Y) + E(X) \cdot E(Y)$$

$$= E(XY) - E(X) \cdot E(Y)$$

X	P(X)
-3	0.1
-1	0.2
0	0.15
1	0.2
2	0.1
3	0.15
5	0.05
8	0.05

Find

$$1) E(X)$$

$$2) V(X)$$

$$3) E(2X+3)$$

$$4) V(2X+3)$$

Soln

$$1) E(X) = \sum_{i=-3}^8 x_i P(X=x_i)$$

$$= -3 \times 0.1 - 1 \times 0.2 + 0 \times 0.15 + 1 \times 0.2 + 2 \times 0.1 + 3 \times 0.15 + 5 \times 0.05 + 8 \times 0.05$$

$$= 1$$

$$E(X^2) = \sum_{i=-3}^8 (x_i)^2 P(X=x_i)$$

$$= 9 \times 0.1 + 1 \times 0.2 + 0 \times 0.15 + 1 \times 0.2 + 4 \times 0.1 + 9 \times 0.15 + 25 \times 0.05 + 64 \times 0.05$$

$$= 7.5$$

$$2) V(X) = E(X^2) - (E(X))^2$$

$$= 7.5 - 1$$

$$= 6.5$$

$$3) E(2X+3) = E(2X) + E(3)$$

$$= 2E(X) + 3$$

$$= 2 \times 1 + 3$$

$$= 5$$

$$4) V(2X+3) = V(2X) + V(3)$$

$$= 4V(X) + 0$$

$$= 4 \times 6.5$$

$$= 26$$

2. Find mean and variance.

$$f(x) = xe^{-x}, x \geq 0$$

Soln

$$E(x) = \int_{-\infty}^{\infty} xf(x) dx$$

$$= \int_0^{\infty} x \cdot xe^{-x} dx$$

$$= \int_0^{\infty} x^2 e^{-x} dx$$

Put

$$x^2 = u$$

$$2x dx = du$$

$$\text{When, } x=0, u=0$$

$$\text{When, } x=\infty, u=\infty$$

$$= \int_0^{\infty} e^{-x} x^{3-1} dx$$

$$= \frac{\Gamma(n)}{(n-1)!}$$

$$= \frac{(3-1)!}{2!}$$

$$= 2$$

$$E(x^2) = \int_0^{\infty} x^2 \cdot xe^{-x} dx$$

$$= \int_0^{\infty} x^3 e^{-x} dx$$

$$= \int_0^{\infty} e^{-x} x^{4-1} dx$$

$$= \frac{\Gamma(4)}{(4-1)!}$$

$$= \frac{3!}{3!} = 3 \times 2 = 6$$

$$V(x) = E(x^2) - (E(x))^2$$

$$= 6 - 2^2$$

$$= 2$$

3. $f(x) = Ax(1-x)^2, 0 \leq x \leq 1$

Find

$E(x)$ and $V(x)$.

Soln

$$E(x) = \int_{-\infty}^{\infty} xf(x) dx$$

$$= \int_0^1 x Ax(1-x)^2 dx$$

$$= A \int_0^1 x^2(1-x)^2 dx$$

$$= A \int_0^1 x^2(1-2x+x^2) dx$$

$$= A \int_0^1 x^2 - 2x^3 + x^4 dx$$

$$= A \left[\frac{x^3}{3} - \frac{2x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$$= A \left[\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right]$$

$$= \frac{A}{30}$$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \int_0^1 x^2 Ax(1-x)^2 dx$$

$$= A \int_0^1 x^3(1-x)^2 dx$$

$$= A \int_0^1 x^3(1-2x+x^2) dx$$

$$= A \int_0^1 x^3 - 2x^4 + x^5 dx$$

$$= A \left[\frac{x^4}{4} - \frac{2x^5}{5} + \frac{x^6}{6} \right]_0^1$$

$$= \frac{A}{60}$$

$$1 = \int_0^1 Ax(1-x)^2 dx$$

$$1 = A \int_0^1 x - 2x^2 + x^3 dx$$

$$1 = A \left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right]_0^1$$

$$1 = A \left[\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right]$$

$$1 = \frac{A}{12}$$

$$\therefore A = 12$$

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$$\begin{aligned}
 V(X) &= E(X^2) - (E(X))^2 \\
 &= \frac{A}{60} - \left(\frac{A}{30}\right)^2 \\
 &= \frac{A}{60} - \frac{A^2}{900} \\
 &= \frac{15A - A^2}{900}
 \end{aligned}$$

for median value

$$\int_0^m f(x) dx = \int_m^1 f(x) dx = \frac{1}{2}$$

$$\int_0^m Ax(1-x)^2 dx = \int_m^1 Ax(1-x)^2 dx = \frac{1}{2}$$

$$\Rightarrow 12 \int_0^m x(1-2x+x^2) dx = \int_m^1 12x(1-2x+x^2) dx = \frac{1}{2}$$

$$\text{or, } \int_0^m (12x - 24x^2 + 12x^3) dx = \int_m^1 (12x - 24x^2 + 12x^3) dx = \frac{1}{2}$$

$$\Rightarrow \left[\frac{x^2}{2} - \frac{24x^3}{3} + \frac{12x^4}{4} \right]_0^m = \left[\frac{x^2}{2} - \frac{24x^3}{3} + \frac{12x^4}{4} \right]_m^1 = \frac{1}{2}$$

$$\Rightarrow \frac{M^2}{2} - \frac{24M^3}{3} + \frac{12M^4}{4} = \frac{1}{2} - \frac{12}{2} + \frac{12}{2} - \frac{M^2}{2} + \frac{24M^3}{3} - \frac{12M^4}{4} = \frac{1}{2}$$

$$\Rightarrow \frac{M^2}{3} - \frac{4M^3}{2} + \frac{M^4}{12} = \frac{1}{12} = \frac{1}{12} - \frac{1}{2} + \frac{1}{2} - \frac{1}{12} = 0$$

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$$\Rightarrow \frac{M^2}{3} - \frac{4M^3}{2} + \frac{M^4}{12} = \frac{1}{12}$$

$$\Rightarrow 6M^2 - 8M^3 + 3M^4 = \frac{1}{2}$$

$$\Rightarrow 6M^4 - 16M^3 + 12M^2 - 1 = 0$$

$$\Rightarrow \left[\frac{M^2}{2} - \frac{2M^3}{3} + \frac{M^4}{4} \right]_{12} = \frac{1}{2}$$

$$\Rightarrow 6M^2 - 8M^3 + 3M^4 = \frac{12}{24}$$

$$\Rightarrow 6M^2 - 8M^3 + 3M^4 = \frac{1}{2}$$

$$\therefore 6M^4 - 16M^3 + 12M^2 - 1 = 0$$

To find first quarter,

$$\text{or, } \int_0^{Q_1} Ax(1-x)^2 dx = \frac{1}{4}$$

$$\text{or, } 12 \int_0^{Q_1} x(1-x)^2 dx = \frac{1}{4}$$

$$\text{or, } 12 \left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right]_0^{Q_1} = \frac{1}{4}$$

$$\text{or, } 12 \left[\frac{Q_1^2}{2} - \frac{2Q_1^3}{3} + \frac{Q_1^4}{4} \right] = \frac{1}{4}$$

$$\text{or, } 6Q_1^2 - 8Q_1^3 + 3Q_1^4 = \frac{12}{48}$$

$$\text{or, } 6Q_1^2 - 8Q_1^3 + 3Q_1^4 = \frac{1}{4}$$

$$\therefore 12Q_1^4 - 32Q_1^3 + 24Q_1^2 - 1 = 0$$

To find third quarter

$$\int_0^3 Ax(1-x)^2 dx = \frac{3}{4}$$

$$\Rightarrow 12 \left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right]_0^3 = \frac{3}{4}$$

$$\text{or, } \frac{Q_3^2}{2} - \frac{2Q_3^3}{3} + \frac{Q_3^4}{4} = \frac{1}{16}$$

$$\text{or, } 6Q_3^2 - 8Q_3^3 + 3Q_3^4 = \frac{3}{4}$$

$$\therefore 12Q_3^4 - 32Q_3^3 + 24Q_3^2 - 3 = 0$$

$$E(Y) = \int_y y f(y) dy$$

$$E(X) = \int_x x f(x) dx$$

$$E(XY) = \int_x \int_y xy f(x,y) dy dx$$

$$f(x,y) = f(x) \cdot f(y) = \text{joint pdf}$$

$$E(X,Y) = \sum \sum xy P(XY)$$

$$P(XY) = P(X) \cdot P(Y)$$

for continuous
r.v

for discrete R.V

$$Q. f(x) = 3x^2 \quad 0 \leq x \leq 1$$

$$f(y) = 2y + 4 \quad 0 \leq y \leq 1$$

find

$$\text{COV}(X,Y) = E(XY) - E(X) \cdot E(Y)$$

Soln

$$f(XY) = 3x^2(2y+4) \\ = 6x^2y + 12x^2$$

$$E(XY) = \int_x \int_y xy (6x^2y + 12x^2) dy dx$$

$$= \int_0^1 \int_0^1 (6x^3y^2 + 12x^3y) dy dx$$

$$= \int_0^1 \left[\frac{6x^3y^3}{3} + \frac{12x^3y^2}{2} \right]_0^1 dx$$

$$= \int_0^1 (2x^3 + 6x^3) dx$$

$$= \int_0^1 8x^3 dx$$

$$= \left[\frac{8x^4}{4} \right]_0^1$$

$$= 2$$

$$E(X) = \int_0^1 x(8x^2) dx$$

$$= \int_0^1 8x^3 dx$$

$$= \left[\frac{8x^4}{4} \right]_0^1$$

$$= 2$$

$$E(Y) = \int_0^1 y(2y+4) dy$$

$$= \int_0^1 2y^2 + 4y dy$$

$$= \left[\frac{2y^3}{3} + \frac{4y^2}{2} \right]_0^1$$

$$= \frac{2}{3} + 2$$

$$= \frac{8}{3}$$

$$COV(XY) = E(XY) - E(X)E(Y)$$

$$= 2 - 2 \times \frac{8}{3}$$

$$= 0$$

Distribution of discrete R.V

- Binomial Distribution

→ only two outcomes $E(X)$ = mean / variance

- Poisson Distribution

Mean / Variance - Rare event

- Negative Binomial Distribution

→ Success in r^{th} trial

* Binomial Distribution =

→ It is widely used distribution and is based on only two outcomes of a R.V. such as pass or fail, Yes or No etc.

When a trial is conducted 'n' times but the prob. of getting success is 'p' then that of failure is 'q'.

Let 'X' be a r.v in a sample space 'S' is said to follow Binomial distribution with parameter 'n' and p i.e. $B \sim (n, p)$ if it's pmf is =

$$P(X=x) = {}^nC_x p^x q^{n-x}, x=0, 1, 2, \dots, n$$

$$= {}^nC_x p^x (1-p)^{n-x}$$

$$p+q=1$$

$$\therefore q=1-p$$

mean = np
variance = npq

$$(x+a)^n = {}^nC_0 x^n a^0 + {}^nC_1 x^{n-1} a^1 + \dots$$

Features of Binomial Distribution

- 'n' no. of trials should be finite.
- Each trial gives only two possible outcomes.
- All trials are independent.
- The prob. of success in any trial is 'p' and is constant for each trial.

Mean of Binomial Distribution

$$E(x) = \sum_{i=1}^n n P(X=x) = \sum_{i=1}^n x \cdot {}^nC_x p^x q^{n-x}$$

$$= \sum_{i=0}^n \frac{n!}{(n-x)! x!} (x) p^n q^{n-x}$$

$$= {}^nC_0 \cdot 0 \cdot p^0 q^{n-0} + {}^nC_1 p^1 q^{n-1} + \dots$$

$$= \sum_{x=0}^n \frac{x \cdot n(n-1)!}{(n-x)! x(x-1)!} p^x q^{n-x}$$

$$= \sum_{x=1}^n \frac{n(n-1)!}{(n-x)! (x-1)!} p^{x-1} \cdot p \cdot q^{n-x}$$

$$= np \left[\sum_{i=1}^n {}^{n-1}C_{x-1} p^{x-1} q^{n-x} \right]$$

$$= np \cdot (p+q)^{n-1} \left[{}^nC_0 p^0 q^n + {}^nC_1 p^1 q^{n-1} + \dots \right]$$

$$= np \cdot 1 \cdot (p+q)^n$$

$$V(x) = E(x^2) - [E(x)]^2 \rightarrow np$$

$$E(x^2) = \sum_{x=0}^n x^2 \cdot P(X=x)$$

$$= \sum_{x=0}^n (x(x-1) + x) \cdot P(X=x)$$

$$= \sum_{x=0}^n (x(x-1) P(X=x) + \sum_{x=0}^n x \cdot P(X=x))$$

$$= \sum_{x=0}^n x(x-1) {}^nC_x p^x q^{n-x} + np$$

$$= \sum_{x=2}^n \frac{x(x-1) \cdot n(n-1)(n-2)!}{(n-x)! x(x-1)(x-2)!} p^{x-2} \cdot p^2 q^{n-x} + np$$

$$= n(n-1)p^2 \left(\sum_{x=2}^n \frac{n-2}{x-2} {}^{n-2}C_{x-2} p^{x-2} q^{n-x} \right) + np$$

$$= n(n-1)p^2 \cdot 1 + np$$

$$= n^2 p^2 - np^2 + np$$

$$\therefore V(x) = E(x^2) - [E(x)]^2$$

$$= n^2 p^2 - np^2 + np - n^2 p^2$$

$$= np(1-p)$$

$$= npq \quad (\because 1-p=q)$$

$$\therefore V(x) = npq$$

Moment generating function

$$M_X(t) = E(e^{tx})$$

$$= \sum_{x=0}^n e^{tx} \cdot P(X=x)$$

$$= \sum_{x=0}^n e^{tx} \cdot {}^n C_x p^x q^{n-x} \left[{}^n C_{n-x} p^{n-x} q^x = (q+p)^n \right]$$

$$= \sum_{x=0}^n (pe^t)^x {}^n C_x q^{n-x}$$

$$= (q+pe^t)^n$$

$$\frac{d}{dt} (q+pe^t)^n = n(q+pe^t)^{n-1} pe^t$$

$$\text{at } t=0$$

$$M'_X(t) \Big|_{t=0} = np = \text{Mean}$$

$$\frac{d^2}{dt^2} (q+pe^t)^n = n(n-1)pe^t(q+pe^t)^{n-2} pe^t + n(q+pe^t)^{n-1} pe^t$$

$$= n(n-1)p^2 e^{2t} (q+pe^t)^{n-2} + npe^t (q+pe^t)^{n-1}$$

$$\text{at } t=0$$

$$M''_X(t) \Big|_{t=0} = n(n-1)p^2 + np$$

$$= n^2 p^2 - np^2 + np$$

$$V(X) = M''_X(t) \Big|_{t=0} - (M'_X(t) \Big|_{t=0})^2$$

$$= n^2 p^2 - np^2 + np - n^2 p^2$$

$$= np(1-p)$$

$$= npq$$

8. An insurance sales man sells policies to five men. All of identical age and good health. According to the actuarial table, the prob. that a man of a particular age will be alive 30 years is $\frac{2}{3}$. Find the prob. that in 30 years

i) All five men

ii) At least one man

iii) At most three men will be lived

Soln.

$$n=5$$

$$\text{prob. of alive } (p) = \frac{2}{3}$$

$$q = 1 - \frac{2}{3} = \frac{1}{3}$$

Here, Acc. to the question, Binomial distribution is used to calculate prob.

$$\text{pmf} = P(X=x) = {}^n C_x p^x q^{n-x}, x=0, 1, \dots, n$$

$$\begin{aligned} \text{i) } P(X=5) &= {}^5 C_5 p^5 q^{5-5} \\ &= {}^5 C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^{5-5} \\ &= 1 \cdot \left(\frac{2}{3}\right)^5 \cdot 1 \\ &= \left(\frac{2}{3}\right)^5 \\ &= 0.132 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(X \geq 1) &= \sum_{x=1}^5 P(X=x) \\ &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &\quad + P(X=5) \end{aligned}$$

$${}^5C_0 = \frac{5!}{(5-0)!0!}$$

$$\begin{aligned} &= 1 - P(X=0) \\ &= 1 - {}^5C_0 p^0 q^{5-0} \\ &= 1 - {}^5C_0 \cdot 1 \cdot \left(\frac{1}{3}\right)^5 \\ &= 1 - 1 \cdot \left(\frac{1}{3}\right)^5 \\ &= 0.995 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= {}^5C_0 p^0 q^{5-0} + {}^5C_1 p^1 q^{5-1} + {}^5C_2 p^2 q^{5-2} \\ &\quad + {}^5C_3 p^3 q^{5-3} \\ &= 5C_0 \cdot 1 \cdot \left(\frac{1}{3}\right)^5 + 5C_1 \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^4 + 5C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^3 \\ &\quad + 5C_3 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^2 \\ &= 0.589 \end{aligned}$$

Q. 12% of the items produced by a machine are defective. What is the prob. that out of 20 sample of items, produced by machine, 5 items were found to be defective.

* find the prob. that atmost 4 items are found to be defective.

Soln

$$n = 20$$

$$p = 0.12$$

$$q = 1 - p = 1 - 0.12 = 0.88$$

Here, Acc to the question, Binomial distribution is used to calculate prob.

$$\text{pmf} = P(X=x) = {}^nC_x p^x q^{n-x}, x=0, 1, \dots, n$$

$$\begin{aligned} \text{i) } P(X=5) &= {}^nC_5 p^5 q^{n-5} \\ &= {}^{20}C_5 (0.12)^5 (0.88)^{20-5} \\ &= 0.056 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(X \leq 4) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &= {}^{20}C_0 p^0 q^{20-0} + {}^{20}C_1 p^1 q^{20-1} + {}^{20}C_2 p^2 q^{20-2} \\ &\quad + {}^{20}C_3 p^3 q^{20-3} + {}^{20}C_4 p^4 q^{20-4} \\ &= {}^{20}C_0 (0.12)^0 (0.88)^{20} + {}^{20}C_1 (0.12)^1 (0.88)^{19} \\ &\quad + {}^{20}C_2 (0.12)^2 (0.88)^{18} + {}^{20}C_3 (0.12)^3 (0.88)^{17} \\ &\quad + {}^{20}C_4 (0.12)^4 (0.88)^{16} \\ &= 0.917 \end{aligned}$$

* Poisson distribution

$$\text{pmf} = P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, x=0, 1, 2, \dots, \infty$$

It is used in rare events.

- No. of Accidents.
- No. of telephone calls arriving at a telephone switch board in unit time.
- No. of suicides reported in a particular time.

$$\text{pmf} = P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, x=0, 1, \dots, \infty$$

$$E(X) = \sum_{x=0}^{\infty} x \cdot P(X=x)$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=1}^{\infty} \frac{e^{-\lambda} \lambda^x}{(x-1)!}$$

$$= \sum_{x=1}^{\infty} \frac{e^{-\lambda} \lambda^{x-1} \lambda}{(x-1)!}$$

$$= e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= e^{-\lambda} \lambda \left[\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \dots \right]$$

$$= e^{-\lambda} \lambda \cdot e^{\lambda}$$

$$= \lambda$$

$$E(X^2) = E[X(X-1) + X]$$

$$= \sum_{x=0}^{\infty} x^2 p(x)$$

$$= \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} + \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=2}^{\infty} \frac{e^{-\lambda} \lambda^x}{(x-2)!} + \lambda$$

$$= e^{-\lambda} \left[\frac{\lambda^2}{1!} + \frac{\lambda^3}{2!} + \frac{\lambda^4}{3!} + \dots \right] + \lambda$$

$$= e^{-\lambda} \lambda^2 \left[\frac{1}{1!} + \frac{\lambda}{2!} + \frac{\lambda^2}{3!} + \dots \right] + \lambda$$

$$= e^{-\lambda} \lambda^2 \cdot e^{\lambda} + \lambda$$

$$= \lambda^2 + \lambda$$

Hence,

$$V(X) = E(X^2) - (E(X))^2$$

$$= \lambda^2 + \lambda - \lambda^2$$

$$\therefore V(X) = \lambda$$

$$\text{Mean} = \text{Variance} = \lambda$$

* Moment Generating function

$$M_x(t) = \sum_{x=0}^{\infty} e^{tx} \cdot p(x=x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} (e^t \lambda)^x \cdot \frac{e^{-\lambda}}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(e^t \lambda)^x}{x!}$$

$$= e^{-\lambda} \left[1 + \frac{e^t \lambda}{1!} + \frac{(e^t \lambda)^2}{2!} + \dots + \infty \right]$$

$$= e^{-\lambda} \cdot e^{e^t \lambda}$$

$$= e^{(-\lambda + e^t \lambda)}$$

$$= e^{\lambda(e^t - 1)} \quad \#$$

$$\frac{d}{dt} (e^{\lambda(e^t - 1)}) = \frac{d}{dt} M_x(t)$$

$$\Rightarrow \log M_x(t) = \lambda(e^t - 1)$$

$$\frac{d}{dt} \log M_x(t) = \lambda(e^t - 1)$$

$$\frac{d}{dt} M_x(t) = \lambda e^t - 0$$

$$\begin{aligned} M_x'(t) &= \frac{d}{dt} e^{\lambda(e^t - 1)} \times \frac{d}{dt} (e^t - 1) \times \frac{d}{dt} e^t \\ &= e^{\lambda(e^t - 1)} \cdot \lambda \frac{d}{dt} (e^t - 1) \cdot e^t \\ &= e^{\lambda(e^t - 1)} \cdot \lambda \cdot e^t \end{aligned}$$

at $t=0$

$$M_x'(t) \Big|_{t=0} = e^{\lambda(1-1)} \cdot \lambda e^0$$

$$= e^0 \cdot \lambda$$

$$= \lambda$$

$\therefore \lambda = \text{Mean}$

$$\frac{d^2}{dt^2} (e^{\lambda(e^t - 1)}) = \frac{d}{dt} (e^{\lambda(e^t - 1)} \cdot \lambda e^t)$$

* Between the hours of 2 and 4 p.m., the average number of phone calls per minute coming into the switch board of a company is 2.5. Find the prob. during one particular minute there will be

- i) no call at all
- ii) Exactly 3 calls
- iii) Atleast 5 calls
- iv) Exactly 4 calls in a 15 minute

$$\rightarrow \text{Pmf } (P(X=x)) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots, \infty$$

$$\text{Average no. of phone calls } (\lambda) = 2.5$$

i) no call at all

$$P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda} = e^{-2.5} = 0.082$$

ii) Exactly 3 calls

$$P(X=3) = \frac{e^{-\lambda} \lambda^3}{3!} = \frac{e^{-2.5} (2.5)^3}{3!} = 0.213$$

iii) Atleast 5 calls

$$P(X \geq 5) = 1 - P(X < 5) = 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)]$$

$$= 1 - [0.082 + \frac{e^{-\lambda} \lambda^1}{1!} + \frac{e^{-\lambda} \lambda^2}{2!} + 0.213 + \frac{e^{-\lambda} \lambda^4}{4!}]$$

$$= 1 - [0.082 + 0.213 + 0.5953] = 0.109$$

iv) Exactly 4 calls in a 15-minute

$$\rightarrow \text{Calls in 15 min} = 2.5 \times 15 = 37.5 = \lambda$$

$$P(X=4) = \frac{e^{-\lambda} \lambda^4}{4!}$$

$$= \frac{e^{-37.5} (37.5)^4}{4!} = 0.549$$

$$= 4.264 \times 10^{-12}$$

Q. A typist made 2.8 mistake on an average per page. Find the prob. that in the page typed by him

- There is no mistake
- There are two or less mistake
- More than 10 mistake
- At least 7 mistake

Soln

$$\rightarrow \text{Pmf } (P(X=x)) = \frac{e^{-\lambda} \lambda^x}{x!}, x=0, 1, 2, \dots, \infty$$

$$\text{Average mistake } (\lambda) = 2.8$$

i) No mistake:

$$P(X=0) = \frac{e^{-2.8} (2.8)^0}{0!} \\ = 0.061$$

ii) There are two or less mistake

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) \\ = 0.061 + \frac{e^{-2.8} \lambda^1}{1!} + \frac{e^{-2.8} \lambda^2}{2!}$$

$$= 0.470$$

iii)

$$P(X \geq 10) = 1 - P(X \leq 10) \\ = 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ + P(X=4) + P(X=5) + P(X=6) + P(X=7) \\ + P(X=8) + P(X=9) + P(X=10)]$$

$$= 1 - \left[\frac{0.470}{1!} + \frac{e^{-\lambda} \lambda^3}{3!} + \frac{e^{-\lambda} \lambda^4}{4!} + \frac{e^{-\lambda} \lambda^5}{5!} + \frac{e^{-\lambda} \lambda^6}{6!} \right. \\ \left. + \frac{e^{-\lambda} \lambda^7}{7!} + \frac{e^{-\lambda} \lambda^8}{8!} + \frac{e^{-\lambda} \lambda^9}{9!} + \frac{e^{-\lambda} \lambda^{10}}{10!} \right]$$

$$= 1 - [0.470 + 0.222 + 0.156 + 0.087 + 0.041 \\ + 0.016 + 0.006 + 0.002 + 0.0004]$$

$$= 1 - 0.999 \\ = 0.001$$

iv)

$$P(X \geq 7) = 1 - P(X < 7)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ + P(X=4) + P(X=5) + P(X=6)]$$

$$= 1 - 0.470 + 0.222 + 0.156 + 0.087 + 0.041 \\ = 0.024$$

* Hypergeometric Distribution

→ Hypergeometric distribution is widely used in the case of sampling without replacement case.

Let 'N' be the total size of the distribution with $N = N_1 + N_2$ where N_1 and N_2 are size of first and 2nd sample drawn.

Let 'n' be the total no. of defective and 'x' be the no. of defective from first sample ' N_1 ' then no. of defective in 2nd sample is $n - x$.

The pmf of hypergeometric distribution is:-

$$P(X=x) = \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, 2, \dots, (N_1 \leq n)$$

$$= \frac{{}^{N_1}C_x \cdot {}^{N_2}C_{n-x}}{{}^NC_n}$$

$$\text{Mean} = E(X) = \sum_{x=0}^{\infty} x P(X=x)$$

$$= \sum_{x=0}^n x \cdot \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} = \sum_{x=0}^n x \cdot \frac{\binom{N+N_2}{n}}{\binom{N}{n}}$$

$$= \sum_{x=0}^n \frac{{}^{N_1}C_x \cdot {}^{N_2}C_{n-x} \cdot x}{{}^NC_n}$$

$$= \sum_{x=0}^n x \cdot \frac{N_1!}{(N_1-x)! x!} \cdot \frac{N_2!}{(N_2-n+x)! (n-x)!} \cdot \frac{N!}{(N-n)! n!}$$

$$= \sum_{x=0}^n \frac{x \cdot N_1(N_1-1)!}{x(x-1)!(N_1-x)!} \cdot \frac{N_2!}{(n-x)!} \cdot \frac{N!}{\binom{N}{n}}$$

$$= \sum_{x=1}^n \left(\frac{N_1-1}{x-1} \right) \cdot \frac{N_2!}{(n-x)!} \cdot \frac{N!}{\binom{N}{n}}$$

$$= N_1 \cdot \frac{(N_1-1+N_2)!}{(n-1)!} \cdot \frac{N!}{\binom{N}{n}} \quad [\because N_1+N_2=N]$$

$$= N_1 \cdot \frac{(N-1)!}{(n-1)!} \cdot \frac{N!}{(N-n)! n!}$$

$$= N_1 \cdot \frac{(N-1)! \cdot x(N-n)! \cdot n!}{N! (N-n)! (n-1)!}$$

$$= N_1 \cdot \frac{(N-1)! n(n-1)!}{N(N-1)!(n-1)!}$$

$$\therefore E(X) = \frac{N_1 \cdot n}{N}$$

N_1 = first sample size

N = Total sample size ($N_1 + N_2$)

n = Total no. of defective

$$E(X^2) = \sum_{x=0}^{\infty} x^2 P(X=x)$$

$$= \sum_{x=0}^n x^2 \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}}$$

$$= \sum_{x=0}^n x(x-1) \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} + \sum_{x=0}^n x \cdot \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}}$$

$$= \sum_{x=2}^n x(x-1) \frac{N_1!}{x!(N_1-x)!} \frac{\binom{N_2}{n-x}}{\binom{N}{n}} + \frac{N_1 \cdot n}{N}$$

$$= \sum_{x=2}^n x(x-1) N_1! \frac{\binom{N_2}{n-x}}{x(x-1)(x-2)!(N_1-x)!} + \frac{N_1 \cdot n}{N}$$

$$= \sum_{x=2}^n \frac{N_1!}{(x-2)!(N_1-x)!} \frac{\binom{N_2}{n-x}}{\binom{N}{n}} + \frac{N_1 \cdot n}{N}$$

$$= \sum_{x=2}^n \frac{N_1(N_1-1)(N_1-2)!}{(x-2)!(N_1-2-(x-2))!} \frac{\binom{N_2}{n-x}}{\binom{N}{n}} + \frac{N_1 \cdot n}{N}$$

$$= N_1(N_1-1) \sum_{x=2}^n \frac{\binom{N_1-2}{x-2} \binom{N_2}{n-x}}{\binom{N}{n}} + \frac{N_1 \cdot n}{N}$$

$$= N_1(N_1-1) \frac{\binom{N_1-2+N_2}{n-2}}{\binom{N}{n}} + \frac{N_1 \cdot n}{N}$$

$$= N_1(N_1-1) \frac{(N-2)!}{(n-2)!} \frac{1}{N!} + \frac{N_1 \cdot n}{N}$$

$$= N_1(N_1-1) \frac{(N-2)!}{(n-2)!} \times \frac{n(n-1)(n-2)!}{N(N-1)(N-2)!} + \frac{N_1 \cdot n}{N}$$

$$= N_1(N_1-1) \frac{n(n-1)}{N(N-1)} + \frac{N_1 \cdot n}{N}$$

$$V(X) = E(X^2) - (E(X))^2$$

$$= \frac{N_1(N_1-1)n(n-1)}{N(N-1)} + \frac{N_1 \cdot n}{N} + \left(\frac{N_1 \cdot n}{N}\right)^2$$

$$= \frac{N_1 \cdot n}{N} \left(\frac{(N_1-1)(n-1)}{(N-1)} + 1 + \frac{N_1 \cdot n}{N} \right)$$

$$= \frac{N_1 \cdot n}{N} \left(\frac{(N_1-1)(n-1) \cdot N + N(N-1) - N_1 \cdot n(N-1)}{N(N-1)} \right)$$

$$= \frac{N_1 \cdot n}{N} \left(\frac{NN_1 - N(n-1) + N^2 - N - N_1 \cdot n(N-1)}{N(N-1)} \right)$$

$$= \frac{N_1 \cdot n}{N} \left(\frac{NN_1 - NN_1 - Nn + N + N^2 - N - N_1 \cdot n(N-1)}{N(N-1)} \right)$$

$$= \frac{N_1 \cdot n}{N} \left(\frac{N^2 - NN_1 + nN_1 - Nn}{N(N-1)} \right)$$

$$= \frac{N_1 \cdot n}{N^2(N-1)} (N(N-N_1) - n(N-N_1))$$

$$= \frac{N_1 \cdot n}{N^2(N-1)} (N-n)(N-N_1)$$

8. From a lot containing 17 items, 7 of which are defective, 6 are chosen at a random. find the prob. that 4 are defective

1. Atmost 4 are defective

2. at least 3 are defective

3. 2 or 3 defective

Soln

Total no. of items, $N = 17$

No. of defective, $n = 7$

First sample, $N_1 = 6$

Then,

$$N = N_1 + N_2$$

$$17 = 6 + N_2$$

$$\therefore N_2 = 11$$

2)

$$P(X \leq 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$= \binom{6}{0} \binom{11}{7-0} + \binom{6}{1} \binom{11}{6} + \binom{6}{2} \binom{11}{5}$$

$$+ \binom{6}{3} \binom{11}{4} + \binom{6}{4} \binom{11}{3}$$

$$= 0.017 + 0.143 + 0.356 + 0.339 + 0.127$$

$$= 0.982$$

$$1) P(X=4) = \frac{\binom{6}{4} \binom{11}{3}}{\binom{17}{7}} = 0.127$$

$$3) P(X \geq 3) = 1 - P(X \leq 2) = 1 - (P(X=0) + P(X=1) + P(X=2)) = 1 - (0.017 + 0.143 + 0.356) = 0.484$$

$$4) P(X=3) \text{ or } P(X=2) = P(X=3) + P(X=2) = 0.339 + 0.356 = 0.695$$

Negative Binomial distribution

It is the distribution of a R.V. where experiment is repeated until getting k^{th} success. It is reciprocal of Binomial distribution, as number of trials are not fixed and prob. of success is fixed (i.e. getting k^{th} success). Each trial are independent. Let we get k^{th} success in last trial then, we get x^{th} failures before getting success and k^{th} success obtain in $(x+k)^{\text{th}}$ trial.

Let 'X' be a discrete R.V. is said to be follow Negative Binomial distribution of parameter 'K' and 'p' is

$$X \sim \text{NBD}(K, p)$$

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x, \quad x = 0, 1, 2, \dots, \quad k \geq 0$$

Mean:

$$E(x) = \sum_{x=0}^{\infty} x \cdot P(X=x)$$

$$= \sum_{x=0}^{\infty} x \cdot \binom{x+k-1}{k-1} p^k q^x$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{(x+k-1)!}{x! (k-1)!} p^k q^x$$

$$= \sum_{x=0}^{\infty} x \cdot \frac{(x+k-1)!}{x(x-1)! (k-1)!} p^k q^x$$

$$= \sum_{x=0}^{\infty} \frac{(x+k-1)!}{(x-1)!(k-1)!} p^k q^x$$

$$= \sum_{x=1}^{\infty} \frac{(x+k-1)!}{(x-1)!(k-1)!} p^k q^{x-1} q$$

$$= \sum_{x=1}^{\infty} \frac{k(x+k-1)!}{(x-1)!(k-1)!} p^{k-1} q^{x-1} q$$

$$= kq \sum_{x=1}^{\infty} \frac{(x+k-1)!}{(x-1)!k!} p^{k+1} p^{-1} q^{x-1}$$

$$= kq \sum_{x=1}^{\infty} \frac{(x+k-1)!}{k!} p^{k+1} q^{x-1}$$

$$\therefore E(x) = \frac{kq}{p}$$

Now,

$$E(x^2) = \sum_{x=0}^{\infty} x^2 p(x=x)$$

$$= \sum_{x=0}^{\infty} x^2 \frac{(x+k-1)!}{(k-1)!} p^k q^x$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{(x+k-1)!}{(k-1)!} p^k q^x + \sum_{x=0}^{\infty} x \frac{(x+k-1)!}{(k-1)!} p^k q^x$$

$$= \sum_{x=1}^{\infty} \frac{x(x-1)(x+k-1)!}{x!(k-1)!} p^k q^x + \frac{kq}{p}$$

$$= \sum_{x=1}^{\infty} \frac{x(x-1)(x+k-1)!}{x(x-1)(x-2)!(k-1)!} p^k q^x + \frac{kq}{p}$$

$$= \sum_{x=2}^{\infty} \frac{(x+k-1)!}{(x-2)!(k-1)!} p^k q^{x-2} q^2 + \frac{kq}{p}$$

$$= \sum_{x=2}^{\infty} \frac{(x+k-1)!}{(x-2)!(k-1)!} p^k q^{x-2} q^2 + \frac{kq}{p}$$

$$= \sum_{x=2}^{\infty} \frac{(x+k-1)!}{(x-2)!(k-1)!} p^{k+2} p^{-2} q^{x-2} q^2 + \frac{kq}{p}$$

$$= \sum_{x=2}^{\infty} \frac{k(k+1)(x+k-1)!}{(x-2)!(k-1)!} p^{k+2} p^{-2} q^{x-2} q^2 + \frac{kq}{p}$$

$$= k(k+1)q^2 \sum_{x=2}^{\infty} \frac{(x+k-1)!}{(x-2)!(k-1)!} p^{k+2} p^{-2} q^{x-2} q^2 + \frac{kq}{p}$$

$$= \frac{k(k+1)q^2}{p^2} + \frac{kq}{p}$$

Then,

$$V(x) = E(x^2) - (E(x))^2$$

$$= \frac{k(k+1)q^2}{p^2} + \frac{kq}{p} - \frac{k^2 q^2}{p^2}$$

$$= \frac{(k^2 + k)q^2 + pkq - k^2 q^2}{p^2}$$

$$= \frac{k^2 q^2 + kq^2 + pkq - k^2 q^2}{p^2}$$

$$= \frac{kq(q+p)}{p^2} = \frac{kq}{p^2}$$

Q. If a boy is throwing stone at a target what is the prob. that his 10th throw is his 5th hit. If prob. of hitting the target at any trial is 0.45.

Soln

$$\text{Success}(k) = 5$$

$$\text{Total trial} = x + k$$

$$10 = x + 5$$

$$\therefore x = 5$$

$$\therefore p = 0.45$$

Then

$$P(X=x) \text{ prob. of hitting target } (p) = 0.45$$

$$q = 1 - p = 0.55$$

Now,

$$P(X=5) = \binom{x+k-1}{k-1} p^k q^x$$

$$= \binom{5+5-1}{5-1} (0.45)^5 (0.55)^5$$

$$= \binom{9}{4} (0.45)^5 (0.55)^5$$

$$= 0.117$$

Q. If the prob. that an applicant will pass a certain job test is 0.4. What is the prob. that the applicant will fail the test fewer than 3 times before he/she passes the test first time.

Soln

$$K = 1$$

$$p = 0.4$$

$$p + q = 1$$

$$0.4 + q = 1$$

$$\therefore q = 0.6$$

Now

prob. of failing the test fewer than 3 times

$$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2)$$

$$= \binom{x+k-1}{k-1} p^k q^x + \binom{x+k-1}{k-1} p^k q^x + \binom{x+k-1}{k-1} p^k q^x$$

$$= \binom{0+1-1}{1-1} (0.4)^1 (0.6)^0 + \binom{1+1-1}{1-1} (0.4)^1 (0.6)^1 + \binom{2+1-1}{1-1} (0.4)^1 (0.6)^2$$

$$= 0.4 + 0.24 + 0.144$$

$$= 0.784$$

Rectangular Distribution / Uniform distribution

→ Let 'X' be a continuous R.V in a sample space 'S' is said to follow rectangular distribution having pdf is:-

$$f(X=x) = \frac{1}{b-a}; a < x < b$$

= 0; otherwise

where, a and b are interval in between the function exist.

f(X=x)



* Probability distribution function

$$F(x) = f(X \leq x)$$

$$= \int_a^x f(x) dx$$

$$= \int_a^x \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \int_a^x dx$$

$$= \frac{1}{b-a} [x]_a^x$$

$$= \frac{1}{b-a} [x-a] = \frac{x-a}{b-a}$$

$$\frac{d}{dx} \frac{x-a}{b-a} = \frac{1}{b-a} \frac{d}{dx} (x-a)$$

$$= \frac{1}{b-a}$$

Mean:

$$E(X) = \sum_{x=0}^{\infty} x \cdot P(X=x)$$

$$= \int_a^b x \cdot f(X=x) dx$$

$$= \int_a^b x \cdot \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \int_a^b x dx$$

$$= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b$$

$$= \frac{1}{b-a} \left[\frac{b^2}{2} - \frac{a^2}{2} \right]$$

$$= \frac{b^2 - a^2}{2(b-a)}$$

$$= \frac{(b-a)(b+a)}{2(b-a)}$$

$$= \frac{b+a}{2}$$

$$E(X^2) = \int_a^b x^2 \cdot f(X=x) dx$$

$$= \int_a^b x^2 \cdot \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \int_a^b x^2 dx$$

$$= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b$$

$$= \frac{1}{b-a} \left[\frac{b^3 - a^3}{3} \right]$$

$$= \frac{b^3 - a^3}{3(b-a)}$$

$$= \frac{(b-a)(b^2 + ba + a^2)}{3(b-a)}$$

$$= \frac{a^2 + b^2 + ab}{3}$$

Variance:

$$V(x) = E(x^2) - (E(x))^2$$

$$= \frac{a^2 + b^2 + ab}{3} - \left(\frac{b+a}{2} \right)^2$$

$$= \frac{a^2 + b^2 + ba}{3} - \frac{(b^2 + 2ba + a^2)}{4}$$

$$= \frac{4a^2 + 4b^2 + 4ba - 3b^2 - 6ba - 3a^2}{12}$$

$$= \frac{a^2 + b^2 - 2ba}{12}$$

$$= \frac{(b-a)^2}{12}$$

* Moment generating function

$$M_x(t) = \int_0^b e^{tx} f(x) dx$$

$$= \int_0^b e^{tx} \cdot \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \int_0^b e^{tx} dx$$

$$= \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_0^b$$

$$= \frac{1}{b-a} \left[\frac{e^{bt} - e^{at}}{t} \right]$$

Diff w.r.t 't'

$$\frac{d}{dt} M_x(t) = \frac{1}{b-a} \frac{d}{dt} \frac{e^{bt} - e^{at}}{t}$$

$$= \frac{1}{b-a} \left(\frac{t(e^{bt} - e^{at}) - (e^{bt} - e^{at})}{t^2} \right)$$

$$= \frac{1}{b-a} \left(\frac{ate^{bt} - bte^{at} - e^{bt} + e^{at}}{t^2} \right)$$

It doesn't satisfy

Q. A random variable 'X' has a uniform distribution over $(-3, 3)$. Show that total probability is equal to 1. Also find distribution function.

Soln.

$$p.d.f = \frac{1}{b-a}; a < x < b$$

$$= 0; \text{ otherwise}$$

Acc. to qns

$$(a, b) = (-3, 3)$$

Then

$$\text{Total prob.} = \int_a^b f(x) dx$$

$$= \int_{-3}^3 \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \int_{-3}^3 dx$$

$$= \frac{1}{b-a} [x]_{-3}^3$$

$$= \frac{1}{b-a} [3+3]$$

$$= \frac{6}{3+3} = 1$$

Now,

$$\text{Distribution function, } = \frac{x-a}{b-a}$$

$$= \frac{x-a}{b-a} = \frac{x+3}{6}$$

* Gamma Distribution

Let 'X' be a r.v of a sample space 'S' is said to follow gamma distribution with parameter α and β is:

$$X \sim G(\alpha, \beta)$$

$$f(X=x) = \frac{e^{-x/\beta} \cdot x^{\alpha-1}}{\beta^{\alpha} \Gamma(\alpha)}; x > 0, \alpha > 0, \beta > 0$$

Properties:

$$1. \Gamma(\alpha) = \int_0^{\infty} e^{-x} x^{\alpha-1} dx$$

$$2. \frac{\Gamma(\alpha)}{a^{\alpha}} = \int_0^{\infty} e^{-ax} x^{\alpha-1} dx$$

$$3. \Gamma(\alpha+1) = \alpha \Gamma(\alpha)$$

Mean:

$$E(X) = \int_0^{\infty} x \cdot f(x) dx$$

$$= \int_0^{\infty} x \cdot \frac{e^{-x/\beta} \cdot x^{\alpha-1}}{\beta^{\alpha} \Gamma(\alpha)} dx$$

$$= \int_0^{\infty} \frac{x^{\alpha}}{\beta^{\alpha} \Gamma(\alpha)} e^{-x/\beta} dx$$

$$= \int_0^{\infty} \frac{e^{-x/\beta} \cdot x^{(\alpha+1)-1}}{\beta^{\alpha} \Gamma(\alpha)} dx$$

$$= \int_0^{\infty} \frac{e^{-x/\beta} \cdot x^{(\alpha+1)-1}}{\beta^{\alpha} \Gamma(\alpha)} dx$$

$$= \frac{\alpha+1}{\left(\frac{1}{\beta}\right)^{\alpha+1}} \cdot \frac{1}{\beta^{\alpha} \Gamma(\alpha)}$$

$$= \frac{\alpha \Gamma(\alpha)}{\left(\frac{1}{\beta}\right)^{\alpha} \frac{1}{\beta}} \cdot \frac{1}{\beta^{\alpha} \Gamma(\alpha)}$$

$$= \alpha \beta$$

$$E(X^2) = \int_0^{\infty} x^2 \cdot f(x) dx$$

$$= \int_0^{\infty} x^2 \cdot \frac{e^{-x/\beta} \cdot x^{\alpha-1}}{\beta^{\alpha} \Gamma(\alpha)} dx$$

$$= \frac{1}{\beta^{\alpha} \Gamma(\alpha)} \int_0^{\infty} e^{-x/\beta} \cdot x^{(\alpha+2)-1} dx$$

$$= \frac{1}{\beta^{\alpha} \Gamma(\alpha)} \frac{\Gamma(\alpha+2)}{\left(\frac{1}{\beta}\right)^{\alpha+2}}$$

$$= \frac{1}{\beta^{\alpha} \Gamma(\alpha)} \frac{1}{\beta^{\alpha}} \cdot \frac{1}{\beta^2}$$

$$= \frac{\beta^2}{\Gamma(\alpha)} \frac{\alpha+1}{\Gamma(\alpha+1)}$$

$$= \frac{\beta^2}{\Gamma(\alpha)} (\alpha+1) \alpha \Gamma(\alpha)$$

$$= \frac{\beta^2 \alpha (\alpha+1)}{\beta^2 \alpha^2 + \alpha \beta^2}$$

Variance:

$$V(X) = E(X^2) - (E(X))^2$$

$$= \beta^2 \alpha^2 + \beta^2 \alpha - \beta^2 \alpha^2$$

$$= \alpha \beta^2$$

* Distribution function (F(x))

$$= f(X \leq x)$$

$$= \int_0^x f(x) dx$$

$$= \int_0^x \frac{e^{-x/\beta} \cdot x^{\alpha-1}}{\beta^{\alpha} \Gamma(\alpha)} dx$$

which is called incomplete gamma function

$$\alpha = \alpha$$

$$\beta = \frac{1}{\lambda}$$

$$\sqrt{\pi} = \sqrt{\frac{1}{2}}$$

$$= \frac{1}{\Gamma(\alpha)} \int_0^x e^{-x} \cdot x^{\alpha-1} dx$$

$$= \int_0^x e^{-x} x dx$$

$$= x \int e^{-x} dx - \left[\int dx (x e^{-x}) \right] dx$$

$$= x \cdot e^{-x} - \left[\int e^{-x} dx \right]$$

$$= [-x e^{-x} - e^{-x}]_0^x$$

$$= -x e^{-x} - e^{-x} + 1$$

$$= 1 - x e^{-x} - e^{-x}$$

- 1) If a random variable has the gamma distⁿ with $\alpha=2$ and $\beta=3$, find mean and S.D of distribution.

Soln

$$\alpha=2$$

$$\beta=3$$

Then

$$\begin{aligned} \text{Mean } (E(X)) &= \alpha\beta \\ &= 2 \times 3 \\ &= 6 \end{aligned}$$

Again

$$\begin{aligned} V(X) &= \alpha\beta^2 \\ &= 2 \times (3)^2 \\ &= 2 \times 9 \\ &= 18 \end{aligned}$$

$$\begin{aligned} \text{S.D} &= \sqrt{V(X)} \\ &= \sqrt{18} \\ &= 4.243 \end{aligned}$$

- 2) Suppose that the life time of a certain kind of computer is random variable 'X' having Gamma distribution with $\alpha=2$, $\beta=6$, find

a) Mean lifetime of computers

b) The prob. that such a computer will last more than 10 years.

Soln

$$\alpha=2$$

$$\beta=6$$

$$p.d.f = \frac{e^{-x/\beta} \cdot x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)} ; \alpha > 0, \beta > 0, x > 0$$

$$\begin{aligned} \text{a) Mean} &= \alpha\beta \\ &= 2 \times 6 \\ &= 12 \end{aligned}$$

b) prob. that a computer will last < 10 yrs

$$\begin{aligned} f(x=10) &= \int_{10}^{\infty} \frac{e^{-x/\beta} \cdot x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)} dx \\ &= \int_{10}^{\infty} \frac{e^{-x/6} \cdot x^1}{6^2 \Gamma(2)} dx \end{aligned}$$

$$= \frac{1}{36} \int_{10}^{\infty} e^{-x/6} \cdot x^1 dx$$

$$= \frac{1}{36} \int_{10}^{\infty} e^{-x/6} x^1 dx$$

$$= \frac{1}{36} \left[\frac{x e^{-x/6}}{-1/6} - \frac{e^{-x/6}}{(-1/6)^2} \right]_{10}^{\infty}$$

$$= \frac{1}{36} \left[0 + 6 \cdot 10 e^{-10/6} + 36 e^{-10/6} \right]$$

$$= \frac{1}{36} \left[60 e^{-10/6} + 36 e^{-10/6} \right]$$

$$= \frac{5}{3} e^{-10/6} + e^{-10/6}$$

$$= \frac{8}{3} e^{-10/6}$$

$$= 0.504$$

* Beta distribution

Let 'X' be a continuous R.V. in a sample space 'S' is said to follow Beta distribution with parameter ' α ' and ' β ' i.e. $X \sim B(\alpha, \beta)$.

$$f(x=x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\beta(\alpha, \beta)}, \quad 0 < x < 1, \beta > 0, \alpha > 0$$

Properties:

$$\beta(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$\int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx = \beta(\alpha, \beta)$$

Mean

$$E(X) = \int_0^1 x \cdot f(x=x) dx$$

$$= \int_0^1 x \cdot \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\beta(\alpha, \beta)} dx$$

$$= \int_0^1 \frac{x^{(\alpha+1)-1}(1-x)^{\beta-1}}{\beta(\alpha, \beta)} dx$$

$$= \frac{1}{\beta(\alpha, \beta)} \cdot \beta(\alpha+1, \beta)$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+1+\beta)}$$

$$= \frac{\Gamma(\alpha+\beta) \cdot \alpha \Gamma(\alpha)}{\Gamma(\alpha) (\alpha+\beta) \Gamma(\alpha+\beta)} = \frac{\alpha}{\alpha+\beta}$$

$$E(X^2) = \int_0^1 x^2 f(x=x) dx$$

$$= \int_0^1 x^2 \cdot \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\beta(\alpha, \beta)} dx$$

$$= \frac{1}{\beta(\alpha, \beta)} \int_0^1 x^{(\alpha+2)-1}(1-x)^{\beta-1} dx$$

$$= \frac{1}{\beta(\alpha, \beta)} \cdot \beta(\alpha+2, \beta)$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+2+\beta)}$$

$$= \frac{\Gamma(\alpha+\beta) \cdot \Gamma(\alpha+1) \cdot \Gamma(\alpha+1)}{\Gamma(\alpha) \Gamma(\alpha+\beta+1) \Gamma(\alpha+\beta+1)}$$

$$= \frac{\Gamma(\alpha+\beta) (\alpha+1) \Gamma(\alpha)}{\Gamma(\alpha) (\alpha+\beta+1) \Gamma(\alpha+\beta+1)}$$

$$= \frac{\Gamma(\alpha+\beta) (\alpha+1) \alpha \Gamma(\alpha)}{\Gamma(\alpha) (\alpha+\beta+1) (\alpha+\beta) \Gamma(\alpha+\beta)}$$

$$= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}$$

$$= \frac{\alpha^2 + \alpha}{(\alpha+\beta)(\alpha+\beta+1)}$$

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$$\begin{aligned}
 V(X) &= E(X^2) - (E(X))^2 \\
 &= \frac{\alpha^2 + \alpha}{(\alpha + \beta)(\alpha + \beta + 1)} - \frac{\alpha^2}{(\alpha + \beta)^2} \\
 &= \frac{\alpha^2 + \alpha - \alpha^2(\alpha + \beta + 1)}{(\alpha + \beta)(\alpha + \beta + 1)(\alpha + \beta)} \\
 &= \frac{\alpha^2 + \alpha - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha + \beta)^2(\alpha + \beta + 1)} \\
 &= \frac{\alpha(1 - \alpha\beta - \alpha^2)}{(\alpha + \beta)^2(\alpha + \beta + 1)} \\
 &= \frac{\alpha^2 + \alpha(\alpha + \beta) - \alpha^2(\alpha + \beta + 1)}{(\alpha + \beta)^2(\alpha + \beta + 1)} \\
 &= \frac{\alpha^3 + \alpha^2\beta + \alpha^2 + \alpha\beta - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha + \beta)^2(\alpha + \beta + 1)} \\
 &= \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}
 \end{aligned}$$

* If the annual proportion of erroneous income tax returns filed with the I.R.S. can be looked upon as a random variable having beta distribution with $\alpha = 2$ and $\beta = 3$. What is the total probability of distribution. Also find mean, variance, S.D and coefficient of variation of distribution.

Soln

$$\alpha = 2$$

$$\beta = 3$$

Now

$$\text{Total prob} = \int_0^1 \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\beta(\alpha, \beta)} dx$$

$$= \int_0^1 \frac{x(1-x)^{3-1}}{\beta(2, 3)} dx$$

$$= \frac{\Gamma(2+3) \cdot \frac{1}{3}}{\Gamma(2)\Gamma(3)} \int_0^1 x^1 x^{2-1} dx$$

$$= \frac{\Gamma(5) \cdot \frac{1}{3}}{\Gamma(2)\Gamma(3)} \left[\frac{x^2 - x^3}{2} \right]_0^1$$

$$= \frac{24}{3} \cdot \frac{1}{3} \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$\text{Total prob} = \int_0^1 \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\beta(\alpha, \beta)} dx$$

$$= \int_0^1 \frac{x^{2-1}(1-x)^{3-1}}{\beta(2, 3)} dx$$

$$= \frac{1}{\beta(2, 3)}$$

$$\beta(2, 3)$$

$$= 1$$

Mean:

$$E(X) = \frac{\alpha}{\alpha + \beta}$$

$$= \frac{2}{2+3} = \frac{2}{5}$$

$$= 0.4$$

$$V(X) = \frac{\alpha \beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

$$= \frac{2 \times 3}{(2+3)^2 (2+3+1)}$$

$$= \frac{6}{25 \times 6}$$

$$= 0.04$$

$$S.D = \sqrt{V(X)}$$

$$= \sqrt{0.04}$$

$$= 0.2$$

$$C.V = \frac{S.D}{\bar{X}} \times 100\%$$

$$= \frac{0.2}{0.4} \times 100\%$$

$$= 50\%$$

* Exponential Distribution

Let 'X' be a random variable in a sample space 'S' is said to follow exponential distribution of parameter ' λ ' with pdf:-

$$f(X=x) = \lambda e^{-\lambda x} dx, x > 0, \lambda > 0$$

Mean: ∞

$$E(X) = \sum_{x=0}^{\infty} x \cdot P(X=x)$$

$$= \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx$$

$$= \lambda \left[\frac{x \cdot e^{-\lambda x}}{-\lambda} - \frac{e^{-\lambda x}}{\lambda^2} \right]_0^{\infty}$$

$$= \lambda \left[0 + 0 - 0 + \frac{1}{\lambda^2} \right]$$

$$= \frac{1}{\lambda}$$

$$E(X^2) = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx$$

$$= \lambda \left[\frac{x^2 \cdot e^{-\lambda x}}{-\lambda} - \frac{2x \cdot e^{-\lambda x}}{\lambda^2} + \frac{2 \cdot e^{-\lambda x}}{(-\lambda^3)} \right]_0^{\infty}$$

$$= \frac{\lambda \cdot 2}{\lambda^3}$$

$$= \frac{2}{\lambda^2}$$

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$$V(X) = E(X^2) - (E(X))^2$$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda^2}$$

$$= \frac{1}{\lambda^2}$$

Q. The mileage in thousands of miles which car owners get with a certain kind of tyres is a r.v. having exponential distribution with $\lambda = 0.0005$. Find the prob. that one of these types will last

- i) At most 10,000 miles
- ii) Anywhere from 16000 to 24000 miles
- iii) At least 30,000 miles

Soln

$$\lambda = 0.0005$$

i) At most 10,000 miles

$$f(X=x) = \int_0^{10000} \lambda \cdot e^{-\lambda x} dx$$

$$= \lambda \int_0^{10000} e^{-\lambda x} dx$$

$$= \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^{10000}$$

$$= 0.0005 \left[\frac{e^{-0.0005x}}{-0.0005} \right]_0^{10000}$$

$$= -e^{-0.0005 \times 10} + e^0$$

$$= 0.00498$$

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ii) Anywhere from 16000 to 24000 miles

$$f(X=x) = \int_{16}^{24} \lambda \cdot e^{-\lambda x} dx$$

$$= 0.0005 \int_{16}^{24} e^{-0.0005x} dx$$

$$= 0.0005 \left[\frac{e^{-0.0005x}}{-0.0005} \right]_{16}^{24}$$

$$= -e^{-0.0005 \times 24} + e^{-0.0005 \times 16}$$

$$= 0.00396$$

iii) At least 30,000 miles

$$f(X=x) = \int_{30}^{\infty} \lambda e^{-\lambda x} dx$$

$$= 0.0005 \left[\frac{e^{-\lambda x}}{-\lambda} \right]_{30}^{\infty}$$

$$= 0.0005 \left[\frac{e^{-0.0005x}}{-0.0005} \right]_{30}^{\infty}$$

$$= -e^{-0.0005 \times 30}$$

$$= 0.9851$$

Normal distribution:-

Let 'X' be a r.v. in a sample space 'S' is said to follow normal distribution with parameter (μ, σ^2) and its pdf is:-

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-1/2\left(\frac{x-\mu}{\sigma}\right)^2}; \quad -\infty < x < \infty, \\ -\infty < \mu < \infty, \sigma > 0$$

$$\text{Mean} = \mu$$

$$\text{Variance} = \sigma^2$$

$$\text{S.D} = \sigma$$

For standardization

$$Z = \frac{X - \mu}{\sigma}$$

$$X = -\infty, Z = -\infty$$

$$X = \infty, Z = \infty$$

$$dz = \frac{dx}{\sigma}$$

$$dx = \sigma dz$$

Now

$$f(z) = \int_{-\infty}^{\infty} f(x) dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2 z^2} \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2 z^2} dz$$

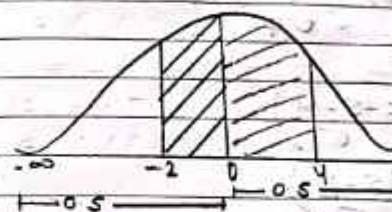
$$= \frac{1}{\sqrt{2\pi}} e^{-1/2 z^2}; \quad -\infty < z < \infty$$

Mean = 0 = μ
Variance (σ^2) = 1

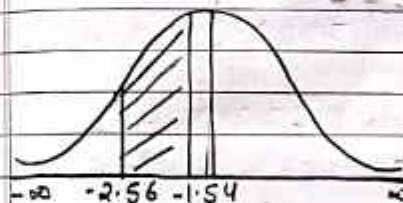
Standard Normal distribution

- Probability
- Value

$$Q. P(-2 \leq x \leq 4) = ?$$

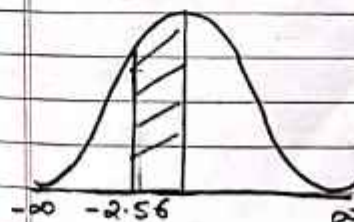


$$Q. P(-2.56 \leq Z \leq -1.54)$$

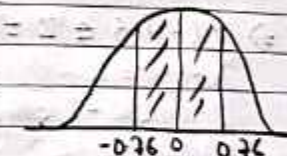


Soln

$$P(-2.56 \leq Z \leq 0) = 0.4948 \quad P(-1.54 \leq Z \leq 0)$$

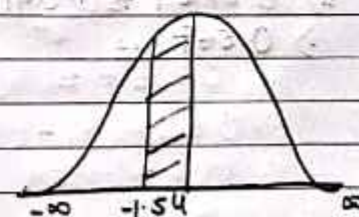


$$P(-2.56 \leq Z \leq 0) = 0.4948$$



$$P(0 \leq Z \leq 0.76)$$

$$P(-0.76 \leq Z \leq 0)$$



$$P(-1.54 \leq Z \leq 0) = 0.4382$$

$$P(-2.56 \leq Z \leq 0) = 0.0566$$

Q) Let 'X' follows a normal distribution with mean 10 and s.d 5. Find the prob. that it lies between 7 to 15.

Soln

$$\mu = 10$$

$$\sigma = 5$$

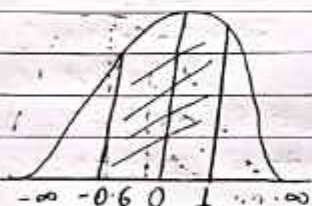
$$P(7 \leq X \leq 15) = \int_7^{15} f(x) dx$$

$$P\left(\frac{7-\mu}{\sigma} \leq \frac{X-\mu}{\sigma} \leq \frac{15-\mu}{\sigma}\right)$$

$$\Rightarrow P\left(\frac{7-10}{5} \leq Z \leq \frac{15-10}{5}\right)$$

$$\Rightarrow P\left(\frac{7-10}{5} \leq Z \leq \frac{15-10}{5}\right)$$

$$\Rightarrow P(-0.6 \leq Z \leq 1)$$



$$\Rightarrow P(-0.6 \leq Z < 0) + P(0 \leq Z \leq 1)$$

$$\Rightarrow 0.2257 + 0.3413$$

$$\Rightarrow 0.567 \text{ \#}$$

i) a) Greater than 22
b) less than 3

Soln

a) Greater than 22

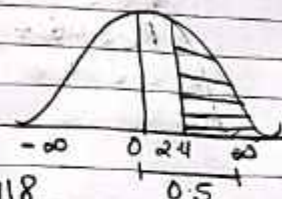
$$P(X \geq 22) = \int_{22}^{\infty} f(x) dx$$

$$P\left(\frac{X-\mu}{\sigma} \geq \frac{22-\mu}{\sigma}\right)$$

$$P\left(Z \geq \frac{22-10}{5}\right)$$

$$P(Z \geq 2.4) = \int_{2.4}^{\infty} f(z) dz$$

$$= 0.5 - P(0 \leq Z \leq 2.4) = 0.5 - 0.4918 = 0.0082$$



b) less than 3

$$P(X \leq 3) = \int_0^3 f(x) dx$$

$$P\left(\frac{X-\mu}{\sigma} \leq \frac{3-\mu}{\sigma}\right)$$

$$P\left(Z \leq \frac{3-10}{5}\right)$$

$$P(Z \leq -1.4)$$

$$= 0.5 - P(-1.4 \leq Z \leq 0)$$

$$= 0.5 - 0.4192$$

$$= 0.0808$$



iii) To between 2 to 5.

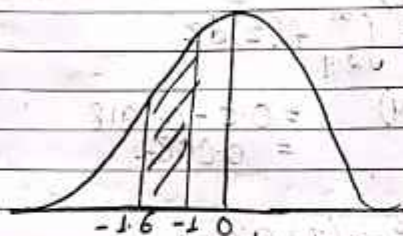
Soln

$$P(2 \leq X \leq 5) = \int_2^5 f(x) dx$$

$$P\left(\frac{2-M}{\sigma} \leq \frac{X-M}{\sigma} \leq \frac{5-M}{\sigma}\right)$$

$$P\left(\frac{2-10}{5} \leq Z \leq \frac{5-10}{5}\right)$$

$$P(-1.6 \leq Z \leq -1)$$



$$P(-1.6 \leq Z < 0) - P(-1 \leq Z < 0)$$

$$0.4452 - 0.3413$$

$$= 0.1039$$

iv) In between 12 to 17

Soln

$$P(12 \leq X \leq 17) = \int_{12}^{17} f(x) dx$$

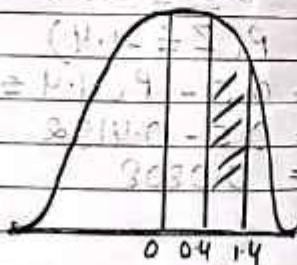
$$P\left(\frac{12-M}{\sigma} \leq \frac{X-M}{\sigma} \leq \frac{17-M}{\sigma}\right)$$

$$P(0.4 \leq Z \leq 1.4)$$

$$P(0 \leq Z < 0.4) - P(0 \leq Z < 1.4)$$

$$= 0.4192 - 0.1554$$

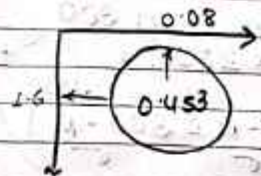
$$= 0.2638$$



Value Calculate

$$\text{Prob} = 0.453$$

$$Z = 1.68$$



2) A normal population of 700 wage earners has a mean income of Rs. 220 per month and variance is 332. Find the no. of person who earn

i) Between Rs. 150 & 220 per month

ii) More than 300

iii) Less than Rs. 100

Soln

i) Between Rs 150 & 220 per month

$$\mu = 220$$

$$\sigma^2 = 332$$

$$\sigma = \text{Rs. } 18.22$$

Then

$$P(150 \leq X \leq 220) = \int_{150}^{220} f(x) dx$$

$$P\left(\frac{150-M}{\sigma} \leq \frac{X-M}{\sigma} \leq \frac{220-M}{\sigma}\right)$$

$$P(-3.84 \leq Z \leq 0)$$



$$= 0.49994 \times 700 = \text{Rs } 349.95$$

ii) More than 300

$$P(X \geq 300) = \int_{300}^{\infty} f(x) dx$$

$$P\left(\frac{X - \mu}{\sigma} \geq \frac{300 - \mu}{\sigma}\right)$$

$$P(Z \geq 4.39) = \int_{4.39}^{\infty} f(z) dz$$

$$0.5 - P(0 \leq Z \leq 4.39)$$

$$0.5 - P(0 \leq Z \leq 3.99) + P(3.99 \leq Z \leq 4.39)$$

$$= 0.5 - 0.49997$$

$$= 0.00003$$

iii) Less than 100

$$P(X \leq 100) = \int_{-\infty}^{100} f(x) dx$$

$$P\left(\frac{X - \mu}{\sigma} \leq \frac{100 - \mu}{\sigma}\right)$$

$$P(Z \leq -6.58)$$

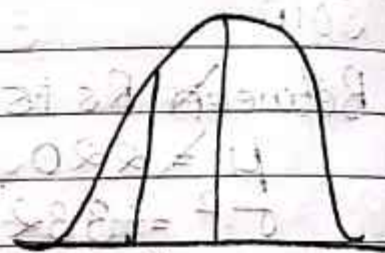
$$0.5 - P(-6.58 \leq Z \leq 0)$$

$$0.5 - P(0 \leq Z \leq 3.99) + P(3.99 \leq Z \leq 6.58)$$

$$= 0.5 - 0.49997 + 0.00003$$

$$= 0.00003 \times 700$$

$$= 0.021$$



Introduction to Statistics

- The word 'statistics' is derived from the Latin word 'status' or Italian word 'statista' or the German word 'statistik' each of which means 'political state'. In ancient times, the government used to collect information about the population, poverty or wealth status of the country.

Statistics is the branch of science that deals with the collection, tabulation, analysis and interpretation of data. In today's world, data is not only numbers it is a group of numbers and information from which the results can be generated.

* Importance and Scope of Statistics

- In modern times statistics is viewed not as a mere device for collecting numerical data but as a means of developing and sound techniques for their handling, analysis and drawing inference.

Some of the importance of statistics in different fields are:

↓ Statistics in Planning

In the modern era, almost all organizations

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in the government or management of business are resorting to planning for efficient working and formulating policy decisions. To achieve those plans the statistical data are handled and analyzed using complex statistical techniques and hence statistics is indispensable to planning.

2. Statistics in Mathematics

Statistics is intimately related and dependent on mathematics. The development of statistics and theories for application to various sciences: social, physical and natural are based on fitting different mathematical models to the observed data.

3. Statistics in Economics

4. Statistics in Biology, Astronomy and medical sciences

5. Statistics in Business

6. Statistics in Engineering

* Application of Statistics in Engineering

Data analysis: Statistics helps engineers analyze data collected from experiments and field studies, enabling them to draw meaningful conclusions.

Quality Control: Statistical methods are essential in quality assurance processes, helping to monitor and improve manufacturing processes through techniques like Six Sigma.

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Decision Making: Engineers use statistical models to make informed decisions based on data rather than intuition, reducing risks associated with engineering projects.

Design Optimization: Statistical techniques, such as Design of Experiments (DOE), help in optimizing designs by evaluating the influence of different variables on performance.

Reliability Engineering: Statistics is crucial in assessing the reliability and life expectancy of components and systems, ensuring safety and effectiveness.

Predictive Modeling: Engineers use statistical models to predict future performance, which is essential for maintenance planning and resource allocation.

Risk Assessment: Statistical tools help in identifying and quantifying risks associated with engineering projects, leading to better management strategies.

Sampling Techniques: Engineers often rely on statistical sampling to make inferences about large populations, saving time and resources in testing and analysis.

Performance Evaluation: Statistical metrics are used to evaluate the performance of systems.

and processes, helping engineers identify areas for improvement.

* Limitations of Statistics

- Statistic is not suited to the study of qualitative phenomena.
- Statistics does not deal with individuals.
- Statistics laws are not exact.
- Statistics is liable to be misused.

* Data Collection and presentation of data

Data Collection

Data is an important tool in statistics. Data means not only numbers, it means information, evidence, and tools which is used to find out the current status as well as to predict the future. There are two types of data based on collecting methods. They are

1. Primary Data

The data that are collected by the researchers themselves to fulfill their own purpose or needs is known as primary data. The sources of collecting primary data are:

- a. Direct personal Interview Method
- b. Indirect personal Interview Method
- c. Mailed Questionnaire method
- d. Schedule sent through enumerator
- e. Data Collection from correspondence

2. Secondary Data

The data are collected by other organizations but researcher use these data to fulfill their need and purposes then those data are called secondary data.

- a. Published and unpublished source
- b. Administrative records
- c. Census, Survey

Presentation of data

It is the method of presenting collected data in a good manner so that the viewer can easily understand and observe the scenario of data and its relation with other variables.

Methods of Presenting data are:

1. Diagrammatic Representation of data
2. Tabulation Representation

1. Diagrammatic Representation of data

Diagrams are the visual representation of the data. Different types of diagrams are used to represent the dataset. Some of them are

- a. Bar graph
- b. Multi-bar graph
- c. Sub-divided bar graph
- d. Percentage sub-divided bar graph
- e. Histogram
- f. Ogive curve

Importance of Diagrammatic Representation

- It makes comparison easy.
- It saves time and effort.
- It has universal applicability.
- It facilitates the location of various statistical measures and established trends.

2. Tabulation Representation

→ It is the process of converting raw data into easily readable form so that we can easily understand and compare datasets. As our data are always in individual series we have to convert them into discrete or continuous as per objectives and needs. These tabulations are used for further analysis.

Importance of Tabulation Representation

- It simplifies the complex information to understand easily.
- It will help to compare related facts and facilitate the computation of various statistical measures like averages, dispersion, correlation etc.
- It presents facts in a nutshell and avoids unnecessary repetitions and explanation are avoided. Moreover, the needed information can be easily located.

• Tabulated data are good for references and they make it easier to present the information in the form of graphs and diagrams.

Limitations of Tabulations

- Tables contain only numerical data and qualitative expression.
- Tables help to draw conclusions where it finds that a normal layman cannot understand properly.

- Q. A normal distribution has mean as 77. Find its standard deviation if 20% of the area under the curve lies to the right of 90.

Soln

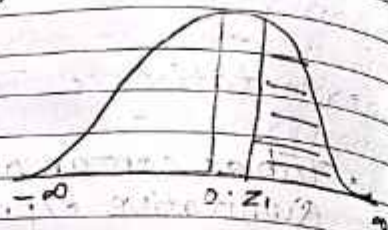
$$\text{Mean } (\mu) = 77$$

Here,

$$P(X \geq 90) = 0.20$$

$$P\left(Z \geq \frac{90 - \mu}{\sigma}\right) = 0.20$$

$$P(Z \geq Z_1) = 0.20$$



$$0.5 - P(0 \leq Z \leq Z_1) = 0.20$$

$$P(0 \leq Z \leq Z_1) = 0.30$$

from standard normal table

$$\therefore Z_1 = 0.84$$

$$\text{or, } \frac{90 - \mu}{\sigma} = 0.84$$

$$\therefore \sigma = 15.47$$

$\therefore$ Its standard deviation is 15.47 if 20% of the area under the curve lies to the right of 90.

- Q. A r.v has a normal distribution with S.D 10. Find its mean if the prob. of r.v. takes on a value less than 80.5 is 0.3264.

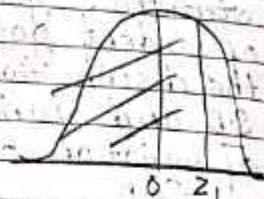
Soln

$$\sigma = 10$$

$$\text{Probability} = 0.3264$$

$$P(X \leq 80.5) = 0.3264$$

$$P\left(Z \leq \frac{80.5 - \mu}{\sigma}\right) = 0.3264$$



$$P(Z \leq Z_1) = 0.3264$$

$$0.5 - P(0 \leq Z \leq Z_1) = 0.3264$$

$$P(0 \leq Z \leq Z_1) = 0.1736$$

from standard normal table

$$\therefore Z_1 = -0.45$$

$$\text{or, } \frac{80.5 - \mu}{10} = -0.45$$

$$\therefore \mu = 85$$

$\therefore$ The mean is 85 if the prob. of r.v. takes on a value less than 80.5.

Q. At a certain examination 10% of the student who appeared for the paper in statistics got less than 30 marks and 97% of student got less than 62 marks. Assuming the distribution to be normal, find mean, S.D and variance. Also find the coefficient of variation of the distⁿ.

Soln

$$P(X \leq 30) = 0.1 \quad \dots i)$$

$$P(X \leq 62) = 0.97 \quad \dots ii)$$

Now,

for eqn i)

$$P(Z \leq \frac{30 - \mu}{\sigma}) = 0.1$$

$$P(Z \leq Z_1) = 0.1$$

$$0.5 + P(0 \leq Z \leq Z_1) = 0.1 \quad \dots \infty$$

$$P(0 \leq Z \leq Z_1) = -0.4$$

from Standard normal table,

$$\therefore Z_1 = -1.29$$

$$\frac{30 - \mu}{\sigma} = -1.29 \quad \dots A)$$

$$\Rightarrow -1.29\sigma + \mu = 30$$

from eqn ii)

$$P(Z \leq \frac{62 - \mu}{\sigma}) = 0.97$$

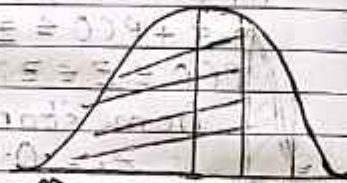
$$P(Z \leq Z_2) = 0.97$$

$$0.5 + P(0 \leq Z \leq Z_2) = 0.97$$

$$P(0 \leq Z \leq Z_2) = 0.47$$

Since, Probability cannot be 've'

$$P(Z_2 \leq Z \leq 0) = 0.47$$



from standard normal table,

$$\therefore Z_2 = 1.89$$

$$62 - \mu = 1.89\sigma$$

$$62 - \mu = 1.89\sigma$$

$$1.89\sigma + \mu = 62 \quad \dots B)$$

from eqn A) & B)

$$\sigma = 10.06$$

$$\mu = 42.98$$

$$V = \sigma^2 = 101.20$$

$$C.O.V = \frac{\sigma}{\mu} \times 100\%$$

$$= \frac{10.06}{42.98} \times 100\%$$

$$= 23.40\%$$

$$= 23.40\%$$

Exams

- 3a) The length of time (in minutes) that a certain lady speaks on the telephone is found to be a random phenomenon with a probability function specified by the probability density function $f(x)$ as:

$$f(x) = Ae^{-x/5} \text{ for } x \geq 0$$

- i) Determine the value of A
 ii) What is the probability that she will talk over the phone is
 a. More than 10 minutes
 b. less than 5 minutes

Soln

Here

$$f(x) = Ae^{-x/5}; x \geq 0$$

$$\text{Total prob} = \int_0^{\infty} Ae^{-x/5} dx$$

$$1 = A \left[\frac{e^{-x/5}}{-1/5} \right]_0^{\infty}$$

$$1 = -5A [e^{-\infty/5} - e^0]$$

$$1 = -5A [0 - 1]$$

$$1 = 5A$$

$$\therefore A = \frac{1}{5}$$

- ii) a) More than 10 min

$$= \int_{10}^{\infty} f(x) dx$$

$$= \int_{10}^{\infty} Ae^{-x/5} dx$$

$$= \frac{1}{5} \left[-5e^{-x/5} \right]_{10}^{\infty}$$

$$= -1 \left[e^{-\infty/5} - e^{-10/5} \right]$$

$$= 0.135$$

- b) less than 5 min

$$= \int_0^5 f(x) dx$$

$$= \int_0^5 Ae^{-x/5} dx$$

$$= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_0^5$$

$$= [-e^{-5/5} + e^0]$$

$$= -0.36 + 1$$

$$= 0.64$$

* Bivariate Distribution

Let 'X' and 'Y' be two r.v. having values $x_1, x_2, \dots, x_m$ and $y_1, y_2, \dots, y_n$ then, the distribution of 'X' and 'Y' is known as bivariate distribution.

The joint probability of 'X' and 'Y' is written as:

$$P(x_i, y_j) = P(X=x_i, Y=y_j) = P(X=x_i \cap Y=y_j)$$

if 'X' and 'Y' are independent then

$$P(X=x_i, Y=y_j) = P(X=x_i) \cdot P(Y=y_j)$$

* Joint Probability mass function:

→ Let 'X' and 'Y' be two discrete r.v. having values $x_1, x_2, \dots, x_m$ and $y_1, y_2, \dots, y_n$ then, Jpmf of 'X' and 'Y' is given as:-

$$P(x_i, y_j) = P(X=x_i, Y=y_j) = \sum_{i=1}^m \sum_{j=1}^n P(X=x_i \cap Y=y_j)$$

if 'X' and 'Y' are independent

$$P(x_i, y_j) = \sum_{i=1}^m \sum_{j=1}^n P(X=x_i) \cdot P(Y=y_j)$$

Jpmf must follow following conditions:-

$$1. P(x_i, y_j) \geq 0$$

$$2. \sum_{i=1}^m \sum_{j=1}^n P(x_i, y_j) = 1$$

Marginal and Conditional prob-mass function

$Y \backslash X$	x_1	x_2	$\dots$	x_i	$\dots$	x_m	Total = $\sum_{j=1}^n P(x_i, y_j)$
y_1	P_{11}	P_{12}	$\dots$	P_{1i}	$\dots$	P_{1m}	$\sum_{j=1}^m P(x_1, y_j)$
y_2	P_{21}	P_{22}	$\dots$	P_{2i}	$\dots$	P_{2m}	$\sum_{j=1}^m P(x_2, y_j)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
y_j	P_{j1}	P_{j2}	$\dots$	P_{ji}	$\dots$	P_{jm}	$\sum_{i=1}^m P(x_i, y_j)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
y_n	P_{n1}	P_{n2}	$\dots$	P_{ni}	$\dots$	P_{nm}	$\sum_{j=1}^m P(x_i, y_n)$
Total	$\sum_{j=1}^n P(x_1, y_j)$	$\sum_{j=1}^n P(x_2, y_j)$	$\dots$	$\sum_{j=1}^n P(x_i, y_j)$	$\dots$	$\sum_{j=1}^n P(x_m, y_j)$	$\sum_{i=1}^m \sum_{j=1}^n P(x_i, y_j)$

* Marginal prob-mass function of 'X' is

$$P_{x_i}(X) = P_{x_i}(X=x_i) = \sum_{j=1}^n P(x_i, y_j)$$

$$= P(x_i, y_1) + P(x_i, y_2) + \dots + P(x_i, y_n)$$

When $i=1$

$$P_{x_1}(X) = P(x_1, y_1) + P(x_1, y_2) + P(x_1, y_3) + \dots + P(x_1, y_n)$$

When $i=2, y=y_1, y_2, \dots, y_n$

$$P_{x_2}(X) = P(x_2, y_1) + P(x_2, y_2) + \dots + P(x_2, y_n)$$

* Marginal prob. mass function of 'Y' is

$$P_y(Y) = P_{y_j}(Y=y_j) = \sum_{i=1}^m P(X_i, Y_j)$$

$$= P(X_1, Y_j) + P(X_2, Y_j) + \dots + P(X_m, Y_j)$$

when $j=1$ or y_1

$$P_{y_1}(Y) = P(X_1, Y_1) + P(X_2, Y_1) + \dots + P(X_m, Y_1)$$

when $j=2$ or y_2

$$P_{y_2}(Y) = P(X_1, Y_2) + P(X_2, Y_2) + \dots + P(X_m, Y_2)$$

* Conditional prob. of 'X' at $Y=1$

$$P(X=x_i / Y=1) = \frac{P(X_i=x_i \cap Y=y_j)}{P_{y_1}(Y)} = \frac{P(X_i, Y_j)}{P_{y_1}(Y)}$$

at $i=1$

$$P(X=x_1 / Y=1) = \frac{P(X_1, Y_1)}{P_{y_1}(Y)}$$

* Conditional prob. of 'Y' at $X=1$

$$P(Y=y_j / X=1) = \frac{P(X=x_i \cap Y=y_j)}{P_{x_1}(X)} = \frac{P(X_i, Y_j)}{P_{x_1}(X)}$$

at $j=1$

$$P(Y=y_1 / X=1) = \frac{P(X_1, Y_1)}{P_{x_1}(X)}$$

Q.

Y \ X	0	1
0	0.28	0.37
1	0.22	0.13

find marginal prob. of 'X' and 'Y'
Soln

Marginal prob. of 'X'

$$P_{x_i}(X=x_i) = \sum_{j=1}^n P(X_i, Y_j)$$

$$= P(X_i, Y_1) + P(X_i, Y_2)$$

At $i=0$

$$P_{x_0}(X) = \sum_{j=0}^1 P(X_0, Y_j)$$

$$= P(X_0, Y_0) + P(X_0, Y_1)$$

$$= 0.28 + 0.22 = 0.50$$

At $i=1$

$$P_{x_1}(X) = \sum_{j=0}^1 P(X_1, Y_j)$$

$$= P(X_1, Y_0) + P(X_1, Y_1)$$

$$= 0.37 + 0.13$$

$$= 0.5$$

Marginal prob. of 'Y' is :-

$$P_{Y_j}(Y) = \sum_{i=1}^m P(X_i, Y_j)$$

when $j=0$

$$P_{Y_0}(Y) = \sum_{i=0}^1 P(X_i, Y_0)$$

$$= P(X_0, Y_0) + P(X_1, Y_0)$$

$$= 0.28 + 0.37$$

$$= 0.65$$

when $j=1$

$$P_{Y_1}(Y) = \sum_{i=0}^1 P(X_i, Y_1)$$

$$= P(X_0, Y_1) + P(X_1, Y_1)$$

$$= 0.22 + 0.13$$

$$= 0.35$$

* Conditional prob. of Y at $x=1$.

$$P(Y=y_j/X=1) = \frac{P(X_1, Y_j)}{P_{X_1}(X)}$$

at $j=0$

$$P(Y=y_0/X=1) = \frac{P(X_1, Y_0)}{P_{X_1}(X)}$$

$$= \frac{0.37}{0.5}$$

$$= 0.74$$

at $j=1$

$$P(Y=y_1/X=1) = \frac{P(X_1, Y_1)}{P_{X_1}(X)}$$

$$= \frac{0.13}{0.5}$$

$$= 0.26$$

* Conditional prob. of X at $Y=1$

$$P(X=x_i/Y=1) = \frac{P(X_i, Y_1)}{P_{Y_1}(Y)} = \frac{P(X_i, Y_1)}{P_{Y_1}(Y)}$$

at $i=0$

$$P(X=x_0/Y=1) = \frac{P(X_0, Y_1)}{P_{Y_1}(Y)}$$

$$= \frac{0.22}{0.35}$$

$$= 0.6286$$

at $i=1$

$$P(X=x_1/Y=1) = \frac{P(X_1, Y_1)}{P_{Y_1}(Y)}$$

$$= \frac{0.13}{0.35}$$

$$= 0.3714$$

$Y \backslash X$	-1	0	1	
-1	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{4}{15}$
0	$\frac{3}{15}$	A	$\frac{1}{15}$	$\frac{4}{15} + A$
1	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{5}{15}$

* Find value of A

* Find $P(X=1/Y=0)$

* Find marginal prob. of 'X' and 'Y'

* Find conditional prob. of 'X' at $Y=0$

Soln

Total Marginal = 1

$$\Rightarrow \frac{4}{15} + \frac{4}{15} + A + \frac{5}{15} = 1$$

$$\Rightarrow \frac{13}{15} + A = 1$$

$$\Rightarrow A = 1 - \frac{13}{15}$$

$$\therefore A = \frac{2}{15}$$

$$2) P(X=1/Y=0)$$

$$P(X=x_i/Y=0) = P(X_i, Y_0)$$

$$= \frac{P_{11}(\cdot)}{P_{1\cdot}(\cdot)}$$

$$= \frac{P_{11}(\cdot)}{P_{1\cdot}(\cdot)}$$

$$= \frac{1}{15}$$

3) Marginal prob. of 'X' is -

$$P_{X\cdot}(X) = \sum_{j=-1}^1 P(X_i, Y_j)$$

At $i=-1$

$$P_{X-1}(X) = \sum_{j=-1}^1 P(X_{-1}, Y_j) = P(X_{-1}, Y_{-1}) + P(X_{-1}, Y_0) + P(X_{-1}, Y_1)$$

$$= \frac{1}{15} + \frac{3}{15} + \frac{2}{15} = \frac{6}{15} = 0.4$$

At $i=0$

$$P_{X0}(X) = \sum_{j=-1}^1 P(X_0, Y_j)$$

$$= P(X_0, Y_{-1}) + P(X_0, Y_0) + P(X_0, Y_1)$$

$$= \frac{2}{15} + \frac{2}{15} + \frac{1}{15} = \frac{5}{15} = 0.33$$

At $i=1$

$$P_{X1}(X) = \sum_{j=-1}^1 P(X_1, Y_j)$$

$$= P(X_1, Y_{-1}) + P(X_1, Y_0) + P(X_1, Y_1)$$

$$= \frac{1}{15} + \frac{1}{15} + \frac{2}{15} = \frac{4}{15} = 0.266$$

Joint pdf (JPdf)

Let 'X' and 'Y' be two continuous R.V. in a certain interval $(-\infty, \infty)$ then, JPdf 'X' and 'Y' is denoted as:-

$$f(x_i, y_j) = f(X=x_i, Y=y_j) \text{ and given as:-}$$

$$f(x_i, y_j) = f(X=x_i, Y=y_j)$$

when 'X' and 'Y' are independent then,

$$f(x_i, y_j) = f(x_i) \cdot f(y_j)$$

It must satisfy certain conditions as:-

1. $f(x_i, y_j) = f(XY) \geq 0$

2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(XY) dy dx = 1$

* Joint prob. distribution function

$$F_{XY}(XY) = f(X \leq x, Y \leq y)$$

$$= \int_{-\infty}^x \int_{-\infty}^y f(XY) dy dx$$

Conversion of distth function to density function

$$f(XY) = \frac{\partial^2 F(XY)}{\partial x \partial y}$$

Marginal probability of 'X'

→ It is denoted by
 $f_X(X) = \int_y f(XY) dy$

Marginal probability of 'Y'

→ It is denoted by
 $f_Y(Y) = \int_x f(XY) dx$

* Conditional prob. of 'X' given 'Y'

$$f_{X/Y}(X/Y) = \frac{\text{JPdf}}{\text{Marginal}} = \frac{f(XY)}{f_Y(Y)}$$

* Conditional prob. of 'Y' given 'X'

$$f_{Y/X}(Y/X) = \frac{f(XY)}{f_X(X)}$$

Q. The JPdf of 'X' and 'Y' is:-
 $f(x, y) = A(x+y)$; $0 < x < 2$, $0 < y < 1$
 Find

1. Value of A
2. Marginal prob. of 'X' and 'Y'.
3. Conditional prob. of 'X' and given $Y=y$
4. Find distribution function.

Soln

Here,

JPdf is given as

$$f(x_i, y_j) = f(X=x_i, Y=y_j)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(XY) dy dx = 1$$

$$\int_0^2 \int_0^1 A(x+y) dy dx = 1$$

$$\Rightarrow A \int_0^2 \left[xy + \frac{y^2}{2} \right]_0^1 dx = 1$$

$$\Rightarrow A \int_0^2 \frac{x+1}{2} dx = 1$$

$$\Rightarrow A \left[\frac{x^2}{2} + x \right]_0^2 = 1$$

$$\Rightarrow A \left[\frac{4}{2} + 1 \right] = 1$$

$$\Rightarrow A \cdot 3 = 1$$

$$\therefore A = \frac{1}{3}$$

2. Marginal prob. of 'X'

$$f_x(x) = \int_y f(xY) dy$$

$$= \int_0^1 \frac{1}{3} (x+y) dy$$

$$= \frac{1}{3} \left[xy + \frac{y^2}{2} \right]_0^1$$

$$= \frac{1}{3} \left[x + \frac{1}{2} \right]$$

$$= \frac{2x+1}{6}$$

Marginal prob. of 'Y'

$$f_y(y) = \int_x f(xY) dx$$

$$= \int_0^2 \frac{1}{3} (x+y) dx$$

$$= \frac{1}{3} \left[\frac{x^2}{2} + xy \right]_0^2$$

$$= \frac{1}{3} \left[\frac{4}{2} + 2y \right]$$

$$= \frac{2+2y}{3}$$

3. Conditional prob. of 'X' given Y.

$$f_{(X/Y)} = \frac{\text{JPdf}}{\text{Marginal}} = \frac{f(XY)}{f_Y(Y)}$$

$$= \frac{1/3(x+y)}{2+2y}$$

$$= \frac{x+y}{2+2y}$$

Conditional prob. of 'Y' given X

$$f_{(Y/X)} = \frac{f(XY)}{f_X(X)}$$

$$= \frac{1/3(x+y)}{2x+1}$$

$$= \frac{2(x+y)}{2x+1}$$

4. Distribution function

$$\int_{-\infty}^x \int_{-\infty}^y f(XY) dy dx$$

$$= \int_0^x \int_0^y \frac{1}{3}(x+y) dy dx$$

$$= \frac{1}{3} \int_0^x \left[yx + \frac{y^2}{2} \right]_0^y dx$$

$$= \frac{1}{3} \int_0^x \left(xy + \frac{y^2}{2} \right) dx$$

$$= \frac{1}{3} \left[\frac{x^2 y}{2} + \frac{xy^2}{2} \right]_0^x$$

$$= \frac{1}{3} \left[\frac{x^2 y}{2} + \frac{xy^2}{2} \right]$$

$$= \frac{1}{6} (x^2 y + xy^2)$$

8.

$$f(X) = Ax^2 \quad 0 < x < 5$$

$$f(Y) = B(y^2+1) \quad 0 < y < 9$$

Find:-

1. $f(XY)$
2. Marginal prob. of 'X' and 'Y'
3. Conditional prob. of 'X' given Y=y and 'Y' given X=x
4. Find joint distⁿ function of 'X' and 'Y'
5. $E(X)$, $V(X)$, $E(Y)$, $V(Y)$

Soln

Here,

$$f(X) = Ax^2 \quad 0 < x < 5$$

$$f(Y) = B(y^2+1) \quad 0 < y < 9$$

Now,

$$1 = \int_0^5 Ax^2 dx$$

$$1 = A \left[\frac{x^3}{3} \right]_0^5$$

$$1 = A \cdot \frac{125}{3}$$

$$\therefore A = 0.024$$

$$1 = \int_0^9 f(y) dy$$

$$\Rightarrow 1 = \int_0^9 B(y^2+1) dy$$

$$\Rightarrow 1 = B \left[\frac{y^3}{3} + y \right]_0^9$$

$$\Rightarrow 1 = B [243 + 9]$$

$$\Rightarrow 1 = B \cdot 252$$

$$\therefore B = \frac{1}{252} = 0.0039$$

$$1. f(XY) = f(X) \cdot f(Y)$$

$$= Ax^2 \cdot B(y^2+1)$$

$$= \frac{1}{10500} x^2 y^2 + x^2$$

2. Marginal prob. of 'X'

$$f_x(x) = \int_y f(XY) dy$$

$$= \int_0^9 \frac{1}{10500} x^2 y^2 + x^2 dy$$

$$= \frac{1}{10500} \left[\frac{x^2 y^3}{3} + x^2 y \right]_0^9$$

$$= \frac{1}{10500} [243x^2 + 9x^2]$$

$$= \frac{3x^2}{125}$$

Marginal prob. of 'Y'

$$f_y(y) = \int_x f(XY) dx$$

$$= \int_0^5 \frac{1}{10500} x^2 y^2 + x^2 dx$$

$$= \frac{1}{10500} \left[\frac{x^3 y^2}{3} + \frac{x^3}{3} \right]_0^5$$

$$= \frac{1}{10500} \left[\frac{125 y^2}{3} + \frac{125}{3} \right]$$

$$= \frac{(y^2+1)}{252}$$

3. Conditional prob. of 'X' given Y=y

$$f_{(x/y)} = \frac{f(XY)}{f_y(y)}$$

$$= \frac{1}{10500} (x^2 y^2 + x^2) \times \frac{252}{y^2+1}$$

$$= \frac{3x^2(y^2+1)}{125(y^2+1)}$$

$$= \frac{3x^2}{125}$$

Conditional prob. of 'Y' given X=x

$$f_{(y/x)} = \frac{f(XY)}{f_x(x)}$$

$$= \frac{1}{10500} (x^2 y^2 + x^2) \times \frac{125}{3x^2}$$

$$= \frac{1}{252} \frac{y^2+1}{x^2}$$

$$= \frac{y^2+1}{252}$$

4. Joint function of 'X' and 'Y'.

$$\int_{-\infty}^x \int_{-\infty}^y f(x,y) dy dx$$

$$= \int_0^x \int_0^y \frac{1}{10500} (x^2 y^3 + x^2 y) dy dx$$

$$= \frac{1}{10500} \int_0^x \left[\frac{x^2 y^4}{4} + \frac{x^2 y^2}{2} \right]_0^y dx$$

$$= \frac{1}{10500} \int_0^x \frac{x^2 y^4 + 2x^2 y^2}{2} dx$$

$$= \frac{1}{10500} \left[\frac{y^5}{5} + \frac{2y^3}{3} \right]_0^x$$

$$= \frac{1}{10500} \left[\frac{x^5}{5} + \frac{2x^3}{3} \right]$$

$$= \frac{1}{94500} [x^5 + 3x^3]$$

5. $E(X) = \int_0^5 x \cdot f(x) dx$

$$= \int_0^5 x \cdot \frac{3}{125} x^2 dx$$

$$= \frac{3}{125} \int_0^5 x^3 dx$$

$$= \frac{3}{125} \left[\frac{x^4}{4} \right]_0^5$$

$$= \frac{3}{125} \times \frac{625}{4}$$

$$= \frac{15}{4}$$

$$E(X^2) = \int_0^5 x^2 \cdot \frac{3}{125} x^2 dx$$

$$= \frac{3}{125} \int_0^5 x^4 dx$$

$$= \frac{3}{125} \left[\frac{x^5}{5} \right]_0^5$$

$$= \frac{3}{125} \times \frac{3125}{5}$$

$$= 15$$

$$V(X) = E(X^2) - (E(X))^2$$

$$= 15 - \left(\frac{15}{4} \right)^2$$

$$= \frac{15}{16}$$

$$E(Y) = \int_0^9 y \cdot f(y) dy$$

$$= \int_0^9 y \cdot \frac{1}{252} (y^2 + 1) dy$$

$$= \frac{1}{252} \int_0^9 y^3 + y dy$$

$$= \frac{1}{252} \left[\frac{y^4}{4} + \frac{y^2}{2} \right]_0^9$$

$$= \frac{1}{252} \left[\frac{6561}{4} + \frac{81}{2} \right]$$

$$= \frac{747}{112}$$

$$E(Y^2) = \int_0^9 y^2 \cdot f(y) dy$$

$$= \int_0^9 y^2 \cdot \frac{1}{252} (y^2 + 1) dy$$

$$= \frac{1}{252} \int_0^9 y^4 + y^3 dy$$

$$= \frac{1}{252} \left[\frac{y^5}{5} + \frac{y^4}{4} \right]_0^9$$

$$= \frac{1}{252} \left[\frac{59049}{5} + \frac{729}{4} \right]$$

$$= \frac{1674}{35}$$

$$V(Y) = E(Y^2) - (E(Y))^2$$

$$= \frac{1674}{35} - \left(\frac{747}{112} \right)^2$$

$$= 3.344$$

Chapter 6

Central Limit theorem

Sum and averages of R.V.

Let $x_1, x_2, \dots, x_n$ be a r.v in a sample space 'S' with mean $\mu_1, \mu_2, \dots, \mu_n$ and variance $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$ then,

The sum of 'n' r.v is denoted as S_n and their averages is denoted as $\bar{X}_n$.

$$S_n = X_1 + X_2 + \dots + X_n = \sum_{i=1}^n X_i \quad \text{--- (i)}$$

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{--- (ii)}$$

Taking expectation on both side on eqn (i) we get

$$E(S_n) = E(X_1 + X_2 + \dots + X_n)$$

$$E(S_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$= \mu_1 + \mu_2 + \dots + \mu_n$$

$$= \sum_{i=1}^n \mu_i$$

Taking expectation on both side on eqn (ii) we get

$$E(\bar{X}_n) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right)$$

$$= \frac{1}{n} \sum_{i=1}^n \mu_i$$

Taking variance on eqn (i) we get

$$\begin{aligned} V(S_n) &= V(X_1 + X_2 + \dots + X_n) \\ &= V(X_1) + V(X_2) + \dots + V(X_n) \\ &= \sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2 \\ &= \sum_{i=1}^n \sigma_i^2 \end{aligned}$$

Again,

Taking variance on both side on eqn (ii) we get

$$\begin{aligned} V(\bar{X}_n) &= V\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) \\ &= \frac{1}{n^2} V(X_1 + X_2 + \dots + X_n) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2 \end{aligned}$$

If $x_1, x_2, \dots, x_n$ follows normal distribution with mean $\mu_1, \mu_2, \dots, \mu_n$ and variance $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$ then their sum S_n follows normal distribution with

$$\text{Mean, } E(S_n) = \sum_{i=1}^n \mu_i$$

and

$$\text{Variance, } V(S_n) = \sum_{i=1}^n \sigma_i^2$$

Similarly, their averages also follow normal distribution with

$$\begin{aligned} \text{Mean, } E(\bar{X}_n) &= E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) \\ &= \frac{1}{n} \sum_{i=1}^n \mu_i \end{aligned}$$

$$\text{and Variance, } V(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2$$

If $x_1, x_2, \dots, x_n$ follows normal distribution with mean ' μ ' and variance ' σ^2 '. Then their sum

$$S(n) = X_1 + X_2 + \dots + X_n$$

and Expectation of S_n is

$$\begin{aligned} E(S_n) &= E(X_1 + X_2 + \dots + X_n) \\ &= E(X_1) + E(X_2) + \dots + E(X_n) \\ &= \mu + \mu + \dots + \mu \\ &= n\mu \end{aligned}$$

Similarly their average $(\bar{X}_n) = \frac{(X_1 + X_2 + \dots + X_n)}{n}$

$$E(\bar{X}_n) = \frac{n\mu}{n} = \mu$$

$$\begin{aligned} V(S_n) &= V(X_1 + X_2 + \dots + X_n) \\ &= V(X_1) + V(X_2) + \dots + V(X_n) \\ &= \sigma^2 + \sigma^2 + \dots + \sigma^2 \\ &= n\sigma^2 \end{aligned}$$

Similarly their average, $\bar{X} = \frac{S}{n}$

$$\begin{aligned} V(\bar{X}_n) &= V\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) \\ &= \frac{1}{n^2} V(X_1 + X_2 + \dots + X_n) \\ &= \frac{1}{n^2} \times n\sigma^2 \\ &= \frac{1}{n} \sigma^2 \end{aligned}$$

If $x_1, x_2, \dots, x_n$ follows normal distribution with mean ' μ ' and variance ' σ^2 ' then their sum also follows normal distribution with

$$E(S_n) = n\mu$$

and

$$V(S_n) = n\sigma^2$$

Similarly their average

$$E(\bar{x}_n) = \mu$$

and

$$V(\bar{x}_n) = \frac{\sigma^2}{n}$$

* Central Limit Theorem

→ This theorem is used to standardize the distribution of R.V.

$$Z = \frac{A - E(A)}{\sqrt{V(A)}} \sim N(0, 1)$$

Properties:

1. If a r.v 'X' follows Binomial distⁿ then its standard variates is,

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} \sim N(0, 1)$$

$$Z = \frac{X - np}{\sqrt{npq}}$$

2. For poisson distⁿ

$$X \sim P(\lambda)$$

then it's standard variates is:-

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} = \frac{X - \lambda}{\sqrt{\lambda}} \sim N(0, 1)$$

Q. A manufacture company produce bulbs that have a length of life follows normal distⁿ with mean 900 and S.D 36.

* Find the prob. that a random sample of 20 bulbs will have average below 820.

* Find the prob. that a random sample of 25 bulbs will have sum above 860.

Here,

$$\text{Mean } (\mu) = 900$$

$$\text{S.D. } (\sigma) = 36$$

$$\text{Variance } (\sigma^2) = 36^2 = 1296$$

for (i)

$$n = 20$$

$$P(\bar{x}_{20} \leq 820) = ?$$

for (ii)

$$n = 25$$

$$P(\sum \leq 860) = ?$$

Now, Converting into standard normal variate using C.L.T

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} \sim N(0, 1)$$

$$Z = \frac{\bar{x}_{20} - E(\bar{x}_{20})}{\sqrt{V(\bar{x}_{20})}}$$

$$= \frac{\bar{x}_{20} - \mu}{\sqrt{\frac{\sigma^2}{n}}}$$

$$= \frac{\bar{x}_{20} - 900}{\sqrt{\frac{1296}{20}}}$$

$$= \frac{\bar{x}_{20} - 900}{\sqrt{64.8}}$$

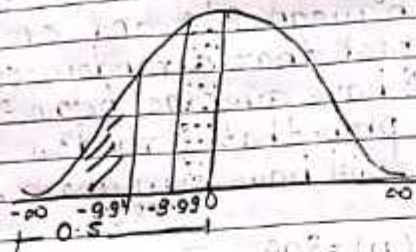
$$= \frac{\bar{x}_{20} - 900}{8.05}$$

$$\text{Then, } P(\bar{x}_{20} \leq 820)$$

$$= P\left(\frac{\bar{x}_{20} - \mu}{\sqrt{\sigma^2/n}} \leq \frac{820 - \mu}{\sqrt{\sigma^2/n}}\right)$$

$$= P\left(Z \leq \frac{820 - 900}{\sqrt{\frac{1296}{20}}}\right)$$

$$P(Z \leq -9.94)$$



$$0.5 - (P(-9.94 \leq Z \leq -3.99) + P(-3.99 \leq Z \leq 0))$$

$$0.5 - (0 + 0.49997)$$

$$0.00003$$

Now, $P(825 \geq 860)$

Now, converting into standard normal variate using C.L.T

$$Z = \frac{825 - E(825)}{\sqrt{V(825)}}$$

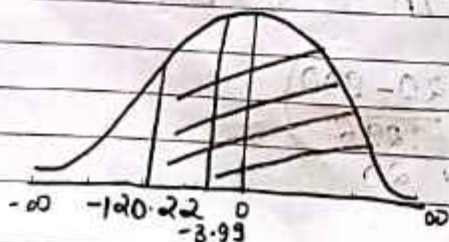
$$= \frac{825 - n\mu}{\sqrt{nr^2}}$$

Then,

$$P(825 \geq 860) = P\left(\frac{825 - E(825)}{\sqrt{V(825)}} \geq \frac{860 - n\mu}{\sqrt{nr^2}}\right)$$

$$= P\left(Z \geq \frac{860 - 25 \times 900}{\sqrt{25 \times 1296}}\right)$$

$$= P(Z \geq -120.22)$$



$$0.5 + (P(-120.22 \leq Z \leq -3.99) + P(-3.99 \leq Z \leq 0))$$

$$= 0.5 + (0 + 0.49997)$$

$$= 0.99997$$

* Estimation

- It is the process of estimating population parameter using sample statistic is known as estimation.

	Population parameter	Sample statistic
Mean	μ	$\bar{X}$
S.D	σ	s
Variance	σ^2	s^2
Proportion of success	P	p
proportion of failure	Q	q
Size	N	n

Point estimation

- Internal estimation

- unbiasedness : $E(\bar{X}) = \mu$

- Sufficiency : $f(x, Q) = f(x) \cdot f(Q)$

- efficiency : $V \propto \frac{1}{e}$

- consistency

$$\alpha = 5\% = 1.96$$

$$\alpha = 10\% = 1.64$$

* Confidence Interval

1. Confidence interval of mean
(Sample size above 30 ($n \geq 30$))

$$\bar{Z} = \frac{\bar{X} - \mu}{\sqrt{\frac{\sigma^2}{n}}} \sim N(0,1)$$

$$\begin{array}{|l} P(\bar{X}) = \mu \\ V(\bar{X}) = \frac{\sigma^2}{n} \end{array} \quad \bar{Z} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$P(-Z_{\alpha/2} \leq \bar{Z} \leq Z_{\alpha/2}) = 1 - \alpha$$

$$\text{or, } P\left(-Z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}\right) = 1 - \alpha$$

$$\text{or, } P\left(-Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$\text{or, } P\left(-Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} - \bar{X} \leq -\mu \leq Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} - \bar{X}\right) = 1 - \alpha$$

$$\text{or, } P\left(-\left(\bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) \leq -\mu \leq -\left(\bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)\right) = 1 - \alpha$$

$$\text{or, } P\left(\left(\bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) \geq \mu \geq \left(\bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)\right) = 1 - \alpha$$

$$\therefore \text{C.I of } \mu \text{ at } \alpha\% \text{ level of significance} \\ = \bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

When 'N' is given,

$$\text{C.I} = \bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$

Sample size above 30 bhaya bhane matra table bata
herinxo

$$\alpha = 1 - 0.95$$

$$= 0.05$$

Q. A normal population has a mean of 10 and S.D 2.5. And 95% C.I of population mean, out of 10000 size, only 900 were taken during study.

Soln

$$\mu = 10$$

$$\text{S.D} = 2.5$$

$$N = 10000$$

$$n = 900$$

C.I of 10 at 5% level of significance

$$\text{C.I} = \bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$

$$= \bar{X} \pm Z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{10000-900}{10000-1}}$$

$$= 10 \pm 1.96 \times 2.5 \times 0.95 \cdot \frac{1}{\sqrt{900}}$$

$$= 9.84$$

$$\text{C.I} = 10 \pm 1.96 \times 2.5 \times 0.95 \cdot \frac{1}{\sqrt{900}}$$

$$= 10.15$$

$$\text{C.I} = (9.84, 10.15)$$

* Confidence interval of population propⁿ

$$P\left(p - z_{\alpha/2} \sqrt{\frac{pq}{n}} \leq p \leq p + z_{\alpha/2} \sqrt{\frac{pq}{n}}\right) = 1 - \alpha$$

when, p is unknown then it's unbiased estimator $\hat{p}$ is used as:-

$$P\left(\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}\right) = 1 - \alpha$$

C.I of mean of sample size below 30 ($n \leq 30$)
 $t_{\alpha/2} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$

Then, C.I is written as:-

$$P\left(\bar{X} - t_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2} \cdot \frac{S}{\sqrt{n}}\right) = 1 - \alpha$$

'S' is sampling

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Confidence Interval for double (X and Y) Two sample

1. When, sample size is greater than 30,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

Now,

1 - α % C.I. is:-

$$P\left((\bar{X} - \bar{Y}) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq (\mu_1 - \mu_2) \leq (\bar{X} - \bar{Y}) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right) = 1 - \alpha$$

$$\therefore \text{C.I} = (\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

When population S.D (σ) is unknown, we use Z-test. In this case

Sample S.D (s) = popⁿ S.D (σ)

When sample size are less than (30)

(1 - α)% C.I of difference of μ_1 and μ_2 is:-

$$= (\bar{X} - \bar{Y}) \pm t_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$S_1 = \sqrt{\frac{1}{n_1 - 1} \left(\sum_{i=1}^n (X_i - \bar{X})^2 \right)}$$

$$S_2 = \sqrt{\frac{1}{n_2 - 1} \left(\sum_{i=1}^n (Y_i - \bar{Y})^2 \right)}$$

Proportion of two sample

$$Z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}} \sim N(0, 1)$$

$$\text{C.I} = ((p_1 - p_2) \pm z_{\alpha/2} \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}})$$

When study area is same for both sample:-

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$\hat{q} = 1 - \hat{p}$$

(1- α)% C.I. is:-

$$= (\hat{p} - \hat{q}) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

$$n = n_1 + n_2$$

When samples are taken from same area and below 30, then,

$$S = \sqrt{\frac{1}{(n_1 + n_2 - 2)} \left(\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n (Y_i - \bar{Y})^2 \right)}$$

and C.I. of mean difference is:-

$$= (\bar{X} - \bar{Y}) \pm t_{\alpha/2} \cdot S \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Find 90% C.I. for the percentage defective in a large lot of material on the basis of random sample of 50 items. The number of defective items found in the sample is 18.

Soln.

$$n = 50$$

$$\text{No. of defective, } x = 18$$

$$\text{Pro. of defective } (p) = \frac{x}{n} = \frac{18}{50} = 0.36$$

$$q = 1 - p$$

$$= 1 - 0.36$$

$$= 0.64$$

90% of C.I.

$$= 0.36 \pm z_{0.05} \sqrt{0.36 \times 0.64}$$

$$= 0.36 \pm 1.64 \times 0.067$$

$$= 0.4718, 0.2501$$

90% C.I.

$$\text{C.I.} = (0.25, 0.47)$$

When σ is not given
Universal 51.

2. Random samples drawn from two countries given the following data relating to the heights of adult males.

Males	Country A	Country B
Mean height (in inches)	67.42	67.25
S.d in inches	2.58	2.50
No. of sample	1000	1200

find C.I

Soln

$$(\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

as popln is unknown, so, $s = \sigma$

$$95\% \text{ of C.I} = (\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$= (67.42 - 67.25) \pm z_{0.025} \sqrt{\frac{2.58^2}{1000} + \frac{2.50^2}{1200}}$$

$$= -0.0434, 0.3834$$

No. of sample	10	12
---------------	----	----

$$95\% \text{ of C.I} = ((\bar{X} - \bar{Y}) \pm t_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}})$$

$$\begin{aligned} \text{Degree of freedom for Country A} &= n_1 - 1 \\ &= 10 - 1 \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{Degree of freedom for country B} &= n_2 - 1 \\ &= 12 - 1 \\ &= 11 \end{aligned}$$

$$\begin{aligned} \text{Dre Degree of freedom} &= n_1 + n_2 - 2 \\ &= 10 + 12 - 2 \\ &= 20 \end{aligned}$$

$$\begin{aligned} 95\% \text{ of C.I} &= (67.42 - 67.25) \pm 2.086 \sqrt{\frac{2.58^2}{10} + \frac{2.50^2}{12}} \\ &= (-2.102, 2.44) \end{aligned}$$

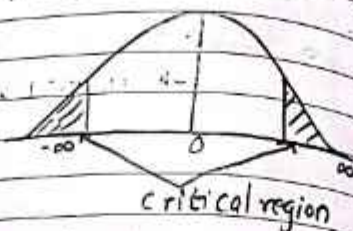
- Q) Random samples drawn from a community given the following data relating to the heights of a adult

Adults	Male	female
Mean ht (in inches)	67.42	67.25
S.d in inches	2.58	2.50
No. of sample	10	12

find 90%, 95% and 99% C.I

Hypothesis Testing

- Level of significance - prob. of rejection area



- degree of freedom $\rightarrow$ No. of freely occupied space.

- Error in hypothesis Testing

	True	False
Accept	$1 - \beta$	$1 - \beta$
Reject	α	$1 - \alpha$

1) Type I error (α)

$\rightarrow$ It is the prob. of rejecting Null hypothesis (H_0) when it is true. It is denoted by ' α ' and can be given as:

$$\alpha = P(\text{Rejecting null hypothesis when it is true})$$

$$= P(\text{Reject } H_0 / H_0)$$

2) Type II error (β)

$\rightarrow$ It is the prob. of accepting Null hypothesis (H_0) when it is false. It is denoted by ' β ' and can be given as:

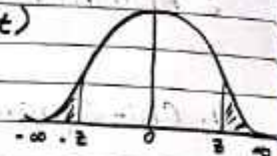
$$\beta = P(\text{Accepting null hypothesis when it is false})$$

$$= P(\text{Accept } H_0 / H_0 \text{ is false})$$

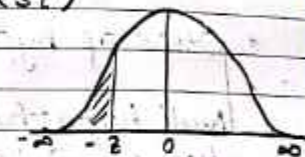
$$= P(\text{Accept } H_0 / H_1)$$

1. Null Hypothesis (H_0) $\rightarrow \mu_1 = \mu_2$

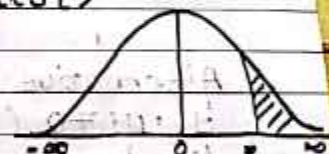
2. Alternative hypothesis (H_1)
 $\rightarrow \mu_1 \neq \mu_2$ (Two tailed test)



$\rightarrow \mu_1 < \mu_2$ (One tailed left test)



$\rightarrow \mu_1 > \mu_2$ (One tailed right test)



- Critical value - Table $\rightarrow$ Tabulated value

- Test statistic

$$Z = \frac{\bar{x} - E(\bar{x})}{S.E(\bar{x})}$$

1. Z-Test - Single Mean? ($n \geq 30$, $S.D$ unknown)

- double Mean

- Single proportion

- double proportion

2. t-test: - Single mean } ($n \leq 30$, Sample S.D known)
- double mean
- paired $\rightarrow$ Before/After

3. ANOVA (F-test) - One way } for more than 2
- Two way } Samples.

4. Chi-square test: (χ^2)

* Steps of hypothesis testing

1. Null hypothesis
 $H_0: \mu = 0$ i.e. there is no significance difference between sample mean and population mean.

2. Alternative hypothesis
 $H_1: \mu \neq 0$ (two tailed test) i.e. There is significant diff between sample mean and population mean.

$H_1: \mu < 0$ (one tailed left) i.e. the sample mean and population mean is less than 0.

$H_1: \mu > 0$ (one tailed Right) i.e. the sample mean and population mean is greater than 0.

3. Test statistic
under H_0 :

$$Z = \frac{t \cdot f(t)}{S.E(t)}$$

$$|Z| = Z \text{ calculated}$$

4. Level of Significance

5. Degree of freedom $\rightarrow n-1$

6. Critical Value

Tabulated value of Z at $\alpha\%$ level of significance.
 $|Z| \leq |Z|_{\alpha}$ or $|Z| < |Z|_{\alpha}$ depending on the tailed test.

7. Decision

$|Z|_{cal} \leq |Z|_{tab}$, we accept H_0 .
otherwise we reject H_0 .

8. Conclusion.

Statistic	Test	Standard Error (S.E.)
1. Mean (' σ ' known, $N \rightarrow \infty$)	Z-Test	$S.E(\bar{x}) = \frac{\sigma}{\sqrt{n}}$
2. Mean (' σ ' known, N -finite)	Z-Test	$S.E(\bar{x}) = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$
3. Mean (' σ ' unknown, $N \rightarrow \infty$)	t-test (if $n \leq 30$) else Z-Test	$S.E(\bar{x}) = \frac{s}{\sqrt{n}}$
4. Mean (' σ ' unknown, N -finite)	t-Test (if $n \leq 30$) else Z-Test	$S.E(\bar{x}) = \frac{s}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$
5. Difference of mean ($\mu_1 - \mu_2$) (σ_1, σ_2 known)	Z-Test	$S.E(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
6. Difference of mean ($\mu_1 - \mu_2$) (σ_1 and σ_2 unknown)	t-test (if $n_1, n_2 \leq 30$) else Z-Test	$S.E(\bar{x}_1 - \bar{x}_2) = s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ for same study area $= \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ for different study area
7. Single proportional (P) when 'P' is known	Z-Test	$S.E(p) = \sqrt{\frac{pq}{n}}$
8. Single proportional 'P' is unknown	Z-Test	$S.E(p) = \sqrt{\frac{pq}{n}}$
9. Difference of proportion ($P_1 - P_2$)	Z-Test	$S.E(p_1 - p_2) = \sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$ for same study area $= \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$ for different study area

Sample mean (μ) = population mean

1. Z-Test for single Mean

1. H_0 : There is no significance difference between sample mean and population mean.
(i.e. $\mu = 0$)

2. H_1 : $\mu \neq 0$ (for two tailed test)
 $\mu < 0$ (for one tailed left test)
 $\mu > 0$ (for one tailed right test)

3. Test statistic, under H_0

$$Z = \frac{\bar{x} - \mu}{S.E(\bar{x})}$$

$$|Z| = Z\text{-cal}$$

4. Level of significance

→ At this test 5% level of significance is used.

5. Degree of freedom

→ The df of this test is $(n-1)$.

6. Critical value

→ The tabulated value of Z at 5% level of significance for two tailed test is 1.96.
i.e. $Z_{0.05} = 1.96$ (two tailed)

7. Decision

$$|Z|_{cal} \leq |Z|_{tab}, \text{ accept } H_0.$$

8. Conclusion

2. Z-Test for double mean

1. H_0 : There is no significance difference between mean of first sample and 2nd sample.
($\mu_1 = \mu_2$)

2. H_1 : $\mu_1 \neq \mu_2$ (for two tailed test)
 $\mu_1 < \mu_2$ (for one tailed left test)
 $\mu_1 > \mu_2$ (for one tailed right test)

3. Test statistic, under H_0

$$Z = \frac{\bar{x} - \bar{y}}{S.E(\bar{x} - \bar{y})}$$

$$|Z| = Z\text{-cal}$$

repeat 4-8

3. Z-test for proportion mean
 1) H_0 : There is no significance difference between sample proportion and population difference (i.e. $P=0$)

2) H_1 : $P \neq 0$ (for two tail)

Q. Experience has shown that 20% of a manufactured product is of the top quality. In one day's production of 400 articles only 50 are top quality. Show that hypothesis of 20% was wrong at 1% level of significance.

Soln

Given

Population Proportion (P) = 20% = 0.2

Sample Size (n) = 400

no. of top quality (x) = 50

$$p = \frac{x}{n} = \frac{50}{400} = \frac{1}{8} = 0.125$$

H_0 : $P = 0.20$ i.e. 20% of a manufactured product is of the top quality.

H_1 : $P \neq 0.20$ i.e. The 20% of manufactured product is of top quality.

Test statistic: under H_0

$$Z = \frac{p - P}{SE(p)}$$

$$= \frac{0.125 - 0.2}{\sqrt{\frac{PQ}{n}}}$$

$$= \frac{0.125 - 0.2}{\sqrt{\frac{0.2 \times 0.8}{400}}}$$

$$= \frac{0.125 - 0.2}{\sqrt{\frac{0.2 \times 0.8}{400}}}$$

$$|Z|_{cal} = 3.75$$

level of significance (α) = 1%

→ At this test 1% level of significance is used.

Degree of freedom
 $\rightarrow n-1 = 399$

Critical value,
 $\rightarrow Z_{0.01} \text{ (two tailed)} = 2.575$

Decision:

$$|Z_{\text{cal}}| > |Z_{\text{tab}}|$$

$$\text{i.e. } 3.75 > 2.575$$

So

H_0 is rejected.

The 20% of manufactured product is not of the top quality.

2. Random sample of 400 male and 600 female were asked whether they would like to have a flyover near their residence. 200 male and 325 female were in the favour of the proposal. Test the hypothesis that propⁿ of male and female in favours of the proposal are same against that they are not, at 5% level.

Soln

Given

$$n_1 = 400$$

$$n_2 = 600$$

$$x_1 = 200$$

$$x_2 = 325$$

$$p_1 = \frac{200}{400} = 0.5$$

$$p_2 = \frac{325}{600} = 0.542$$

H_0 :- $p_1 = p_2$ i.e. There is no significance between the propⁿ of male and female in the favour of flyover proposal.

H_1 :- $p_1 \neq p_2$ i.e. There is significance between the propⁿ of male and female in the favour of flyover proposal.

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$= \frac{400 \times 0.5 + 600 \times 0.542}{400 + 600}$$

$$= 0.5252$$

$$\hat{q} = 1 - \hat{p}$$

$$= 1 - 0.5252$$

$$= 0.4748$$

test statistic : under H_0

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$= \frac{0.5 - 0.542}{\sqrt{0.5252 \times 0.4748 \left(\frac{1}{400} + \frac{1}{600} \right)}}$$

$$= 1.3125$$

$$|Z|_{\text{cal}} = 1.3125$$

Level of significance (α) = 5%

$Z_{5\%}$ of two tailed test = 1.96

∴ Decision:

Here,

$$|Z|_{cal} < |Z|_{tab}$$

$$\text{i.e. } 1.3125 < 1.96$$

So, H_0 is accepted

$$\therefore Df = n_1 + n_2 - 2 = 400 + 600 - 2 = 998$$

∴ There is no significant difference between proportion of male and female in favour of proposal.

3. A manufacturing company produce 10000 metal rods with S.D 2.56 Kgs and average weight was found to be 70 Kgs. A sample of 400 is taken with average 50.70 Kgs. Test their claims on weight of the metal rod.

Here -

$$N = 10000$$

$$\text{popl}^n \text{ S.D } (\sigma) = 2.56 \text{ Kgs.}$$

$$\text{popl}^n \text{ mean } (\mu) = 70 \text{ Kgs.}$$

$$\text{Sample Size } (n) = 400$$

$$\text{Sample mean } (\bar{X}) = 50.70 \text{ Kgs.}$$

$H_0: \mu = 70$ i.e. The average weight of the metal rod is 70 Kgs.

$H_1: \mu \neq 70$ i.e. (two tailed) The average weight of the metal rod is not 70 Kgs.

Test statistic: Under H_0

$$Z = \frac{\bar{X} - \mu}{SE(\bar{X})}$$

$$= \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

$$= \frac{50.70 - 70}{\frac{2.56}{\sqrt{400}} \sqrt{\frac{10000-400}{10000-1}}}$$

$$|Z|_{cal} = 153.88$$

Level of significance $(\alpha) = 5\%$

$$|Z|_{tab} \text{ at } 5\% = 1.96$$

$$d.f = n - 1 = 400 - 1 = 399$$

Decision:

Here,

$$|Z|_{cal} > |Z|_{tab}$$

$$\text{or, } 153.88 > 1.96$$

So, H_0 is rejected.

∴ The average weight of metal rod is not 70 Kgs.

4. A company produce two items A and B of size 400 and 550 respectively. The average length of A and B was found to be 4.76 and 3.73 inches along with S.D of 0.73 inches and 0.746 inches. Test whether the average length of A is greater than B or not at 5% and 1% level of significance.

Soln

$$n_1 = 400$$

$$n_2 = 550$$

$$\bar{x}_1 = 4.76$$

$$\bar{y} = 3.73$$

$$s_1 = 0.73$$

$$s_2 = 0.746$$

H_0 :- $\mu_A = \mu_B$ i.e. there is no significance betⁿ the average length of A and B.

H_1 :- $\mu_A > \mu_B$ (one tailed right test) i.e. there is significance betⁿ the average length of A & B.

Test statistics under H_0

$$Z = \frac{\bar{x} - \bar{y}}{SE(\bar{x} - \bar{y})}$$

$$SE(\bar{x} - \bar{y}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$= \sqrt{\frac{(0.73)^2}{400} + \frac{(0.746)^2}{550}}$$

$$= \sqrt{\frac{0.5329}{400} + \frac{0.5565}{550}}$$

$$= \sqrt{0.001332 + 0.001012}$$

$$= \sqrt{0.002344} = 0.0484$$

$$|Z|_{cal} = 21.27$$

level of significance (α_1) = 5%

$|Z|_{tab}$ at 5% of significance = 1.96

level of significance (α_2) = 1%

$|Z|_{tab}$ at 1% of significance = 2.575

$$D.f = n_1 + n_2 - 2$$

$$= 400 + 550 - 2$$

$$= 948$$

Decision.

$$|Z|_{cal} > |Z|_{tab}$$

$$\text{i.e. } 21.27 > 1.96$$

So, H_0 is rejected

∴ The average length of A is greater than the average length of B at 5% level of significance

$$|Z|_{cal} > |Z|_{tab}$$

$$\text{i.e. } 21.27 > 2.575$$

So, H_0 is rejected

∴ The average length of A is greater than the average length of B at 1% level of significance

t-test

1. for single mean

test statistic under H_0 :

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

critical value = $t_{\alpha/2, n-1}$

where, α = level of significance

$$n-1 = d.f$$

2. for double mean

test statistic under H_0 :

$$t = \frac{\bar{X} - \bar{Y}}{\frac{s}{\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}}$$

$$s = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$$

if individual series is given then

$$s_1^2 = \frac{1}{n_1 - 1} \sum (x_i - \bar{x})^2$$

$$s_2^2 = \frac{1}{n_2 - 1} \sum (y_i - \bar{y})^2$$

$$D.f \Rightarrow n_1 + n_2 - 2$$

Critical value = $t_{\alpha, n_1 + n_2 - 2}$

v. imp

3. Paired t-test

$H_0: d = 0$ i.e. There is no significant difference in knowledge before & after training. [OR the training is not effective].

$$H_1: d \neq 0$$

$$d < 0$$

$$d > 0$$

Test statistic under H_0 :

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

$$s_d^2 = \frac{1}{n-1} \sum d^2 - n \bar{d}^2$$

$$d = X - Y = \text{before} - \text{after}$$

→ level of significance
 $\alpha = ?$, $t_{\alpha} = ?$

→ Degree of freedom
 $\Rightarrow n-1$

→ Critical value
 $t_{\alpha, n-1} =$

→ Decision
 $|t|_{cal} \leq |t|_{tab}$ accept H_0 else reject H_0

→ conclusion

8. The score of 10 candidates prior and after training are given below:-

Prior	After	Is training effective?
84	90	
48	58	
36	56	
37	49	
54	62	
69	81	
83	84	
96	86	
90	84	
6	75	

Soln

$H_0: d=0$ i.e. the training is not effective.

$H_1: d \neq 0$ i.e. the training is effective.

Test statistics: under H_0

$$t = \frac{\bar{d}}{S_d/\sqrt{n}}$$

X	Y	d = X - Y	d ²
84	90	-6	36
48	58	-10	100
36	56	-20	400
37	49	-12	144
54	62	-8	64
69	81	-12	144
83	84	-1	1
96	86	10	100
90	84	6	36
6	75	-10	100
		$\Sigma d = -63$	$\Sigma d^2 = 1125$

$$s^2 d = \frac{1}{n-1} \left(\sum_{i=1}^n d^2 - n \bar{d}^2 \right)$$

Here,

$$\bar{d} = \frac{\Sigma d}{n} = \frac{-63}{10} = -6.3$$

$$S_d^2 = \frac{1}{9} (1125 - 10 \times (-6.3)^2)$$

$$= 80.9$$

$$S_d = 8.99$$

$$\therefore t = \frac{-6.3}{8.99/\sqrt{10}}$$

$$= -2.21$$

$$\therefore |t|_{cal} = 2.21$$

→ Level of significance (α) = 5%

→ Degree of freedom

$$\rightarrow n-1 = 10-1 = 9$$

→ Critical value

$$t_{0.05, 9} \text{ (two tailed test)} = 2.262$$

→ Decision

$$|t|_{cal} \leq |t|_{tab}$$

$$2.21 \leq 2.262$$

It accept H_0 .

The training is not effective.

- Chi-square test (χ^2)
- use for qualitative data / categorical variables.

Gender	Smoking habits			Total
	Yes	No	Maybe	
Male	a	b	c	D
female	d	e	f	E
Total	A	B	C	N

H_0 : There is no significant association / relationship between gender and smoking habits.

H_1 : There is significant association / relationship between gender and smoking habits.

Test statistic: under H_0

$$\chi^2 = \sum_{i=1}^n \frac{(O-E)^2}{E}$$

Expected value	Observed Value	(O-E)	(O-E) ²	(O-E) ² /E
$a^e = \frac{D \times A}{N}$	a	$a - a^e$	$(a - a^e)^2$	$\frac{(a - a^e)^2}{a^e}$
$b^e = \frac{B \times D}{N}$	b	$b - b^e$	$(b - b^e)^2$	$\frac{(b - b^e)^2}{b^e}$
$c^e = \frac{C \times D}{N}$	c	$c - c^e$	$(c - c^e)^2$	$\frac{(c - c^e)^2}{c^e}$
$d^e = \frac{E \times A}{N}$	d	$d - d^e$	$(d - d^e)^2$	$\frac{(d - d^e)^2}{d^e}$
$e^e = \frac{E \times B}{N}$	e	$e - e^e$	$(e - e^e)^2$	$\frac{(e - e^e)^2}{e^e}$
$f^e = \frac{E \times C}{N}$	f	$f - f^e$	$(f - f^e)^2$	$\frac{(f - f^e)^2}{f^e}$
			$\Sigma = ?$	

$$\rightarrow \chi^2_{cal} =$$

$$\rightarrow \text{Level of significance} = \alpha$$

$$\rightarrow \text{Degree of freedom} \rightarrow (r-1) \times (c-1)$$

* for 2×2

Gender	Smoking habits		Total
	Yes	No	
Male	a	b	(a+b)
Female	c	d	(c+d)
Total	(a+c)	(b+d)	(a+b+c+d) = N

H_0 : There is no significant association / relationship between gender and smoking habits.

H_1 : There is significant association / relationship between gender and smoking habits.

Test statistic :- under H_0

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

Note :-

If a, b, c, d < 5, we use correction method.

$$\chi^2 = \frac{N(|ad-bc| - \frac{N}{2})^2}{(a+b)(b+d)(c+d)(a+c)}$$

$$\rightarrow \text{level of significance} = \alpha$$

$$\rightarrow \text{Degree of freedom} = 1$$

→ Critical value
 $\chi^2_{tab} =$

→ Decision,
 $\chi^2_{cal} \leq \chi^2_{tab}$; accept H_0 , else reject H_0 .

* Test the significance association:-

Gender	Disease	
	Abs	Presence
Male	25	27
Female	40	35

Soln

Gender	Disease		Total
	Abs	Presence	
Male	25	27	52
Female	40	35	75
Total	65	62	127

H_0 : There is no significant association, between gender and Disease.

H_1 : There is significant association, between gender and disease.

Test static: under H_0

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

$$= \frac{127(25 \times 35 - 40 \times 27)^2}{52 \times 75 \times 65 \times 62}$$

$$= 0.3395$$

→ Level of significance, $(\alpha) = 5\%$

→ D.f = 1

→ Critical Value
 $\chi^2_{tab} = 3.841$

→ Decision,

$$\chi^2_{cal} \leq \chi^2_{tab}$$

$$0.3395 \leq 3.841$$

It accept H_0 :

∴ There is no-significant association, between gender and disease.

2. Caste & Smoking habit

	Yes	No	Total
Chettni	27	40	67
Brahmin	10	15	25
Newar	25	27	52
Magar	24	20	44
Tharu	30	26	56
Others	28	30	58
Total	144	158	302

Soln

H_0 : There is no significant association between caste and smoking habit.

H_1 : There is significant association between caste and smoking habit.

Test statistic: under H_0
 $\chi^2 = \sum_{i=1}^n \frac{(O-E)^2}{E}$

	Expected Value	Observed Value	$(O-E)$	$(O-E)^2$	$(O-E)^2/E$
1. $a = 144 \times 67 / 302$ $= 31.947$	27	27	-4.947	24.473	0.766
2. 35.053	40	40	4.947	24.473	0.698
3. 11.921	10	10	-1.921	3.690	0.3095
4. 13.079	15	15	1.921	3.690	0.282
5. 24.795	25	25	0.205	0.042	0.0017
6. 27.205	27	27	-0.205	0.042	0.0015
7. 20.98	24	24	3.02	9.12	0.495
8. 23.98	20	20	-3.02	9.12	0.396
9. 26.70	30	30	3.3	10.9	0.407
10. 29.30	26	26	-3.3	10.9	0.371
11. 27.66	28	28	0.34	0.119	0.0043
12. 30.34	30	30	-0.34	0.119	0.0039
					$\chi^2_{cal} = 3.6759$

$\rightarrow \chi^2_{cal} = 3.6759$

$\rightarrow$ Level of significance (α) = 5%

$\rightarrow$ Degree of freedom (Df) = $(6-1) \times (2-1)$
 $= 5$

$\rightarrow \chi^2_{tab/\alpha} = 0.05, 5 = 11.07$

Decision-

$\chi^2_{cal} < \chi^2_{tab}$, So, H_0 is accepted.

There is no significant association between caste and smoking habit.

8.	Smoking habit	Disease observe	preserve
	Yes	4	27
	No	40	30
		44	57

Here, one cell has value lower than 4 so,

$$\chi^2_{cal} = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

H_0 : There is no significant between smoking habit and disease observation.

H_1 : There is significant relationship between smoking habit and disease observation.

Test statistic: under H_0

$$\chi^2_{cal} = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

$$= \frac{101(1120 - 40 \times 27)^2}{31 \times 70 \times 44 \times 57}$$

$$= 15.351$$

$\rightarrow$ Level of significance (α) = 5%
 $\rightarrow$ Degree of freedom (Df) = 1

$$\chi^2_{tab \times 0.05, 1} = 0.841$$

Here $\chi^2_{tab} < \chi^2_{cal}$ so, H_0 is rejected.

∴ There is significant association between smoking habit and disease.

4. Alcohol Consumption

Health issues	Yes	No	
bad	60	120	180
good	125	110	235
	185	230	415

Soln:

H_0 : There is no significant relationship between health issues & Alcohol Consumption.

H_1 : There is significant relationship between health issues & Alcohol Consumption.

Test statistic: under H_0

$$\chi^2_{cal} = \frac{N(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$$

$$= \frac{415(60 \times 110 - 125 \times 120)^2}{180 \times 235 \times 185 \times 230}$$

$$= 16.27$$

Level of significance (α) = 5%

Degree of freedom (D.F) = 1

$$\chi^2_{tab, \alpha = 0.05, 1} = 3.841$$

Here,

$\chi^2_{tab} < \chi^2_{cal}$, so, H_0 is rejected

∴ There is significant relationship between health issues and alcohol consumption.

5. In a study, out of 100 males, 40 had diseases and 250 female, 150 did not had disease. Test the association between gender and the disease presence and absence at 5% level of significance.

Soln

Given,

	Disease		
Gender	Yes	No	
Male	40	60	100
Female	100	150	250
	140	210	350

H_0 : There is no significant relationship between gender and disease.

H_1 : There is significant relationship between gender and disease.

Test statistic under H_0 :

$$\chi^2_{cal} = \frac{N(ad-bc)^2}{(a+b)(b+d)(a+c)(c+d)}$$

$$= \frac{350(40 \times 150 - 100 \times 60)}{100 \times 150 \times 140 \times 210}$$

$$= 0$$

level of significance (α) = 5%

Degree of freedom (D.f) = 1

$$\chi^2_{tab, \alpha=0.05, 1} = 3.841$$

Here,

$\chi^2_{tab} > \chi^2_{cal}$, so, H_0 is accepted.

$\therefore$ There is no significant difference between gender and disease.

* ANOVA

→ for more than two samples

X	Y ₁	Y ₂	...	Y _j	...	Y _n	Total	Mean
1	y ₁₁	y ₁₂	...	y _{1j}	...	y _{1n}	T _{1.}	$\bar{y}_{1.}$
2								
3								
...								
i	y _{i1}	y _{i2}		y _{ij}		y _{in}	T _{i.}	$\bar{y}_{i.}$
...								
m	y _{m1}	y _{m2}		y _{mj}		y _{mn}	T _{m.}	$\bar{y}_{m.}$
Total	T _{.1}	T _{.2}		T _{.j}		T _{.n}	G	$\bar{y}_{..}$
Mean	$\bar{y}_{.1}$	$\bar{y}_{.2}$		$\bar{y}_{.j}$		$\bar{y}_{.n}$		

for Row

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_m$$

H_1 : At least one μ_i is different

for column

$$H_0: \mu_1 = \mu_2 = \dots = \mu_n$$

H_1 : At least one μ_j is different.

Test statistic: under H_0

Source of Variation	Sum of square	degree of freedom	Mean sum of square	F-cal	F-tab
Row	SSR	m-1	MSR = SSR/m-1	$F_R = \frac{MSR}{MSE}$	$F_{0.05(1,2)}$
Column	SSC	n-1	MSC = $\frac{SSC}{n-1}$	$F_C = \frac{MSC}{MSE}$	
Error	SSE	nm-m-n+1	MSE		
Total	TSS	nm-n			

Correction factor (C.F) = $\frac{G^2}{N}$

(Total sum of square)

$$TSS = \sum_{i=1}^m \sum_{j=1}^n y_{ij}^2 - C.F$$

SSR (sum of square of Row)

$$= \sum_{i=1}^m \frac{T_i^2}{n_j} - C.F$$

$$SSC = \sum_{j=1}^n \frac{\bar{T}_j^2}{m_i} - C.F$$

Numerator → horizontal
Denominator → vertical

8. Samsung company product items

	A	B	C	D	E	F	G
A	27	24	60	27	24	45	50
B	26	23	30	35	30	35	52
C	50	15	25	33	45	65	61
D	35	27	23	36	35	60	30
E	20	26	22	30	36	32	20
F	26	20	10	29	40	30	10

1) Test whether there is significant difference in mean weight of Samsung company items.

2) Test whether there is significant difference in mean weight of CG company items.

3) Test whether there is significant difference in mean weight of Samsung company and CG company.

Soln
Given

Samsung company

	A	B	C	D	E	F	G	Total	Mean
A	27	24	60	27	24	45	50	257	38.71
B	26	23	30	35	30	35	52	231	33
C	50	15	25	33	45	65	61	294	42
D	35	27	23	36	35	60	30	246	35.14
E	20	26	22	30	36	32	20	186	26.57
F	26	20	10	29	40	30	10	165	23.57
Total	184	185	170	190	210	267	223	1379	
Mean	30.67	22.5	28.33	31.67	35	44.5	31.6		

1) for Samsung Company

$$H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F = \mu_G$$

H_1 : Atleast one μ_i is different.

2) for CG Company

$$H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F$$

H_1 : Atleast one μ_j is different

3) for Samsung company

$$H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F = \mu_G$$

H_1 : Atleast one μ_i is different.

$$H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F$$

H_1 : Atleast one μ_j is different

Test statistic under H_0 :

$$\text{Correction factor (C.F)} = \frac{G^2}{N} = \frac{1379^2}{7 \times 6}$$

$$= 45277.16$$

$$TSS = \left(\sum_{i=1}^m \sum_{j=1}^n y_{ij}^2 - \text{C.F} \right)$$

$$= 27^2 + 24^2 + 60^2 + \dots + 40^2 + 30^2 + 10^2 - 45277.16$$

$$= 17327.83$$

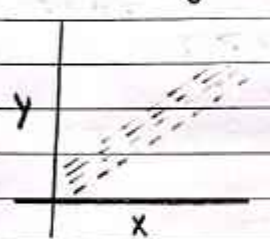
$$SSR = \left(\frac{\bar{y}_A^2}{7} + \frac{\bar{y}_B^2}{7} + \frac{\bar{y}_C^2}{7} + \frac{\bar{y}_D^2}{7} + \frac{\bar{y}_E^2}{7} + \frac{\bar{y}_F^2}{7} \right) - \text{C.F}$$

Correlation and Regression

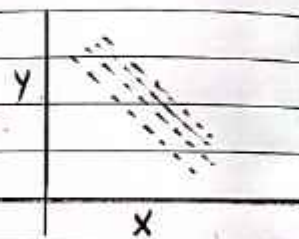
Properties of correlation

- correlation lies in between -1 to 1.
- Independent of unit.
- Independent of change of origin and scale.
- If value of correlation lies near to -1, it indicates high degree of negative association and near to 1 indicates high degree of positive correlation.

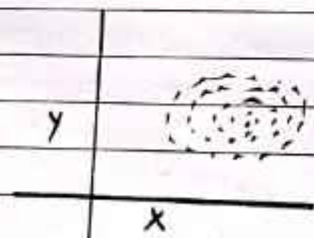
1. Scatter diagram



+ve correlation



-ve correlation



No correlation

2. Pearson's Correlation

$$r = \frac{n \sum XY - \sum X \cdot \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

Case If:

1. $r = 0$, no correlation.
2. r lies near 1, high degree of +ve correlation.
3. r is +ve, show +ve correlation.
4. r lies near -1, high degree of -ve correlation.
5. r is -ve shows -ve correlation.

* Regression

→ It is the method of obtaining unknown value with the help of known value by using their linear relationship.

The simple linear regression model is:-

$$Y = \alpha + \beta X \quad \text{--- (i)}$$

Where,

Y = dependent variable

X = Independent variable

α = Intercept

β = Slope

Normal eqns of eqn (i) are:-

$$\sum Y = n\alpha + \beta \sum X \quad \text{--- (ii)}$$

$$\sum XY = \alpha \sum X + \beta \sum X^2 \quad \text{--- (iii)}$$

$$\Rightarrow n\alpha = \sum Y - \beta \sum X$$

$$\therefore \alpha = \frac{\sum Y}{n} - \beta \frac{\sum X}{n}$$

$$\therefore \alpha = \bar{Y} - \beta \bar{X}$$

$$\Sigma XY = \left(\frac{\Sigma Y}{n} - \beta \frac{\Sigma X}{n} \right) \Sigma X + \beta \Sigma X^2$$

$$\Sigma XY = \frac{\Sigma X \Sigma Y}{n} - \beta \frac{(\Sigma X)^2}{n} + \beta \Sigma X^2$$

$$n \Sigma XY = \Sigma X \cdot \Sigma Y - \beta (\Sigma X)^2 + n \beta \Sigma X^2$$

$$n \Sigma XY - \Sigma X \cdot \Sigma Y = \beta (n \Sigma X^2 - (\Sigma X)^2)$$

$$\therefore \beta = \frac{n \Sigma XY - \Sigma X \cdot \Sigma Y}{n \Sigma X^2 - (\Sigma X)^2}$$

b_{yx} = Regression coeff of Y on X.

$$= \frac{n \Sigma XY - \Sigma X \cdot \Sigma Y}{n \Sigma Y^2 - (\Sigma Y)^2}$$

Similarly,

$$b_{xy} = \frac{n \Sigma XY - \Sigma X \cdot \Sigma Y}{n \Sigma Y^2 - (\Sigma Y)^2}$$

$$r = \pm \sqrt{b_{yx} \cdot b_{xy}}$$

1. Regression eqn of 'Y' on 'X' is:

$$Y - \bar{Y} = b_{yx} (X - \bar{X})$$

2. Regression eqn of 'X' on 'Y' is:

$$X - \bar{X} = b_{xy} (Y - \bar{Y})$$

Coefficient of determination (R^2)

$$R^2 = 1 - \frac{SSR}{TSS}$$

TSS

$$TSS = TSS - SSR$$

TSS

$$SSR = \frac{SSR}{TSS}$$

TSS

observed

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

$\hat{Y}_i \rightarrow$ estimated
 $Y \rightarrow \bar{\alpha} + \bar{\beta} X$

$$TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

$\bar{Y} \rightarrow$ mean

Q)

X	Y	Find
24	25	1. Regression coefficient of Y on X and X on Y.
60	55	2. Find correlation coefficient between 'X' and 'Y'.
25	35	3. Find Regression eqn of Y on X, when X=49.
45	60	4. Find coefficient of determination.
35	45	
36	33	
38	20	
40	26	

Soln

X	Y	X ²	Y ²	XY
24	25	576	625	600
60	55	3600	3025	3300
25	35	625	1225	875
45	60	2025	3600	2700
35	45	1225	2025	1575
36	33	1296	1089	1188
38	20	1444	400	760
40	26	1600	676	1040
$\Sigma X = 303$ $\Sigma Y = 299$ $\Sigma X^2 = 12391$ $\Sigma Y^2 = 12665$ $\Sigma XY = 12038$				

1) Regression coeff of 'Y' on 'X'

$$b_{YX} = \frac{n \Sigma XY - \Sigma X \cdot \Sigma Y}{n \Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{8 \times 12038 - 303 \times 299}{8 \times 12391 - (303)^2}$$

$$= 0.7798 \approx 0.78$$

Regression coeff of 'X' on 'Y'

$$b_{XY} = \frac{n \Sigma XY - \Sigma X \cdot \Sigma Y}{n \Sigma Y^2 - (\Sigma Y)^2}$$

$$= \frac{8 \times 12038 - 303 \times 299}{8 \times 12665 - (299)^2}$$

$$= 0.4788 \approx 0.48$$

$$r = \sqrt{0.78 \times 0.48}$$

$$= 0.611$$

$r = 0.611$ means, there exist high degree of +ve correlation between 'X' and 'Y'.